



Microfluidic tools for biophysical sorting and characterization of fungal spores

Fabiola Elena Chacón Chalé

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Fabiola Elena Chacón Chalé. Microfluidic tools for biophysical sorting and characterization of fungal spores. Cellular Biology. Université Claude Bernard - Lyon I, 2025. English. <NNT : 2025LYO10309>. <tel-05524027>

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**THESE de DOCTORAT DE
L'UNIVERSITE CLAUDE BERNARD LYON 1**

Ecole Doctorale N°160

Electronique Electrotechnique et Automatique (EEA)

Discipline :

Ingénierie pour le vivant

Soutenue publiquement le 12/12/2025, par :

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**Microfluidic tools for biophysical sorting and
characterization of fungal spores**

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Acknowledgments

I am sincerely thankful to all those who have accompanied me during this journey. This work would not have been possible without their support, so I would like to take this opportunity to thank everyone who contributed in any way.

First, I would like to express my deepest gratitude to my supervisors, Magalie, Jean-François, and Christophe. Thank you for trusting me with this project, for your continuous guidance and encouragement, and for always making time for our Monday morning meetings. Your insightful feedback, patience, and support have been invaluable in shaping both this thesis and my development as a researcher.

I would like to sincerely thank Thérèse and Gwennou for agreeing to review my manuscript in detail and for providing their valuable comments and insights. I also thank the rest of the jury, Myriam, Francesco, and Marie, for their thoughtful questions, constructive feedback, and the engaging discussions during my defense.

In addition, I am sincerely grateful to Marie-Charlotte, who assisted me during the AFM characterization. I am thankful for her mentoring, guidance, and willingness to answer all my questions. I would like to thank the Nanolyon platform staff for their support and for providing training on the equipment.

I wish to acknowledge "la région Auvergne-Rhône-Alpes" for funding the BOTRY-PATH project (Pack Ambition Recherche 2021), and CONIDIA Coniphy for their participation in the project.

I am grateful to my colleagues for making each day enjoyable and for the opportunity to always learn new things about different cultures. I am especially grateful to the "Bureau de la Passion" for all the wonderful moments shared both inside and outside the office, including the karaoke sessions and all the laughter we shared together.

Finally, I would like to thank my family and friends back in Mexico. They were my first support on this journey, keeping me updated on everything happening at home and sharing long video calls that always lifted my spirits. I am also grateful to the friends I met here Diana and Orquídea, who patiently listened to the same stories and experiments countless times and were always there for me.

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List of acronyms

μOS Microfluidic Optical Stretcher.

AFM Atomic Force Microscopy.

AsPFF Asymmetric Pinched Flow Fractionation.

BFF Branch Flow Fractionation.

BMA Bone Marrow Aspirate.

CM Clausius-Mossotti factor.

COC cyclic olefin copolymer.

DEP dielectrophoresis.

DLD Deterministic Lateral Displacement.

EDTA Ethylenediaminetetraacetic acid.

EIS Electrical Impedance Spectroscopy.

FACS fluorescence-activated cell sorting.

FCW fungal cell wall.

FSC Forward-scattered light.

GFP green fluorescent protein.

IFC impedance flow cytometry.

InvOLS inverse optical lever sensitivity.

MA Micropipette Aspiration.

MACS magnetic-activated cell sorting.

MAP Laboratory of Microbiology, Adaptation, and Pathogenics.

MF microfiltration.

MSCs Mesenchymal Stem Cells.

nDEP negative dielectrophoresis.

NF nanofiltration.

NLS nuclear localization signal.

PBS phosphate-buffered saline.

PCR Polymerase Chain Reaction.

pDEP positive dielectrophoresis.

PEG Poly Ethylene Glycol.

PEO polyethylene oxide.

PFF Pinched Flow Fractionation.

PGMEA Propylene Glycol Methyl Ether Acetate.

PMMA poly(methyl methacrylate).

PSD power spectral density.

RBCs red blood cells.

RO reverse osmosis.

SEM scanning electron microscopy.

SSC Side-scattered light.

UF ultrafiltration.

WT wild-type strain.

Introduction and context

B. cinerea is a necrotrophic fungus that can infect over 200 plant species of economical importance, such as ornamental and nursery plants, vegetables, and fruits, among others [1]. Because of its role in the development of grey mold, *B. cinerea* is the leading biological cause of post-harvest diseases in plants. It is ranked as the second "World's top 10 fungal pathogens in molecular plant pathology", preceded only by the rice pathogen *Magnaporthe oryzae* [2]. Worldwide, *B. cinerea* infections in grapes, vegetables, and berries have a significant economic impact [3]. Annual losses are estimated between \$10 to \$100 billion dollars [4].

Infection by *B. cinerea* is initiated by the landing of a spore (Fig. 1.1 shows a *B. cinerea* spore) on the plant tissue, its germination, and the development of a penetration structure. Following plant tissue colonization, the disease spreads when the fungus reproduces asexually. The spores generated through this reproduction are called conidia. The conidia are carried in several ways, including by wind, water, insects, machinery, and workers' hands. Plants are affected during both pre- and post-harvest stages. Symptoms of the infection include leaf spot, blight, stem canker or rot, and damping-off of young seedlings [1]. Currently, the use of conventional fungicides is the most common strategy to control and kill the fungus. However, these non-selective fungicides can be harmful to the environment, food, and human health [5]. Another concern about these products is that they promote the emergence of new fungicide-resistant strains. For this reason, restrictions on the application of fungicides have increased in recent years. Hence, the importance of developing new and safer strategies to control gray mold [6].

Some approaches to control the fungus use antimicrobials from bio-sources, chemicals that specifically attack the fungus's activity, or chemicals that promote a defense response in the hosts [7]. Developing targeted fungicides requires understanding and identifying the specific genes and proteins necessary for pathogenesis [8]. Furthermore, by comprehending the molecular biology of the fungus, one can find possible targets for creating new antifungals while limiting potential risks to human health and the environment [9]. In line with this objective, biologists needed and have developed

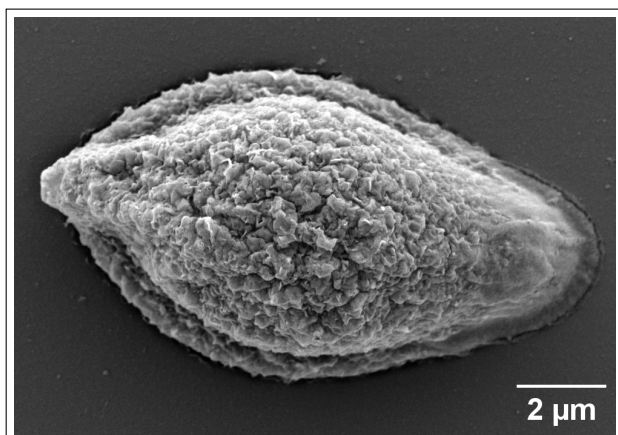


Figure 1.1: SEM image of a *B. cinerea* spore

protocols to create mutant strains of *B. cinerea*, which is explained in the following section.

1.1 Mutant production in *B. cinerea*

Information about the mechanisms controlling the interaction between plants and *B. cinerea* has increased over the last 20 years, but much remains unknown. One way to study this interaction is to identify and characterize the genetic bases responsible for fungal virulence. In recent years, scientists at the Laboratory of Microbiology, Adaptation, and Pathogenics (MAP) UMR5240 - Lyon 1, have produced mutant strains of *B. cinerea* to understand which genes and their corresponding proteins take part in the infection process. Besides contributing to our general knowledge of necrotrophic fungi, results are expected to provide new information useful for the development of fungicides with a site-specific mode of action. Such fungicides promise to reduce harmful effects on the environment. The protocol developed at MAP for mutant production consists mainly of three steps:

1. **Gene modification:** it is the modification of the fungal DNA by inserting DNA cassettes through a process called transformation.
2. **Genetic purification:** it consists of the sorting of fungal spores (conidia) produced by the genetically modified strains to select and isolate a descendant with homogeneous DNA material. This is required because the fungal cells in *B. cinerea* contain multiple nuclei, and in many cases, the gene modification does not occur in all of them. The effect of that modification cannot be assessed or evaluated as long as one or more non-modified (wild-type / parental) nuclei are present alongside the modified nuclei in the fungal cells.
3. **Mutant characterization:** this step is the study of the modification of the infection process caused by the mutation.

1.1.1 Gene modification

The first step in the mutant production protocol is gene modification. A DNA cassette containing a selection marker must be inserted into the fungal genome, either randomly or at a chosen place (genetic locus). To target the DNA cassette to a specific

locus, two DNA sequences with a sequence complementary to the DNA sequence of the genetic locus of interest must flank the antibiotic resistance gene (this is possible in *B. cinerea* because its entire genome sequence is known). Technically, the introduction of the DNA cassette inside the fungal cells is performed by transformation.

1.1.1.1 Protoplast-mediated transformation

The protoplast-mediated transformation is widely used because of its simplicity and rapidity in the gene insertion process [10], and a schematic version of it is presented in Figure 1.2. The first step is an enzymatic treatment of young filaments of *B. cinerea* to break the fungal cell wall (FCW). This digestion of the FCW causes the budding of the cell membrane below and the liberation of round biological particles called **protoplasts**. Some of these protoplasts contain all the necessary intracellular material to regenerate a live fungal cell later, once again surrounded by a newly regenerated FCW. In the meantime, and assuming that a proper osmotic pressure is maintained to avoid the bursting of the protoplasts, the latter can be manipulated to uptake exogenous DNA fragments. This is achieved by using a solution containing mainly three compounds, first CaCl_2 which helps the neutralization of the highly negatively charged DNA molecule to cross the negatively charged cell. Second, the amphiphilic Poly Ethylene Glycol (PEG) helps the crossing of the hydrophobic lipid bilayer that constitutes the protoplast membrane [11]. Third, the designed DNA cassette of interest. Following washes, the transformed protoplasts are plated on Petri dishes that contain nutrients (carbon, nitrogen, sulfur, phosphate) and the antibiotic corresponding to the resistance gene inserted into the DNA cassette. Hence, the genetically modified protoplast—expressing the antibiotic resistance gene—can obtain nutrients to reconstruct their FCW and initiate the typical filamentous growing process that characterizes most fungi. In contrast, the non-modified protoplast (the large majority of them) will not generate any fungal colony. The antibiotic-resistant fungi selected through this process are called **transformants**.

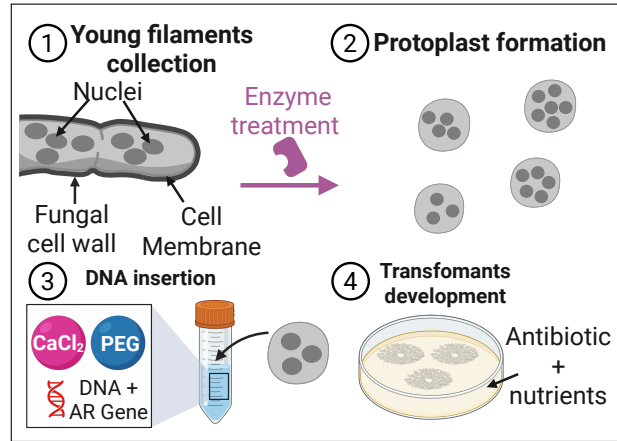


Figure 1.2: Protoplast-mediated transformation process. The first step is the collection of young filaments, followed by the enzymatic degradation of their fungal cell wall to obtain "naked" cells called protoplasts. The protoplasts are kept under controlled osmotic pressure and are mixed with the modified DNA cassette containing an antibiotic resistance gene (AR gene), together with CaCl₂ and PEG to favor the formation of a complex able to cross the protoplast membrane. After several washes, the protoplasts are plated on a Petri dish containing nutrients and the antibiotic corresponding to the resistance gene used in the genetic manipulation. Only the protoplasts that have inserted in their genome the new DNA material will survive, regenerate a cell wall, and grow into a typical fungal colony. These resistant fungi are called transformants. Adapted from Nielsen et al., 2019 [11].

1.1.1.2 Bacterium-mediated transformation

The bacterium-mediated transformation is a process in which the DNA cassette designed to mutate the gene of interest is transferred into the fungus by a bacterium called *Agrobacterium tumefaciens* [12], as illustrated in Figure 1.3. Firstly, these bacteria are transformed for them to uptake the DNA cassette. The bacterial transformation protocol is simpler than the one described above for *B. cinerea* and relies on a heat shock step. Secondly, the transformed bacteria are plated on a Petri dish containing nutrients and spores of the fungus. After about two days of co-culture, during which the interaction between the bacteria and young fungal cells emanating from the germinated spores takes place, the cells are transferred to new plates containing one antibacterial molecule and the appropriate antibiotic to select the fungal transformants (the fungal cells that successfully inserted the DNA cassette into their genome). The fungal colonies developing on these new plates are collected.

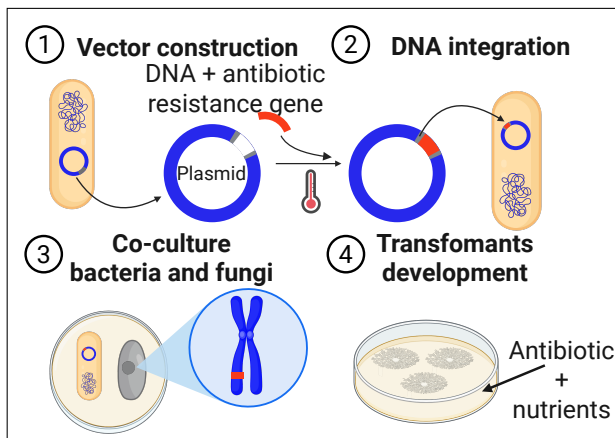


Figure 1.3: Bacterium-mediated transformation process. Adapted from Nielsen et al., 2019 [11].

1.1.2 Genetic purification

Most filamentous fungi contain multiple nuclei in each cell and therefore multiple copies of their genome (one genome per nucleus) [13]. As a consequence, fungal transformants that have integrated the DNA cassette into one copy of their genome still contain non-mutated copies of their genome. Under these conditions, the expression of the mutation will depend on many factors. It will depend on the mutation itself (dominant or recessive), the growing conditions (presence or absence of the selective antibiotic), or the gene involved (expressed throughout the fungal development or at certain stages only).

Genetic purification is a step required to eliminate all the non-mutated copies of the transformants' genome and to obtain a fungal strain with homogeneous DNA material. Such strain can then be called a mutant strain. To do so, the transformants need to be grown in vitro until asexual reproduction (about 10 days) to collect spores (conidia). These spores contain 1 to 8 nuclei, several of them then need to be isolated, and the fungal colonies they will produce need to be tested to check for their genetic homogeneity. The process is repeated until a mutant is obtained.

The spore suspension is diluted, and individual spores need to be isolated. There are two approaches to achieve this isolation, regardless of the transformation method employed:

1. In the first method, as described in the previous section, the spores are plated on a Petri dish containing nutrients and antibiotics. After two to three days of incubation, very small colonies can be detected at the surface of the nutrient solid medium, using a microscope, and can be transferred, manually and individually, onto new dishes using a scalpel or similar tool.
2. In the second method, a commercial device with micro-channels leading to micro-wells is used. Flushed with a calibrated solution of the transformants spores, each well will contain 0, 1 or 2 spores, according to a Poisson distribution. By using a microscope, this allows for the identification and selection of wells that contain only a single spore, which can then be transferred, manually and individually, onto a Petri dish using a pipette.

About three days later, the fungal filaments of the newly formed individual colonies are collected, and a DNA analysis is performed on them. This analysis, called Polymerase Chain Reaction (PCR), detects 1) whether the modified DNA is present and 2) whether non-modified DNA is still present. Two scenarios are possible, depending on the outcome of the test. If the test detects that there is a heterogeneous DNA material (not all nuclei have the mutant gene), then **scenario 1** (in blue Figure 1.4) will occur. Spores are produced again (about 10 days), collected, and the cycle re-starts. Scenario 1 is a looping cycle that continues to loop until the detection of homogeneous DNA is achieved, which can take from two weeks at the minimum up to twelve months. Alternatively, if the test detects homogeneous DNA material, then **scenario 2** (in red in Figure 1.4) will occur. In this case, the mutant fungus is considered pure and ready to be characterized. One last 10-day culture will provide a batch of spores from that mutant, which can be conserved at -80°C.

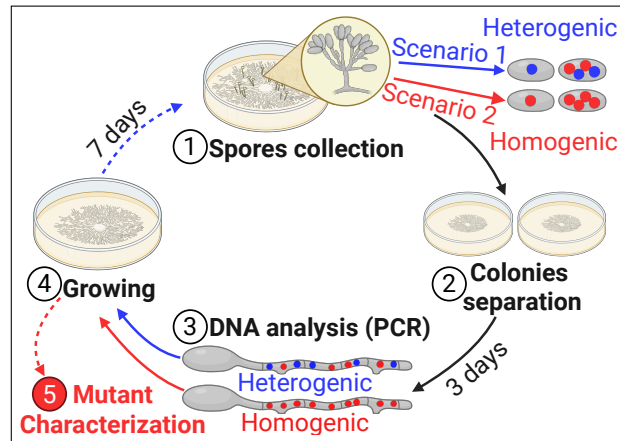


Figure 1.4: Purification process. After the collection of spores, they are sorted and manually isolated to generate new fungal colonies. These colonies are then genetically analyzed using a PCR reaction designed to detect the presence of the modified DNA and that of the non-modified DNA. Two scenarios are possible: the new colonies all contain heterogeneous DNA and the whole process needs to be repeated (blue path), or at least one of the new colonies contains homogeneous DNA and the now isolated mutant can be studied (red path).

1.1.3 Mutant characterization

The last step of the protocol elaborated at the biology laboratory is the mutant characterization. Primarily, it consists of observing and quantifying the effects of the mutation compared to the wild-type strain. Different mutations can cause different modifications in the infection process. The characterization can be done in two ways:

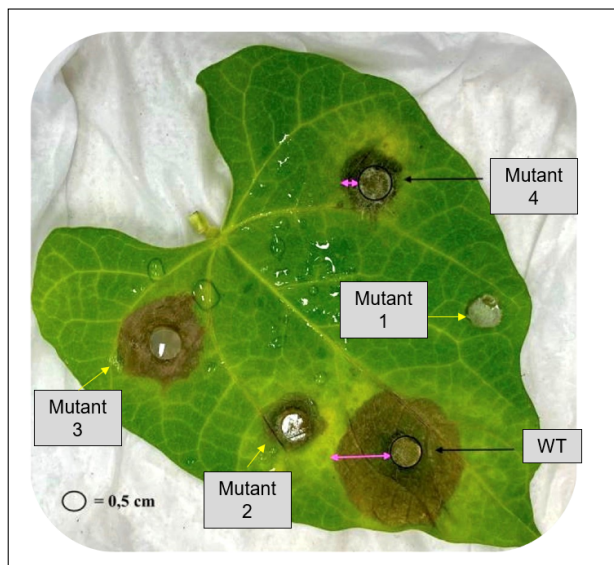


Figure 1.5: Leaf infected with *B. cinerea*. A bean leaf was exposed to the wild type and four different mutant strains of *B. cinerea*. The variation in virulence can be observed. The most harmful is the wild-type strain, while the less virulent strain is mutant 1.

- Observing the **plant/fungus interaction**: spores or filaments of the mutants are deposited onto the surface of a plant to observe any change in virulence in the following days. An example of the effect of a mutation on the pathogenicity of *B. cinerea* can be seen in Figure 1.5. The brown circumference represents the infected area. The bigger the brown circumference, the more harmful the strain (wild type in this case).
- Observing the **fungus** by itself (growth, reproduction, etc): the mutant fungus

is inoculated onto a Petri dish to develop. The rate of its radial growth can be measured over time. Following exposure to near-UV light, the production of conidia can also be quantified, for example.

The different observations enable a better understanding of which genes are relevant to the infection process. For instance, some genes can play no role in the fungal growth, while having a significant role in the infection process of the host.

1.2 Objectives of the PhD project

Clearly, the study of mutants is an efficient way to characterize the importance of a gene in the infectious process of *B. cinerea*. However, the main hurdle to the current protocol is the "**genetic purification**" step, which is highly time-consuming. The shortest time to obtain a pure colony is two weeks, but in general, the purification process lasts 3 months at best. The BOTRYPATH project, in which my thesis project fits, proposed an alternative to tackle this issue using microfluidic approaches, based on biophysical analysis. It also proposed to bring new ways to characterize the mutants. Biophysical approaches are selected because they are label-free and do not compromise the viability of particles or cells. As mentioned earlier, it is important to emphasize that a spore of *B. cinerea* contains up to eight nuclei, each of which bears versions of the gene of interest under the wild type or mutated form. Therefore, the most straightforward method to obtain homogeneous DNA material is to select spores that contain a singular nucleus.

For this reason, the first objective of the project was to generate a microfluidic system capable of separating mononuclear spores from multinuclear spores according to their biophysical properties. Concerning the biophysical characterization of the mutants, we proposed to record biophysical signatures of the fungal cell wall to bring information on the mechanical and electrical properties of this essential component. This has not been addressed so far and will improve our knowledge of how this component is remodeled during fungal development, as well as how it can be affected by mutations disrupting genes involved in its synthesis or dynamics.

In essence, the two principal objectives of the thesis were first to achieve a label-free sorting of *B. cinerea*'s spores based on mechanical (size) properties. This sorting will speed up the genetic purification step. The second objective was to set up appropriate microfluidics tools to characterize the biophysical properties of fungal cell walls and to study the correlation between the wild-type and the mutant signatures.

Résumé du Chapitre 1

Ce chapitre est principalement divisé en trois parties. Tout d'abord, il introduit *B. cinerea* et la pertinence de cette étude. Ensuite, il décrit la production de mutants selon le protocole actuellement en cours au laboratoire MAP. Enfin, il présente les objectifs de ce projet de doctorat.

Pour commencer, *B. cinerea* est brièvement décrit. Il est classé comme le deuxième pathogène fongique le plus important en pathologie moléculaire végétale. Ce champignon nécrotrophe cause des pertes économiques importantes à l'échelle mondiale, car il peut infecter une grande variété de cultures.

Actuellement, la stratégie la plus courante pour lutter contre ce champignon est l'utilisation de fongicides chimiques. Cependant, ces composés peuvent être nuisibles à l'environnement, à la sécurité alimentaire et à la santé humaine. Cela souligne l'urgence de développer des méthodes alternatives pour lutter contre la pourriture grise (infection par *B. cinerea*). Parmi les stratégies prometteuses figurent l'utilisation de fongicides biosourcés ou ciblés.

Pour développer de telles solutions, il est essentiel de comprendre les gènes et les protéines impliqués dans la pathogénie. C'est précisément l'objectif des chercheurs du laboratoire MAP. Afin d'identifier les gènes clés, il est nécessaire de générer et d'étudier des souches mutantes.

La deuxième section décrit le protocole utilisé pour développer les souches mutantes, qui se compose des étapes suivantes :

1. **Modification génétique** : Cette étape consiste à modifier l'ADN fongique en insérant des cassettes d'ADN, soit de manière aléatoire, soit à un locus génétique spécifique. Ce processus est connu sous le nom de transformation. La transformation peut être réalisée de deux manières différentes : par transformation médiée par les protoplastes ou par transformation médiée par des bactéries.
2. **Purification génétique** : Une fois les spores modifiées génétiquement, l'étape suivante consiste à purifier l'échantillon. Cela signifie sélectionner uniquement les conidies (spores fongiques) ayant un ADN homogène (tous les noyaux contenant les mêmes gènes) et éliminer celles ayant un ADN hétérogène (noyaux avec un contenu génétique différent).

Pour ce faire, une analyse de l'ADN (PCR) doit être réalisée sur des colonies individuelles. Afin de garantir une croissance séparée des colonies, les spores

peuvent être isolées de deux manières : en diluant l'échantillon et en le déposant sur une boîte de Petri (de manière à ce que les colonies ne se touchent pas), ou en utilisant un dispositif commercial permettant de séparer les particules individuellement. Une fois que les colonies ont suffisamment poussé pour permettre la récolte des hyphes (environ 3 jours), la surface peut être raclée et une analyse PCR peut être effectuée sur les filaments fongiques.

3. **Caractérisation des mutants** : Après la purification, la dernière étape consiste à évaluer l'impact de la mutation sur le processus d'infection. Cela implique d'observer et de quantifier les effets de la mutation par rapport au type sauvage. La caractérisation peut être réalisée de deux manières : en étudiant l'**interaction plante/champignon**, ou en analysant le **champignon lui-même** (par exemple, la croissance, la reproduction, la morphologie).

Ce protocole est bien établi ; cependant, la principale difficulté réside dans l'étape de purification génétique. Comme mentionné précédemment, il est essentiel d'obtenir un échantillon avec un ADN homogène ; sinon, le processus ne peut pas passer à l'étape suivante.

Si les résultats de la PCR révèlent que les noyaux contiennent un ADN hétérogène, le processus de purification doit être répété. Cela implique de séparer à nouveau l'échantillon, de laisser les colonies se développer, de racler la surface, puis d'effectuer une PCR sur les filaments. Ce cycle peut se poursuivre pendant plusieurs semaines — généralement autour de deux mois — jusqu'à ce que la PCR confirme que l'échantillon contient un ADN homogène.

De plus, la troisième étape repose généralement sur des observations visuelles. Il serait donc intéressant de mesurer des paramètres biophysiques permettant de mieux comprendre ce qui se passe lors de l'infection et quelles différences existent entre les souches de type sauvage et les mutants.

Tous les points précédemment mentionnés mènent aux principaux objectifs de cette thèse :

1. Fournir une technique de tri microfluidique sans marquage, basée sur les propriétés biophysiques, afin d'accélérer le processus de purification génétique.
2. Fournir des outils pour caractériser les propriétés biophysiques des parois cellulaires fongiques et étudier la corrélation entre les signatures du type sauvage et des souches mutantes.



Part 1

Genetic purification in microfluidics

State of the art: sorting techniques

As discussed in the previous chapter, properly visualizing the effect of *B. cinerea*'s gene modifications in the plant-spore interaction requires a spore suspension with homogeneous DNA. Nevertheless, the current method to obtain such a suspension is long and tedious. Therefore, it is important to develop faster and more efficient procedures that allow the identification and isolation of the desired subpopulation. Cell sorting techniques aim to enrich or isolate specific cell types accurately based on different properties, such as **biochemical** [14], **mechanical** [15], **electrical** [16], **optical** [17], or **magnetic** [18]. These properties are briefly described in the following paragraphs.

Biochemical cell sorting takes advantage of differences in the biochemical properties of cells, either on their surface or within the cell. For example, specific biomolecules such as proteins, lipids, carbohydrates, or nucleic acids can be identified through antibody or aptamer interactions, which are usually coupled with fluorescent labels for detection and separation [19, 20]. The most well-known technique in this category is fluorescence-activated cell sorting (FACS), which is considered the gold standard [21]. This method allows high specificity in identifying and sorting target cells from a complex or heterogeneous population.

Similarly, magnetic-activated cell sorting (MACS) relies on the magnetic properties of particles attached to cells. These particles can be labeled either directly (with a primary antibody-conjugated microbead) or indirectly (using a primary antibody and a secondary antibody-conjugated microbead) [22]. By applying a magnetic field gradient, magnetically labeled cells can be separated from the rest of the sample [23].

In contrast to biochemical or magnetic methods, mechanical cell sorting offers a different approach based on physical properties rather than molecular markers. Cell sorting based on intrinsic mechanical properties is particularly useful when cells present similar molecular biomarkers or when detection without any antibody binding is desired [24]. Generally, these sorting systems are label-free methods that separate cells based on size, shape, or deformability [25]. Such characteristics vary depending on the cell type or physiological state [26]. This mechanical approach offers a fast and

non-invasive alternative for applications in both research and clinical diagnostics [27].

Electrical properties represent another intrinsic feature that can be exploited for cell sorting without molecular labeling. Key parameters include polarizability, electrical impedance, and surface charge, which vary among different cell types [28, 29]. A widely used technique based on these properties is dielectrophoresis (DEP), where a non-uniform electric field induces a force on polarized cells, enabling their manipulation and separation. DEP offers a non-invasive, label-free alternative that is particularly effective for distinguishing cells with subtle electrical differences [30, 31].

Optical properties offer an additional intrinsic basis for label-free cell discrimination. These include autofluorescence, light scattering, and absorption, which are directly related to the cell size, granularity, and internal structure composition (*e.g.*, mitochondria) [32, 33]. Techniques like flow cytometry use these intrinsic optical signals or combine them with fluorescent labeling to measure parameters such as Forward-scattered light (FSC) or Side-scattered light (SSC), allowing for the distinction and sorting of specific cell types with high speed and accuracy [34, 35].

As described in the previous paragraphs, cell sorting methods can rely on various intrinsic cellular properties or externally labeled biomarkers. Selecting the most suitable approach depends on the biological question and the cell type of interest. In this project, the goal is to isolate spores with homogeneous genetic material. Since *B. cinerea* produces spores that may contain one or multiple nuclei; the most straightforward way to ensure DNA homogeneity is to select spores with a single nucleus. Based on this objective, not all sorting approaches are appropriate. Techniques such as FACS and MACS are not suitable because these spores do not naturally exhibit magnetic or fluorescent properties. Moreover, external labeling is not an option because it could alter their physiology or even damage the spores. Therefore, the most relevant methods for this application are based on mechanical, electrical, and optical properties, which will be discussed in the following sections. These methods will be grouped into two main categories: passive and active sorting techniques. For each category, different technologies will be introduced and their working principles explained, with an emphasis on their potential for spore sorting to identify the most suitable approach for our objectives.

2.1 Passive sorting methods

Passive cell sorting methods do not apply external forces; instead, they rely on hydrodynamic forces exerted on cells as they flow through microchannels with specific geometries [36]. Separation is achieved based on cell size, morphology, or deformability. Passive microfluidic technologies offer precise manipulation, low-cost fabrication, simple integration, and high throughput capabilities [37]. This section provides an overview of four representative passive cell sorting methods: microfiltration, pinched flow fractionation, deterministic lateral displacement and inertial sorting.

2.1.1 Microfiltration

Membrane filtration is a size-based separation process that can separate solid immiscible particles from liquids or gases and dissolved solutes in liquids or gases in a mixture. It consists of a membrane that acts as a barrier, selecting particles that pass through it and particles that remain on the surface. It can be classified based on the size of the membrane pores. This classification is illustrated in Fig. 2.1. These include microfiltration (MF), ultrafiltration (UF), nanofiltration (NF), and reverse osmosis (RO), each suited to a specific size range of particles [38, 39].

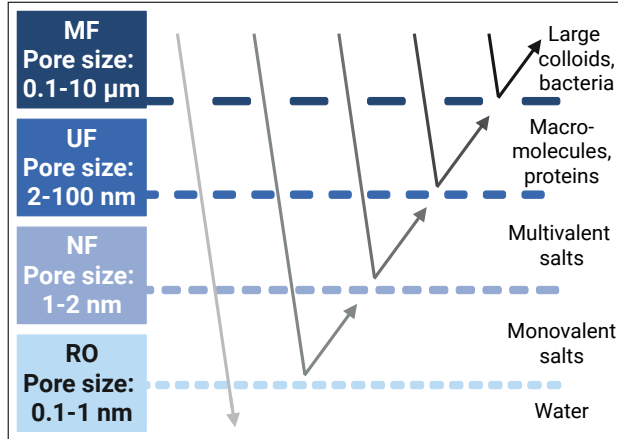


Figure 2.1: Classification of membrane filtration based on their pore size. Adapted from Yang et al. [38].

Among the different membrane filtration techniques, MF is particularly suitable for separating particles in the micron range (approximately 0.1 to 10 µm). Since the spores of *B. cinerea* fall within this size range, the discussion in the following paragraphs will focus on MF rather than other filtration techniques with smaller pore sizes. This section will examine the principles, applications, and relevance of MF for spore sorting.

The pore diameter (d_p) is a key parameter to determine the separation efficiency in microfiltration. Particles larger than d_p are retained on the membrane surface while smaller particles pass through. Therefore, the selection of the appropriate pore size is crucial and depends on the target particles [40]. The flux (J) through an MF membrane can be related to several factors: the pore diameter, the number of pores in the membrane (N), the porosity (ϵ) that is the fraction of the membrane volume occupied by the pores, the applied pressure (P_T), the viscosity of the solvent (μ) and the membrane thickness (Δx). This relationship can be described by the Hagen-Poiseuille equation, which models laminar flow through cylindrical pores:

$$J = \frac{\epsilon d_p^2 P_T}{32 \Delta x \mu} \quad (1)$$

$$\epsilon = \frac{N \pi d_p^2}{4} \quad (2)$$

it can be observed that larger pore diameters allow higher fluxes but reduce the ability of the membrane to retain smaller particles. Therefore, it is important to balance between achieving a high flux and maintaining efficient separation. This equation makes three assumptions. First, all pores are cylindrical and parallel to each other;

second, the fluid is Newtonian, and the flow follows a laminar regime; and third, the fluid velocity at the pore walls is zero (no-slip condition), meaning that the fluid does not slide along the pore surface [41].

In addition to the theoretical understanding of fluid flow in microfiltration, it is also important to consider the operational configurations in which MF is applied [42]. MF can be performed in two configurations: dead-end and crossflow (illustrated in Fig. 2.2). In dead-end filtration, the initial solution crosses vertically through the membrane, accumulating particles on the membrane surface. Dead-end filtration is mostly used in applications where the particle concentration is low. In contrast, crossflow filtration flows parallel across the membrane filter instead of into the filter, allowing only a part of the fluid to pass through the membrane. This mode helps reduce the fouling by the continuous flow that sweeps away the retained particles. It is mainly applied when the solution has a high particle concentration [43].

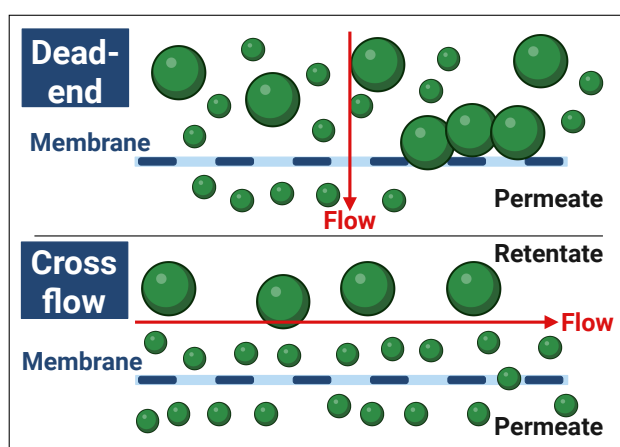


Figure 2.2: Filtration modes: Dead-end (top) and crossflow (bottom). Adapted from Tsi-branska et al.[43].

In terms of performance, the advantages of MF include the low-pressure operation required, generally lower than 2 bar. Therefore, the equipment cost is reduced considerably. During the MF process, there is no change in pH or ionic strength, an advantage while working with biological samples. The principal drawback of microfiltration is particle fouling, which refers to accumulating particles either on the membrane surface or within the porous, gradually clogging it. Over time, this leads to cake layer formation, where retained particles accumulate on the membrane surface. In both cases, the flux and the separation efficiency are significantly reduced [44, 45].

Among its many industrial applications, MF can be used to separate bacteria and spores from milk [46], treat domestic water supplies, and manage industrial wastewater. In the case of fungi, MF has been used effectively to concentrate samples. For instance, Todero et al. [47] applied microfiltration membranes to concentrate a fermented broth containing *Phoma* sp.. This process increased the concentration of phytochemicals—bioactive compounds that can harm weeds, enhancing their potential as bioherbicides. Such an approach offers an environmentally friendly alternative to chemical herbicides by promoting the use of naturally derived fungal compounds.

Similarly, Dey et al. [48] combined MF with chemical coagulation to treat effluent from water-based paint production, targeting microbial contamination by bacteria and fungi. Using a 0.2 μm membrane, the system effectively removed microorganisms, resulting in permeate free of biological contaminants. This process reduced water

consumption by 55% and enabled water reuse in paint manufacturing.

These examples highlight MF applications involving microbial particles, such as fungal spores. The following section explores other passive sorting approaches based on fluid dynamics, particularly within microfluidic systems.

2.1.2 Pinched flow fractionation

The first microfluidic separation technique to be discussed is Pinched Flow Fractionation (PFF), a size-based method that operates under laminar flow conditions.

In this technique (Fig. 2.3), a suspension containing spherical particles and a buffer fluid are continuously injected into a narrow section of the microchannel, known as the pinched segment. By adjusting the flow rates of both fluids, the particle suspension is directed toward one sidewall of the pinched segment. As a result, the centers of the particles align with the flow and settle at a distance of one particle radius from the wall, regardless of their size. Immediately after this region, the channel widens abruptly. As the particles enter the broadened segment, flow forces act differently on them depending on their sizes: larger particles are displaced toward the center of the channel, while smaller particles remain closer to the sidewall [49, 50].

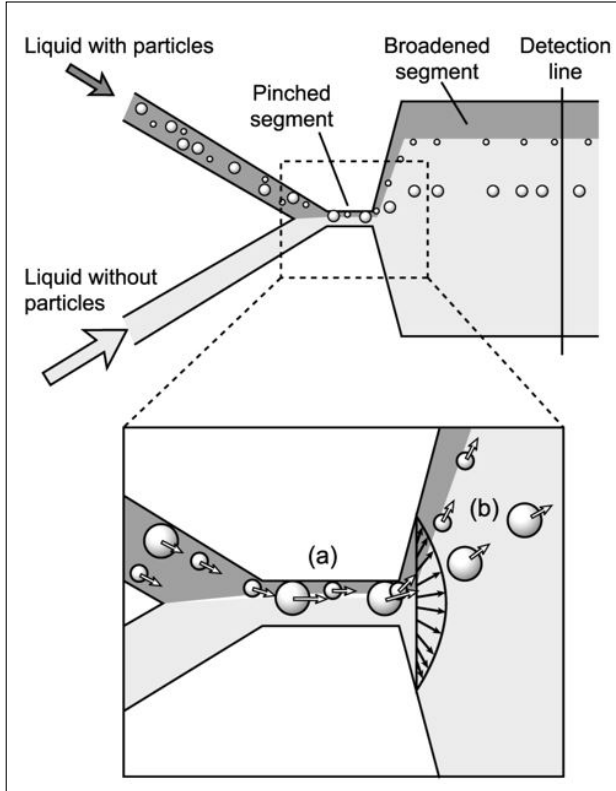


Figure 2.3: Schematic of the PFF device. It contains two inlets: one for the liquid with particles, and one for the liquid without particles, a pinched segment, where particles are focused toward the sidewall, regardless of their size, and a broadened segment, where particles are displaced to different positions within the channel according to their size [51].

To describe PFF, Yamada et al. [51] proposed a model derived from the Stokes equation and the principle of mass conservation. This model relates the distance from the center of a particle to the channel wall in the broadened segment (y_{PFF}) to the particle's radius (r_p):

$$y_{PFF} = \frac{w_b}{w_p} r_p \quad (3)$$

Here, w_b and w_p denote the widths of the broadened and pinched segments, respectively. This one-dimensional model relies on several assumptions: (1) the flow is laminar, (2) the particle is fully pinched before entering the broadened segment, and (3) the width-to-height ratio is large ($w_p \gg h$ and $w_b \gg h$).

Moreover, while w_p and w_b are key parameters in the separation process, the channel height (h) plays a significant role in minimizing particle fouling. It is also important to note that the model neglects both particle-fluid and particle-particle interactions [52].

While the model proposed by Yamada et al. [51] provides valuable insight into particle positioning and separation in the broadened segment, it does not fully address the limitations related to particle size resolution, as it was the first model of its kind. Since then, the PFF technique has continued to evolve.

For instance, it has been observed that the range of particle sizes that can be effectively separated is primarily determined by the flow rate ratio in the inlet channels, whereas the separation efficiency depends on the geometry of both the pinched and broadened segments. One of the main limitations of the original design is that particles significantly smaller than the channel width cannot be efficiently separated. To overcome this issue, an asymmetric version of PFF, known as Asymmetric Pinched Flow Fractionation (AsPFF), was introduced—most notably by Takagi et al. [53]. They demonstrated that AsPFF enables the separation of much smaller particles by modifying the geometry of the outlet channels, such as by shortening one outlet to change the hydraulic resistance. This illustrates some of the improvements and the adaptability of PFF-based techniques to a broader range of applications and particle sizes.

Among the main advantages of PFF is the extreme simplicity of its working principles. Srivastav et al. [50] have shown that PFF is an effective method for separating microspherical particles. However, a major drawback is the high dilution of the sample at the outlet, primarily caused by the continuous introduction of the buffer fluid. This dilution can be problematic in applications that require highly concentrated samples.

PFF has been successfully applied in fungal studies. For example, Park et al. [54] developed a microfluidic device that combines PFF and Branch Flow Fractionation (BFF) to separate fungal spores of *C. cladosporioides* from eukaryotic cells (NIH3T3 fibroblasts), achieving a separation efficiency of approximately 90%. Additionally, they tested the system with spores from two fungal strains, *A. niger* (2–5 μm) and *A. alternata* (10–15 μm), also obtaining around 90% separation efficiency.

Although there are relatively few studies applying PFF to fungal spores, it is important to note that spores are not typically spherical particles. This raises the question of whether PFF can be used to separate such irregularly shaped objects. Timary et al. [55] addressed this by using *E. coli* cells, both filamentous (elongated) and ovoid, in a PFF device. They concluded that particle trajectories in PFF are strongly influenced by both the shape and size of the particles, as well as the flow rate. Their results showed high sample purity and enrichment.

Beyond PFF, other passive microfluidic techniques have been developed for particle separation. One prominent example is Deterministic Lateral Displacement (DLD), a method that enables high-resolution and continuous separation based on particle size.

The following section explores the working principles and applications of DLD.

2.1.3 Deterministic Lateral Displacement

Deterministic Lateral Displacement (DLD) is a hydrodynamic method that enables the separation of particles according to their size. It was first described in 2004 by Huang et al. [56], where they demonstrated that DLD achieves separate particles with a resolution down to 10 nm. DLD gives a high resolution because the separation is based on the repetitive action of the particles.

A DLD device consists of an array of micropillar obstacles inside a microchannel [57]. Each subsequent row is shifted laterally at a set distance ($\Delta\lambda$) from the precedent, creating a periodic flow pattern. According to theory, particle separation is achieved when particles larger than a defined critical diameter (D_c) interact with the micropillars, resulting in consistent lateral displacement with each interaction. Larger particles exhibit a trajectory defined by the displacement angle (θ), resulting from the lateral row shifting. Alternatively, the smaller particles do not experience lateral displacement, so they continue moving along the direction of the fluid flow without deviation [58, 59].

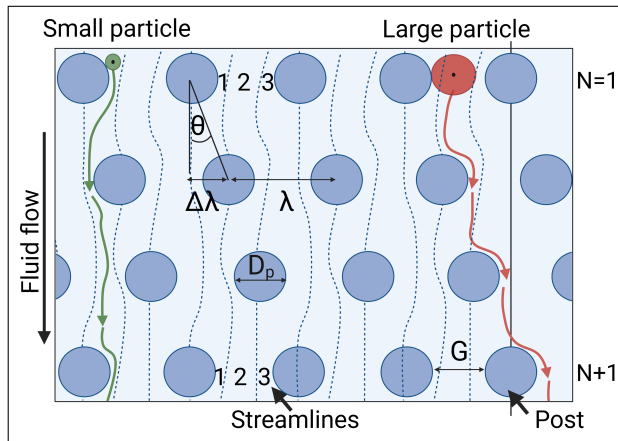


Figure 2.4: Schematic of Deterministic Lateral Displacement principle. Adapted from Davis et al. [59].

Designing a DLD microdevice involves taking into account a series of parameters (Fig. 2.4). The spacing or gap between posts is represented by the letter G and the post diameter by D_p [60]. The sum of G and D_p is the distance λ that represents the distance from center to center of two consecutive pillars in the same row. Each successive row presents a lateral shifting $\Delta\lambda$ caused by the displacement angle θ and the vertical distance between each row is G . The rows present a period N which is related to λ and $\Delta\lambda$ [61]:

$$N = \frac{\lambda}{\Delta\lambda} \quad (4)$$

In addition, the period N also refers to the number of streamlines between two consecutive pillars in the same row. In Fig. 2.4, the array has a period of $N = 3$, which means there are three streamlines between each pair of pillars (illustrated between 2 consecutive pillars with dashed blue lines). In this example, it can be observed that as the flow moves through the array, the streamlines shift their positions. In the

first row, the streamlines between two consecutive pillars are numbered 1, 2, and 3 from left to right. In the second row, within the same pillar columns, the streamlines are shifted so that the first streamline (from left to right) corresponds to streamline 2 instead of streamline 1. This pattern continues until the $(N + 1)$ -th row, where the first streamline returns to be streamline 1, as illustrated in Fig. 2.4. This behavior is important for particle separation. Looking again at the figure, the center of mass of the red particle lies outside the width of the first streamline. As a result, at each pillar, the particle is laterally displaced to the next streamline. This behavior is known as the **bumping or displacement mode**. In contrast, the green particle's center of mass is within the first streamline, so it follows a path that alternates slightly but remains within the same streamline. This is known as the **zigzag mode** [56].

Later, Inglis et al. [62] defined the critical diameter (D_c) as the diameter that determines which path a particle will follow, and it is given by

$$D_c = 2\beta \quad (5)$$

where β represents the width of the first streamline. To calculate β and D_c , they integrated the flow profile $u(x)$, from zero at the post edge to β . It is worth noting that although each streamline carries the same fluid flux, they do not necessarily have the same width. For more details on how the width was obtained, readers are encouraged to consult the original paper.

They also stated that β can be expressed in terms of the gap G and the row shift fraction (also known as the array slope) ε , leading to:

$$D_c = 2\eta G\varepsilon \quad (6)$$

Here, η is a parameter that accounts for the non-uniformity of the flow through the gap. The row shift fraction ε can be described as the inverse of the period N :

$$\varepsilon = \frac{\Delta\lambda}{\lambda} = \frac{1}{N} = \tan\theta \quad (7)$$

They compared their experimental results with those of Huang et al. [56]. Fig. 2.5 shows the outcomes of these experiments. This information was useful for selecting the parameters of the DLD arrays, as it clearly illustrates the relationship between particle diameter, gap size, and row shift fraction.

In his work, Davis [63] derived an empirical formula to approximate the critical diameter D_c :

$$D_c = 1.4G\varepsilon^{0.48} \quad (8)$$

From this equation, it can be inferred that the most influential parameters are the gap G and the row shift fraction ε , which, as previously mentioned, can be expressed in terms of other array parameters.

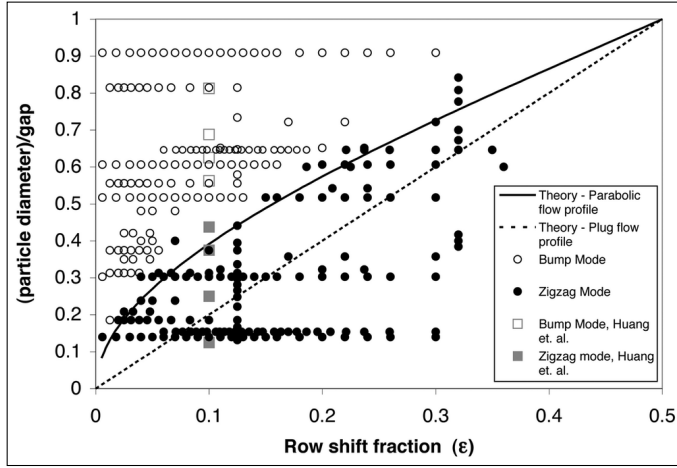


Figure 2.5: Experimental results from Huang et al. [56] (white dots) and Inglis et al. [62] (black dots). The graphs show the relationship between the particle diameter-to-gap ratio and the row shift fraction.

Various pillar shapes have been explored in order to optimize particle separation performance and reduce issues such as clogging or low throughput. The geometry of the pillars can influence the critical diameter and sorting efficiency. Below are some of the shapes that have been studied:

- **Triangular pillars:** Loutherback et al. [64] implemented triangular pillars and observed that this geometry reduced clogging in the array (Fig. 2.6 (A)). This is because triangular pillars allow a larger gap between posts for the same critical diameter as circular pillars. A larger gap also permits a higher flow rate, which increases the overall throughput. They suggested that the pillar size should be larger than the gap.
- **Regular polygonal pillars:** The same study [64] also examined regular polygon shapes with 3 to 11 sides (Fig. 2.6 (B)). They compared pillars with sharp vertices (even-numbered polygons) to those with a combination of flat edges and sharp vertices (odd-numbered polygons). They found that regular polygons with a combination of flat edges and sharp vertices (e.g., odd-numbered sides) tend to produce smaller critical diameters than those with only sharp vertices (e.g., even-numbered sides), for the same gap and ϵ . However, as the number of sides increases, the vertices become less pronounced, and the critical diameter gradually approaches that of a circular post.
- **Inverse L-shaped pillars:** Jiang et al. [65] studied inverse L-shaped pillars using both rigid particles and red blood cells (RBCs) (Fig. 2.6 (C)). They developed a predictive formula for the critical diameter in such arrays. These pillars promote localized deformation of RBCs, leading to different trajectories depending on particle stiffness. This enables sorting based on cell deformability.
- **I-shaped pillars:** Similarly, Zeming et al. [66] designed I-shaped posts to separate non-spherical and deformable particles. These pillars induce rotational movements, increasing the hydrodynamic diameter of the cells and improving separation performance. Later, Ranjan et al. [67] implemented various configurations of I-shaped pillars (Fig. 2.6 (D)) to further enhance the separation performance.

DLD devices can also achieve multi-size particle separation by incorporating arrays

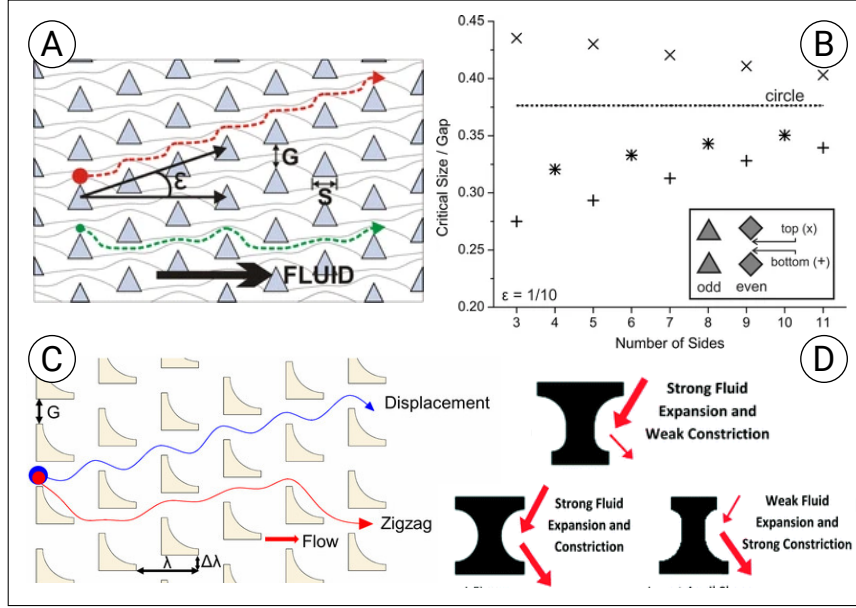


Figure 2.6: Different post shapes implemented in DLD arrays. (A) Triangular and (B) regular polygonal pillars used by Loutherback et al. [64]; (C) inverse L-shaped pillars used by Jiang et al. [65]; and (D) I-shaped pillars used by Ranjan et al. [67].

with sequentially decreasing critical diameters (D_c). Davis [63] proposed three designs for this purpose: the multiple array, the chirped array, and the cascade array. In the multiple array configuration (Fig. 2.7 A), the device includes sections with different gap sizes. This design is more prone to clogging, as each successive array has a smaller gap. The chirped array (Fig. 2.7 B) addresses this issue by varying the row shift fraction instead of the gap, thereby reducing the risk of clogging. Finally, the cascade array (Fig. 2.7 C) increases the separation range by combining several DLD arrays and adding microfluidic channels.

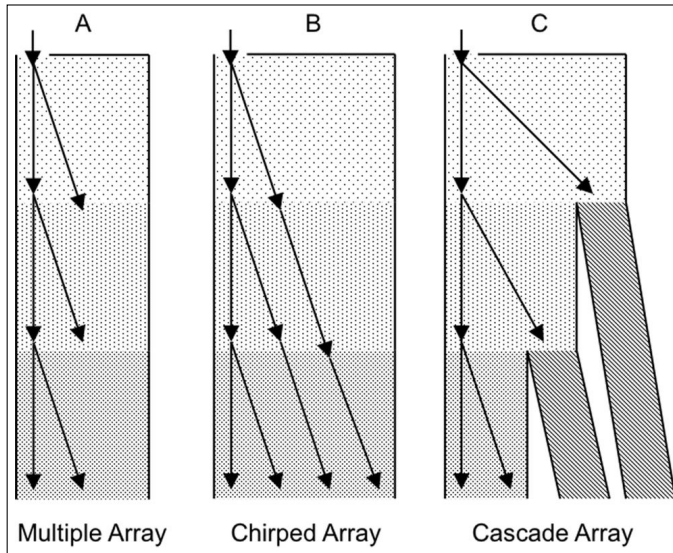


Figure 2.7: Multi-size particle separation in DLD arrays. A) Multiple Array, B) Chirped Array, and C) Cascade Array [63].

DLD technology offers high separation resolution and can distinguish particles not only by size but also by shape, making it suitable for sorting non-spherical particles. For example, Sherbaz et al. [60] implemented a DLD device to separate cells based on their morphology—specifically, *E. coli* minicells (spherical) and their rod-shaped

parental counterparts—achieving high separation efficiency.

However, DLD devices have some limitations: microfabrication is more complex compared to other microfluidic platforms, and throughput is relatively low. The issue of low throughput can be partially addressed by parallelizing the device or increasing the channel height [68]. For instance, Zen et al. [69] developed a multichip DLD system capable of processing raw Bone Marrow Aspirate (BMA) and separating Mesenchymal Stem Cells (MSCs), including leukocytes and red blood cells (RBCs). Their system processed 2.5 mL of raw BMA in approximately 20 ± 5 minutes, demonstrating its potential as an effective downstream method for cell therapy applications.

Other limitations of DLD include mixed motion and sidewall effects. Mixed motion refers to the gradual transition of particles from the zigzag mode to the bumping mode as particle size increases, rather than a sharp switch between the two modes [56].

It has been observed that near the interface between the array and the sidewall, the flow can be disturbed, locally altering the theoretical critical size. As a result, particles tend to follow the sidewalls instead of their intended trajectories, reducing separation efficiency. This issue can be mitigated by focusing the particle stream away from the sidewalls, thereby minimizing sidewall effects [70]. In cases where the sample stream is focused near one sidewall, Inglis [71] proposed modifying the sidewall by adding pillars to prevent flow lane disturbances. This modification makes the sidewall irregular rather than straight. For a device that separates particles from left to right, the gap between the left sidewall and the first post (G_L) can be expressed as

$$G_L = G\sqrt{\frac{n}{N}} \quad (9)$$

where n is the row number. On the other side, the gap between the last post and the right sidewall (G_R) is calculated as

$$G_R = G\sqrt{2 - \frac{n}{N}} \quad (10)$$

DLD has also been applied to the separation of fungal spores. For example, Inglis et al. [72] developed a microfluidic device based on DLD to efficiently separate *Aspergillus* spores (approximately 4 μm in diameter) from debris smaller than 3 μm or larger than 5 μm . The resulting sample showed a two- to threefold increase in the spore-to-debris ratio, demonstrating the device's capability to effectively reduce unwanted particles in the suspension.

These applications highlight the versatility of DLD in biological contexts, particularly when high resolution and label-free separation are required. Despite its complexity and low throughput, DLD remains a robust and well-established passive technique for size-based particle sorting.

Moving beyond separation methods that operate in the laminar and low Reynolds number regime, the next section introduces inertia-based microfluidic cell sorting.

This technique exploits fluid dynamics at higher Reynolds numbers to enable efficient particle separation.

2.1.4 Inertia-based cell sorting

Traditionally, microfluidic systems are considered to operate only in the Stokes flow regime, where the Reynolds number (Re) approaches zero. The Reynolds number is defined as $Re = \rho_f UH / \mu$, where ρ_f is the fluid density, U is the average velocity, H is the channel dimension, and μ is the dynamic viscosity of the fluid. In this regime, the influence of fluid inertia is typically neglected [73].

However, the importance of microfluidics operating in an intermediate Reynolds number range ($1 < Re < 100$) — between the Stokes regime and the turbulent regime ($Re > 1000$) — has been growing. Since this range still corresponds to laminar flow, it allows for controllability of fluid and particles. In this intermediate range, both inertia and viscosity influence the flow, so effects such as inertial migration and secondary flows become relevant. These two inertial effects will be explained in the following paragraphs [74].

- **Inertial migration:** This phenomenon introduces order to randomly dispersed particles at the entrance of a straight channel. As the particles travel through the channel, they migrate laterally toward specific equilibrium positions after a long enough distance. This effect is particularly useful in microfluidic systems where the particle size is comparable to the characteristic dimension of the microchannel [75].

Theoretical analysis of inertial migration suggests that two main opposing forces are responsible for this behavior: 1) the shear-gradient lift force (F_{LS}) and 2) the wall-induced lift force (F_{LW}). The shear-gradient lift force arises from the curvature of the fluid's velocity profile and pushes particles away from the channel centerline. The wall lift force, on the other hand, results from interactions between the particles and the channel wall, pushing them away from the wall. The final inertial equilibrium position of the particles is determined by the balance between these two forces [75].

- **Secondary flow:** This typically occurs in curved channels or in straight channels with disturbance obstacles. In a curved channel, secondary flow is generated by a pressure difference between the center and the near-wall regions. This pressure difference results from a mismatch in fluid momentum at different radial positions: particles near the center have higher momentum (they move faster) than those near the walls, causing them to move outward. To conserve mass, the fluid near the top and bottom walls moves inward, forming two counter-rotating vortices known as **Dean vortices** [76].

Secondary flow provides several advantages. For example, Dean vortices can change the equilibrium positions of particles by applying an additional viscous drag force (F_D) perpendicular to the main flow. This makes it possible to separate particles of different sizes, due to the balance between inertial lift forces (F_L) and the drag from the secondary flow, both of which are size-dependent. Furthermore, the enhanced mixing caused by Dean vortices can reduce the re-

quired channel length, allowing particles to reach their equilibrium positions more quickly and making the device more efficient [77, 78]. This principle is illustrated in Fig. 2.8

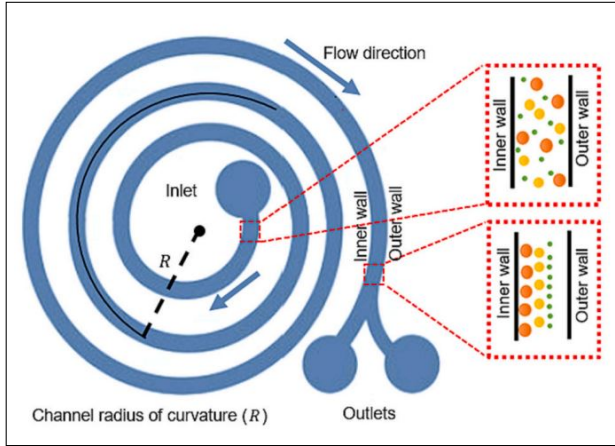


Figure 2.8: Schematic of particle flow in a spiral microchannel. At the inlet, particles are randomly distributed, while at the outlet, they are focused at distinct equilibrium positions depending on their sizes. [78].

Inertia-based cell sorting offers several advantages, including label-free operation and high throughput. However, one of its main limitations is the difficulty in separating particles of similar size [79]. To overcome the limitations of inertial separation for particles of similar size, Shin et al. [80] have used antibody-coated beads to increase the effective size of the target cells, thereby facilitating their separation.

Inertial microfluidics has proven effective in several biological applications. Fuchs et al. [81] studied a method to separate fungal cells (*Candida albicans*) from white blood cells in lysed blood samples, where red blood cells were chemically removed, using inertial microfluidics. Their approach yielded samples that were 2.86 times more concentrated, which can significantly aid in the rapid diagnosis of fungal infections. Similarly, Lu et al. [82] developed an inertial microfluidic device with microsquare-embedded channels to separate *Candida* species (specifically *C. glabrata* and *C. albicans*) from blood samples, achieving high recovery (>80%) and purity (>92%).

In summary, inertial microfluidics offers a versatile and label-free approach for particle separation. Following this, the next section will discuss active microfluidic techniques, which utilize external fields such as electric or optical forces to manipulate particles, providing a complementary set of tools within the broader microfluidics techniques.

2.2 Active sorting methods

Contrary to passive techniques, active sorting methods require the application of an external field—either electrical or optical—to achieve separation. In this section, we introduce three typical active sorting strategies: dielectrophoresis and impedance-based cell sorting, which rely on the electrical properties of particles, and optical-based cell sorting, which depends on their optical characteristics.

2.2.1 Dielectrophoresis

Dielectrophoresis (DEP) refers to the movement of polarized cells (or other dielectric particles) in a dielectric medium when subjected to a nonuniform electric field [83]. When exposed to an electrical field, a biological cell exhibits an induced dipole coming from charge reorganization. In uniform electric fields, the electrostatic forces on opposite ends of the dipole are equal, resulting in no net movement. In contrast, nonuniform fields produce unequal forces on either side of the cell, generating a net DEP force that can induce translational motion [84]. Remarkably, this occurs without the need to pretreat the cell samples. The motion of cells or particles induced by DEP is strongly influenced by their dielectric properties and those of the surrounding medium. The intensity of the DEP force also depends strongly on the particle radius. As a result, DEP is an effective technique for separating and sorting individual cells based on their dielectric properties and/or size, as they exhibit distinct dielectrophoretic behaviors in electric fields [85].

More specifically, the DEP force is given by:

$$F_{DEP} = 2\pi\epsilon_m r^3 \text{Re}[CM](\nabla E^2) \quad (11)$$

where ϵ_m is the permittivity of the medium, r is the radius of the particle, and ∇E^2 is the gradient of the square of the electric field.

Here, $\text{Re}[CM]$ is the real part of the Clausius-Mossotti factor (CM), a polarization factor that determines the direction of the force. It can be defined as:

$$CM = \frac{\epsilon_p^* - \epsilon_m^*}{\epsilon_p^* + 2\epsilon_m^*} \quad (12)$$

where ϵ_m^* is the complex permittivity of the medium and ϵ_p^* is the complex permittivity of the particles. Based on the sign of the CM, DEP can be classified as negative dielectrophoresis (nDEP) or positive dielectrophoresis (pDEP). When a cell is less polarizable than the surrounding medium ($\text{Re}[CM] < 0$), nDEP repels the cell from regions of high electric field. Conversely, when a cell is more polarizable than the medium ($\text{Re}[CM] > 0$), pDEP attracts the cell toward the high-field region [86, 87], as illustrated in Fig. 2.9. Since the dielectric properties of the cells and the permittivity of the suspending medium (mainly water with a $\epsilon \approx 80$ when the frequency is below 1 GHz) cannot be altered, the only way to adjust the CM factor is by changing the medium conductivity. Typically, cells are suspended in a low-conductivity medium, which enhances pDEP (attraction toward the electrodes) while minimizing electrical currents that could cause thermal damage or water electrolysis [88].

Dielectrophoresis offers several advantages. It is a moderate to high throughput label-free technique that enables the manipulation of particles based on their dielectric properties and size, and it has demonstrated high efficiency. However, DEP also faces multiple challenges. One limitation is the device complexity, as the integration and design of electrodes are critical to achieving high separation and purity. Additionally, the electric field parameters must be carefully optimized to ensure high specificity

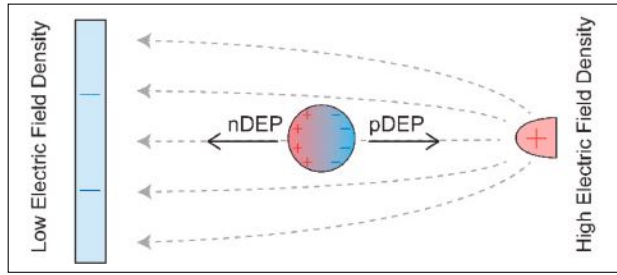


Figure 2.9: Schematic of negative dielectrophoresis (nDEP) and positive dielectrophoresis (pDEP) [87].

[89, 90, 91]. Joule heating remains a significant issue, as it can lead to local temperature increases. This, in combination with high electric field exposure, may compromise cell viability [92].

Dielectrophoresis has been widely used to separate biological samples. For example, in 1994, Markx et al. [93] applied dielectrophoretic forces generated by electrodes in a small chamber to efficiently separate viable from non-viable *Saccharomyces cerevisiae* (yeast) cells. Later, Jiang et al. [94] combined dielectrophoresis with a digital microfluidic (DMF) platform to successfully manipulate and separate yeast cells.

Of particular interest for fungal applications, Wang et al. [95] developed a two-stage microfluidic system to separate airborne disease spores from greenhouse crops directly from gas flow, using DEP for this purpose. The spores tested were *Botrytis cinerea* (16 μm), *Pseudoperonospora cubensis* (25 μm), and *Podosphaera xanthii* (30 μm), achieving separation efficiencies of 75.7%, 83.8%, and 89.4%, respectively, at their specific outlets.

In addition to dielectrophoresis, electrical impedance-based techniques have also been investigated for cell characterization and separation. These methods rely on measuring the electrical response of particles as they pass through microelectrodes, offering a different perspective based on their intrinsic electrical properties. In the following section, the principles behind impedance-based techniques will be explored, and how they have been applied to analyze and separate biological particles.

2.2.2 Impedance sorting

Impedance cytometry is a label-free technique for single-cell detection that distinguishes cells based on their electrical properties [96]. The system typically includes a microfluidic chip, patterned electrodes, and external detection circuits. As cells flow through a microchannel, they pass over the electrodes. These electrodes enable the application of an alternating voltage, inducing an alternating current through the probed cell. Measuring this current at a given frequency provides the cell's complex electrical impedance at that frequency. If the cell flows rather than remains steady over the electrodes –impedance flow cytometry (IFC)– the measured impedance varies over time, which can be exploited for cell detection (Coulter principle). Conversely, if the cell stays steady over the electrodes long enough –Electrical Impedance Spectroscopy (EIS)– measurements can be performed across a range of frequencies, yielding its impedance spectrum [97].

Depending on the frequency, different cellular and subcellular features can be assessed:

- At 0.1–1 MHz, impedance mainly reflects cell size.
- At 2–10 MHz, impedance reveals details about cell morphology, based on measured capacitance.
- Between 10–30 MHz, impedance of the membrane decreases, and the current probes the cytoplasm conductivity and permittivity, and it provides insights into the organelles.
- At even higher frequencies, it is possible to extract information about the nucleus.

These frequency ranges are representative and may vary depending on the cell type and the conductivity of the medium [98]. Figure 2.10 illustrates the specific components that can be analyzed at different frequency ranges.

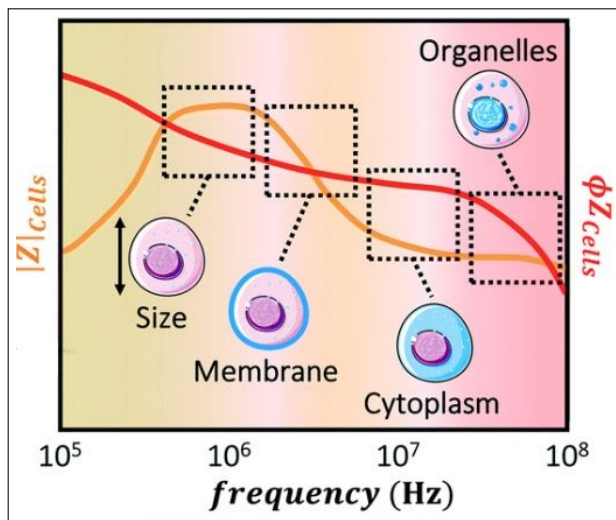


Figure 2.10: Schematic of the spectra of impedance magnitude and phase [98].

In summary, IFC involves several key steps: detecting cells and reference particles, calibrating the system by comparing cell signals to known references, extracting impedance features, and linking them to biophysical properties. Finally, it enables the identification of cell subtypes by highlighting differences between them and other detected particles [99].

To enable cell sorting, IFC must be integrated with an actuation mechanism capable of manipulating selected particles toward designated outlets. This process requires real-time signal processing to trigger the actuator based on the electrical properties of each passing cell. For instance, Lefevre et al. combined impedance flow cytometry with dielectrophoresis to achieve particle separation [100]. Other types of actuators that have been explored include acoustic forces, piezoelectric elements, and microvalves [101].

One of the main advantages of impedance flow cytometry is its high throughput and sorting efficiency. As previously discussed, it can also extract dielectric properties from subcellular structures. However, a key limitation is the complexity involved in translating impedance measurements into meaningful biophysical properties of the particles. This requires time-consuming calibration and extensive data processing [102].

Impedance flow cytometry has been applied to discriminate cells based on their dielectric properties. For example, Haandbaek et al. [103] used an impedance flow cytometer to differentiate wild-type yeast cells from mutants by analyzing the size and distribution of vacuoles in the intracellular fluid at frequencies around 250 MHz. Similarly, Xie et al. [104] employed this technique to distinguish yeast cells at different budding stages, while Liu et al. [105] used it to differentiate budding yeast cells from non-budding ones. Impedance measurements have also been used to discriminate between bacterial species, such as *Bacillus licheniformis* and *Bacillus mycoides*, as demonstrated by Beck et al. [106].

Having explored two electric field-based sorting techniques—dielectrophoresis and impedance flow cytometry—the next section turns our attention to an alternative approach that relies on the optical properties of cells.

2.2.3 Optical sorting

The last active sorting technique presented in this chapter is optical sorting, specifically optical lattices. An optical lattice can be defined as a periodic light structure created by the interference of multiple laser beams. When two beams of equal intensity intersect, they produce a one-dimensional (1D) periodic potential. Introducing a third beam generates a two-dimensional (2D) lattice, while the interference of five well-arranged beams results in a three-dimensional (3D) lattice with a centered tetragonal configuration. The lattice properties—such as geometry, spacing and symmetry—can be tuned by adjusting the laser wavelength and the angles at which the beams intersect [107].

Optical lattices can be used to guide or deflect particles based on their physical properties. The sorting principle relies on the interaction between the particles and the three-dimensional optical lattice. This interaction depends on the optical polarizability of the particles, which typically increases with size. As a result, this technique enables the label-free sorting of particles not only by size but also by their refractive index [107].

This approach was first demonstrated by MacDonald et al. [108] in a study where a five-beam optical lattice was used to deflect targeted cells at a fixed angle (typically 45°) into specific microfluidic channels. Non-targeted particles continued along the main flow path. This method has proven to be versatile and tunable: by adjusting the wavelength of the laser and the angle of orientation of the interfering beams, the lattice geometry can be modified, thereby allowing control over the sorting resolution and particle deflection [107].

A schematic of the sorting system employed by MacDonald et al. [108] is shown in Fig. 2.11. First, a mixture of particles is injected into the main inlet of the microfluidic device (Fig. 2.11, item 2), while a buffer solution is simultaneously introduced through a secondary inlet (also shown in Fig. 2.11, item 1). As the suspension flows through the sorting region, known as fractionation chamber—where a three-dimensional optical lattice, specifically a body-centered tetragonal (b.c.t.) configuration, is embedded—the target cells are deflected from their original trajectory into lateral collection channels (Fig. 2.11, item 3). In contrast, non-target cells continue along the main flow direction

without deviation (Fig. 2.11, item 4).

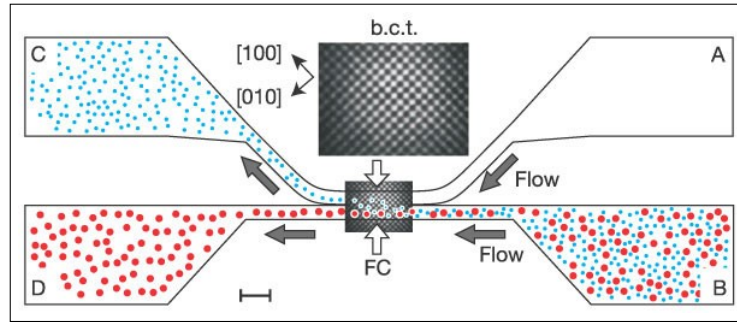


Figure 2.11: Schematic of microfluidic sorting systems based on optical lattice. A) Buffer inlet, B) Sample inlet, C) Collection channel of targeted particle, D) Waste collection channel [108].

The optical lattice sorting technique presents several advantages and limitations. On the one hand, it is a powerful, non-invasive, and label-free optical method. It has demonstrated high sorting efficiency and can be integrated into microfluidic devices. Moreover, it enables the manipulation of multiple cells simultaneously. In contrast to impedance-based methods, which require an additional actuation step to deflect target particles after detection, optical lattices apply direct, contactless forces, eliminating the need for external actuators. The technique also offers high tunability and versatility for different applications [107, 109].

On the other hand, the technique also presents certain drawbacks. At high particle concentrations, particle–particle interactions can reduce sorting efficiency. The system requires sophisticated and expensive optical setups, which may limit its accessibility. Additionally, particles with similar refractive indices are difficult to distinguish. Finally, heating from the laser beams may affect cell viability, especially for sensitive biological samples [108, 110].

To our knowledge, there are relatively few studies focusing on fungal spores using optical lattices; however, these lattices have been successfully applied to yeast cells. Optical lattices have demonstrated effective separation of particles based on size. For example, Jakl et al. [111] developed a sorting system that differentiates between yeast cell spores and living yeast cells: larger particles (living yeast cells) continue straight along the channel, while smaller particles (dormant yeast spores) are deflected into separate paths. Additionally, Shi et al. [112] investigated the use of optical lattices to study bacterial cells.

With the overview of different sorting techniques completed, the next section will provide a comparative analysis of both passive and active sorting methods. Their strengths and weaknesses will be discussed to guide the selection of the most suitable method for our application.

2.3 Balance sheet

Several sorting techniques were introduced in this chapter. The choice of which technique to use depends entirely on the application. To recap, our goal is to isolate spores of *B. cinerea* that contain homogeneous DNA material. The most straightforward way

to achieve this is by targeting spores with only one nucleus. Our hypothesis assumes that there is a size difference between single-nucleated and multinucleated spores. However, since the spores come from the same strain and sample, we do not expect a large size difference; instead, this difference is likely to be quite small. Moreover, we do not expect the spores to exhibit any optical differences related to the number of nuclei, which makes optical-based sorting techniques unsuitable for this application.

Therefore, we need a sorting system with high separation efficiency and high resolution (the ability to distinguish particles of similar size). Additionally, this microdevice will be used by biologists at the MAP laboratory, so it must be user-friendly and not require complex and expensive equipment.

In this context, passive sorting techniques are preferred because they rely mainly on hydrodynamic forces and do not require additional equipment. Among passive methods, Deterministic Lateral Displacement offers the high resolution necessary for this project. We also selected DLD because we do not need high throughput or high particle concentration, unlike other applications. Table 2.1 presents a summary and comparison of the sorting techniques.

In the next chapter, we will focus on the design and implementation of a DLD sorting device to separate mononucleated spores from multinucleated ones, thereby enriching the sample at the outlet with homogeneous DNA material.

Table 2.1: Sorting techniques comparison

Technique	Principle	Efficiency	Throughput	Marker	Pros	Cons	References
Micro filtration	Micro membranes	Moderate	Moderate to High	Size	+ Simple and easy to operate	Clogging	[38, 44]
PFF	Particle pinching driven by hydrodynamic flow	High	High	Size	Simple design and fabrication	Limited separation of particles with very similar sizes	[51, 53, 113]
DLD	Particles undergo differential displacement in tilted pillar array	High	Low to Moderate	Size	High separation resolution	Array clogging	[56, 62, 64, 71]
Inertial	Inertial lift and Dean drag forces	High	High	Size	Rapid and robust	Low separation resolution	[75, 79, 114]
DEP	Interaction between electrical polarization and a non-uniform electric field	High	Moderate to High	Electrical properties	Selective and precise contactless manipulation	Complex design	[89, 90, 91, 115]
Impedance	Impedance changes in electrical field	High	High	Impedance properties	Extraction of subcellular features and properties	Requires advance signal processing	[96, 98, 101]
Optical	Interference laser beams creating optical forces	High	Moderate to High	Size, refractive index	Dynamically reconfigurable	Expensive equipment	[107, 108]

Résumé du Chapitre 2

Ce chapitre a présenté l'état de l'art des différentes techniques de tri. La discussion a commencé par une description des différentes propriétés pouvant être utilisées comme paramètres pour la séparation cellulaire. La technique de référence est le tri cellulaire activé par fluorescence (FACS), qui exploite les différences dans les propriétés biochimiques des cellules [21]. Le tri cellulaire activé par magnétisme (MACS), quant à lui, repose sur les propriétés magnétiques des particules attachées aux cellules cibles [22]. Les deux techniques nécessitent généralement un marquage externe pour différencier les populations cellulaires et ne sont donc pas compatibles avec notre cahier des charges puisque dans ce projet, les techniques sans marquage sont à privilégier. En effet, les spores de *Botrytis cinerea* ne présentent pas de propriétés intrinsèquement fluorescentes ou magnétiques, et le marquage externe n'est pas envisageable en raison des contraintes biologiques imposées par les chercheurs du laboratoire MAP. Le marquage pourrait potentiellement compromettre la viabilité des spores. Par conséquent, les méthodes basées sur les propriétés biophysiques sont préférées, car elles préservent l'intégrité biologique des spores tout en éliminant le besoin de marquage.

En revanche, les méthodes de tri basées sur les propriétés biophysiques — telles que les caractéristiques de taille et de forme, mécaniques, électriques ou optiques — sont considérées comme sans marquage, car elles ne nécessitent aucune forme d'étiquetage externe [24, 29, 32].

Les techniques reposant sur les propriétés biophysiques pour trier les cellules sont présentées en deux catégories : les méthodes de tri passif et les méthodes de tri actif. En raison du grand nombre d'approches et de publications disponibles dans la littérature, l'état de l'art présenté ici vise à fournir au lecteur un aperçu général plutôt qu'une revue exhaustive.

Les méthodes passives de tri n'appliquent pas de forces externes pour effectuer la séparation. Elles reposent plutôt sur les forces hydrodynamiques agissant sur les cellules lorsqu'elles s'écoulent à travers des microcanaux de géométries spécifiques [36]. Les techniques abordées dans cette section incluent la microfiltration, le fractionnement par flux pincé, le déplacement latéral déterministe (DLD) et le tri cellulaire inertiel.

La filtration sur membrane est un procédé de séparation basé sur la taille, dans lequel une membrane agit comme une barrière qui permet sélectivement à certaines particules, plus petites que la taille des pores, de passer tout en retenant les autres plus grandes à sa surface [41]. Cette technique peut être classée selon la taille des pores

de la membrane, qui est donc un paramètre critique. La microfiltration est particulièrement pertinente dans notre cas, car elle fonctionne à l'échelle du micromètre — comparable à la taille des spores de *B. cinerea* [38].

Il existe deux modes de fonctionnement courants : la filtration frontale (dead-end) et la filtration tangentielle (crossflow). Dans la filtration frontale, la solution s'écoule perpendiculairement à travers la membrane. Un inconvénient de ce mode est l'accumulation de particules à la surface de la membrane, ce qui conduit à son colmatage. Dans la filtration tangentielle, le fluide s'écoule parallèlement à la membrane, ne permettant qu'à une partie du liquide de la traverser. Cette configuration réduit l'accumulation de particules à la surface de la membrane [43].

Les principaux avantages de la microfiltration incluent son fonctionnement à basse pression (typiquement en dessous de 2 bar), ce qui réduit considérablement le coût des équipements. Cependant, un inconvénient majeur est le colmatage progressif de la membrane dû à l'accumulation de particules à sa surface ou en son sein. [44].

Le fractionnement par flux pincé (PFF) est une technique microfluidique qui sépare les particules en fonction de leur taille. Dans cette méthode, une suspension de particules et une solution tampon sont co-injectées dans une section étroite d'un microcanal. En conséquence, les particules sont focalisées par le flux vers l'un des bords de cette zone resserrée. Lorsque le flux atteint une section élargie, les lignes de courant divergent et les particules sont séparées selon leur taille : les particules plus grandes sont déplacées vers le centre du canal, tandis que les plus petites restent plus proches de la paroi latérale [49, 50].

Le principal avantage de cette technique réside dans sa simplicité et son fonctionnement passif. Cependant, une limitation est son efficacité réduite pour séparer des particules nettement plus petites que la largeur du canal. Pour remédier à ce problème, Takagi et al. [53] ont introduit la fractionation par flux pincé asymétrique (AsPFF), qui améliore la résolution pour les petites particules.

Cette technique a été appliquée avec succès par Park et al. [54] pour séparer les spores fongiques de *Cladosporium cladosporioides* des cellules eucaryotes NIH3T3 fibroblastes, atteignant une efficacité de séparation d'environ 90%.

Le déplacement latéral déterministe (DLD) est une technique microfluidique qui permet un tri cellulaire haute résolution basé sur la taille des particules. Le système se compose d'un réseau de piliers, où chaque rangée consécutive est décalée latéralement d'une distance fixe. Les particules plus grandes qu'un seuil spécifique, appelé diamètre critique, sont déviées latéralement lorsqu'elles interagissent avec les piliers. En revanche, les particules plus petites que ce diamètre critique suivent les lignes d'écoulement du fluide et ne sont pas déviées [56].

Pour améliorer l'efficacité de séparation, augmenter le débit et réduire le colmatage, diverses formes de piliers ont été développées. Celles-ci incluent des piliers triangulaires, des piliers polygonaux réguliers, des piliers en forme de L inversé, et des piliers en forme de I [64, 65, 67].

Le principal avantage du DLD est sa haute résolution — jusqu'à environ 10 nm. Cepen-

dant, il présente aussi certaines limites, telles qu'un design plus complexe comparé à la fractionation par flux pincé (PFF) et un débit relativement faible [68].

Le DLD a été utilisé avec succès pour séparer des spores fongiques. Par exemple, Inglis et al. [72] ont démontré la séparation efficace des spores de *Aspergillus* (d'un diamètre d'environ 4 μm) de débris plus petits que 3 μm et plus grands que 5 μm .

Le tri cellulaire basé sur l'inertie est la dernière méthode passive présentée dans ce chapitre. Cette technique opère dans une plage intermédiaire de nombres de Reynolds ($1 < Re < 100$), située entre les régimes d'écoulement de Stokes et turbulent. Dans ce régime, les forces inertielles et visqueuses influencent l'écoulement. Bien que l'écoulement reste laminaire — permettant ainsi un contrôle précis des trajectoires des particules — la migration inertielle et les écoulements secondaires deviennent significatifs [74].

Lors de la migration inertielle, deux forces de portance principales et opposées agissent sur les particules : la force de portance associée au gradient de cisaillement (F_{LS}), qui découle de la courbure du profil de vitesse et repousse les particules du centre du canal ; et la force de portance induite par la paroi (F_{LW}), qui résulte des interactions particule-paroi et repousse les particules des parois du canal [75].

Les écoulements secondaires, souvent observés dans les canaux courbes, sont générés par des différences de pression entre le centre et les régions proches des parois. Pour conserver la masse, le fluide près des parois supérieures et inférieures du canal se déplace vers l'intérieur, formant deux tourbillons contrarotatifs connus sous le nom de vortex de Dean.

Le principal avantage du tri inertiel est son débit plus élevé comparé à des techniques comme le DLD. Cependant, sa résolution est relativement faible, ce qui le rend moins efficace pour séparer des particules de tailles proches [79].

Ensuite, nous avons exploré les méthodes de tri actives, qui nécessitent l'application d'un champ externe pour réaliser la séparation des particules. Nous nous sommes concentrés sur trois techniques : la diélectrophorèse, le tri cellulaire basé sur l'impédance, et les réseaux optiques. Les deux premières reposent sur les propriétés électriques des cellules, tandis que la dernière exploite leurs caractéristiques optiques intrinsèques.

La diélectrophorèse (DEP) désigne le mouvement de particules polarisées dans un milieu diélectrique soumis à un champ électrique non uniforme [83]. Il existe deux types de réponses diélectrophorétiques : la DEP positive (pDEP) et la DEP négative (nDEP). Lorsqu'une cellule est plus polarisable que le milieu environnant, la pDEP provoque un déplacement de la cellule vers les régions de fort gradient du champ électrique. À l'inverse, si la cellule est moins polarisable que le milieu (généralement de faible conductivité), la nDEP repousse la cellule loin de ces régions [86].

Cette technique présente plusieurs avantages, notamment une haute efficacité et un débit modéré à élevé. Cependant, elle comporte aussi certains inconvénients : la conception du système est complexe, l'intégration des différents composants peut être difficile, et la viabilité cellulaire peut être compromise en raison de l'effet Joule et de

l'exposition à des champs électriques élevés [89, 90, 91, 92].

Wang et al. [95] ont mis en œuvre la DEP pour séparer directement des spores fongiques en suspension dans un échantillon d'air issu de serres de culture. Les spores testées étaient *Botrytis cinerea* (16 μm), *Pseudoperonospora cubensis* (25 μm) et *Podosphaera xanthii* (30 μm), atteignant des efficacités de séparation de 75,7%, 83,8% et 89,4%, respectivement, à leurs sorties désignées.

La deuxième technique active présentée est la **cytométrie d'impédance**, qui distingue également les cellules en fonction de leurs propriétés électriques [96]. Dans cette méthode, un courant alternatif (CA) est appliqué, et les variations du signal électrique sont mesurées à mesure que chaque cellule traverse la zone de détection. La forme et l'amplitude du signal dépendent de facteurs tels que la tension appliquée, l'impédance du système et — surtout — les propriétés intrinsèques de la cellule (par exemple, son volume et son comportement diélectrique).

Différentes caractéristiques cellulaires peuvent être analysées à différentes plages de fréquences : la taille des cellules à 0,1–1 MHz, la capacité membranaire à 2–10 MHz, la conductivité cytoplasmique à 10–30 MHz, et les propriétés nucléaires à des fréquences supérieures à 30 MHz [98].

Les avantages de la cytométrie d'impédance incluent un débit élevé et une excellente efficacité de tri. Cependant, l'une de ses principales limites réside dans la complexité de la traduction des mesures d'impédance en paramètres biophysiques significatifs [102]. De plus, un actionneur externe est nécessaire pour manipuler physiquement les particules sélectionnées [100].

La dernière technique active de tri présentée est le tri optique, en particulier l'utilisation des **réseaux optiques**. Cette méthode sépare les particules en fonction de leurs propriétés optiques. Un réseau optique est une structure lumineuse périodique générée par l'interférence de plusieurs faisceaux laser [107]. Le tri est réalisé grâce à l'interaction entre les particules et le réseau optique tridimensionnel. Cette interaction est principalement régie par la polarizabilité optique des particules, qui augmente généralement avec la taille et l'indice de réfraction.

Le tri optique présente plusieurs avantages : c'est une technique puissante, non invasive, sans marquage, permettant la manipulation simultanée de plusieurs cellules. De plus, elle ne nécessite pas d'actionneurs externes [107, 109]. Cependant, cette technique présente également des limites. À des concentrations élevées de particules, l'efficacité du tri peut diminuer. De plus, le dispositif est souvent coûteux, ce qui limite son accessibilité. Enfin, ses performances diminuent en cas de particules ayant des indices de réfraction similaires [108, 110].

Ce chapitre a présenté plusieurs techniques de tri, en soulignant que le choix de la méthode la plus appropriée dépend fortement de l'application spécifique. Ici, l'objectif principal est d'isoler des spores de *B. cinerea* contenant un ADN homogène en ciblant les spores mononucléées. Ces spores sont supposées être seulement légèrement plus petites que les spores multinucléées, ce qui nécessite une méthode de tri à haute efficacité et haute résolution pour distinguer des particules de taille similaire. De plus, la méthode doit être conviviale et économique afin de répondre aux besoins du labora-

toire MAP.

Les techniques passives de tri sont préférées pour cette application, avec le Déplacement Latéral Déterministe (DLD) qui se distingue comme la méthode la plus appropriée. Le DLD répond aux critères nécessaires en termes de résolution et d'efficacité de séparation. En outre, puisque notre application ne requiert ni un débit élevé, ni le traitement de concentrations élevées de particules, le DLD devient encore plus adapté.

DLD-Based low nucleus content spores enrichment

In the previous chapter, it was established that Deterministic Lateral Displacement (DLD) would be the most suitable technique for our cell sorting application. This chapter presents the full development process of this technique, starting with the materials and methods used for device fabrication and experimental procedures. The results section then reports on the sorting performance using both calibration beads and the target biological sample: spores of *B. cinerea*. This includes an assessment of sorting efficiency with beads, the viability of the sorted spores, and the overall effectiveness of the DLD system when applied to *B. cinerea*. Finally, the chapter concludes with a discussion of the findings and their implications.

3.1 Materials and Methods

This section outlines the procedures and materials used throughout the study, including sample preparation, size measurement protocols, device fabrication, and experimental setup.

3.1.1 Samples preparation

For all DLD experiments, the preparation of bead and spore suspensions followed a standardized protocol to ensure consistent and reproducible conditions. The bead suspension will be used as model particles to evaluate device performance, while the spore suspensions are the primary focus of this study. Spores represent more complex objects for sorting, as they are not spherical but rather teardrop-shaped.

3.1.1.1 Microbeads preparation

The experiments were carried out using two different bead suspensions. The first suspension contains a mixture of monodispersed polystyrene beads with diameters of 2

μm (Polybead[®] Carboxylate Microspheres, Polysciences Inc.) and 10 μm (Supelco[®], Sigma Aldrich) at a ratio of 2:1. This solution will be referred to as a **2-10 μm bead suspension** throughout the text. In contrast, the second suspension, named **6 μm bead suspension**, contained polystyrene copolymer beads (Thermo Scientific[™]) with a broader diameter distribution centered around 6 μm . It was selected for its relevance mimicking the continuous size distribution of spores. This distribution is a representative model, allowing the evaluation of the device's performance under conditions similar to those encountered in spores sorting applications.

The phosphate-buffered saline (PBS) containing 1% (v/v) Tween 20 was filtered using a 0.22 μm filter before adding and therefore diluting the bead suspension to a concentration up to 4.4×10^6 for both bead suspensions. This PBS-Tween 20 solution will be the buffer in all future experiments and referred to as **buffer solution** throughout the chapter unless stated otherwise.

3.1.1.2 Spores preparation

The different spore suspensions used in this chapter were genetically modified as described in the following paragraphs.

Transformation of *Agrobacterium tumefaciens*

The protocol was adapted from publication [116]. A stock of competent *A. tumefaciens* cells was used. Briefly, 100 ng of plasmid DNA were directly added to 100 μl of cells, followed by a heat shock at 37°C for 5 minutes. Subsequently, 1 ml of TNK YE medium was added, and the culture was incubated for 3 hours at 28°C before being plated on medium 6 supplemented with spectinomycin (250 $\mu\text{g}/\text{ml}$) and kanamycin (50 $\mu\text{g}/\text{ml}$) (Sigma). Transformants appeared after 3 days of incubation at 28°C.

Transformation of *Botrytis cinerea*

The protocol was adapted from publication [116] with minor modifications. First, a solution of frozen fungal spores (2×10^7 spores/mL, 20% glycerol) was used instead of fresh conidia. One *A. tumefaciens* colony containing the appropriate transformation vector was grown overnight at 28°C with agitation (200 rpm) in 20 ml of liquid medium containing 1% bacto-peptone broth (LB) and 0.5% yeast extract, supplemented with spectinomycin (250 $\mu\text{g}/\text{ml}$) and kanamycin (50 $\mu\text{g}/\text{ml}$).

Then, the culture was diluted to an OD600 of 0.15 in 20 ml of TNK YE, and the bacteria were grown under the same conditions to an OD600 of 0.4–0.5. Then, 100 μl of bacterial suspension and 100 μl of conidia were mixed and incubated for 5 minutes at room temperature. The mixture was subsequently diluted 1:10 in IM medium and spread onto TNK YE solid medium over the surface of a cellophane sheet.

After 2 days of co-culture at 21°C, the cellophane sheets were transferred to solid TNK YE medium containing hygromycin B (70 $\mu\text{g}/\text{ml}$) and cefotaxime (200 μM). Transformants emerged after 4–5 days of incubation at 21°C and were individually transferred to a medium containing 0.5% glucose, 2% malt extract, 0.1% bacto-peptone, 0.1% casein hydrolysate, 0.1% yeast extract, and 0.02% RNA (pH 5.0), supplemented with hygromycin B (75 $\mu\text{g}/\text{ml}$) and cefotaxime (200 μM).

Following the genetic modification step (transformation), the transformants were grown

for 3–5 days in the dark at 21°C on solid medium containing the required nutrients. After 3–5 days of exposure to white–blue light (which triggers reproduction), spores were collected by gently scraping the surface of the gel and suspended in 0.1% Tween 20 in water. The suspension was filtered through a 40 µm filter to remove large filaments. This preparation, referred to as the **original solution**, was stored at 4°C and used within 24 hours.

Before use, the original solution was subject to a second filtration through a 15 µm filter to ensure finer removal of any remaining filaments or very large spores that could clog the device. Subsequently, the solution was centrifuged (spinning at 3835 rpm, 1000 × g for 5 min), and the supernatant was removed. The resulting pellet was resuspended in buffer solution to reach a final concentration of up to 3.6×10^6 spores/mL, after which the suspension was referred to as the **15 µm spore suspension**. This concentration was optimized, since higher spore concentrations led to faster clogging of the device.

3.1.2 Beads size measurement

Bead size measurement is critical, as any inaccuracies are particularly significant when determining the DLD cut-off, as they will directly be used to determine the sorting efficiency. While a fluorescence microscope (DMI8) is typically employed for this task, it presents a limitation due to the constraint in the focal plane. The particle size measurement becomes inaccurate when the focus is not properly set at the equator. However, this issue does not arise with scanning electron microscopy (SEM), which allows more precise and reliable measurements, ensuring greater accuracy in determining the DLD cut-off.

As mentioned above, the 6 µm bead suspension has a broad diameter distribution centered around 6 µm; however, the standard deviation is not provided in the manufacturer’s datasheet. For this reason, the measurement protocol focuses on these particles rather than on the 2–10 µm bead suspension, for which the size difference is huge, allowing their easy identification under the microscope. The process is explained in detail below:

3.1.2.1 Size measurement protocol

1. **Sample preparation.** After running the **6 µm bead suspension** through the DLD device, the samples from the inlet as well as for both outlets were collected. Both bead suspensions were centrifuged, and the supernatant was carefully removed. Deionized water was added, and the beads were then resuspended. This procedure was repeated three times to ensure the buffer solution was completely washed. Then, an 8 µm droplet was placed onto a glass slide and let to dry for at least 4 hrs.
2. **Sample coating and Imaging.** Using the sputter coater (Cressington, model 208HR), the two samples were coated with a 15 nm layer of platinum-palladium (80/20%) to make them conductive for SEM imaging (Tescan mira3 fe-sem).
3. **Size detection.** Using a code developed in MATLAB, the bead sizes were measured from SEM images (Fig. 3.1). The process began with the loading and pre-

processing of the images. The contrast was adjusted, and then the image was converted into a binary format, where bright objects were highlighted in white. An erosion step was applied to remove small imperfections and improve individual bead detection. The algorithm then detected the bright circular beads in the processed image. For each detected circle, its radius in pixels was calculated and converted into micrometers using a calibration value. This information was stored for further analysis.

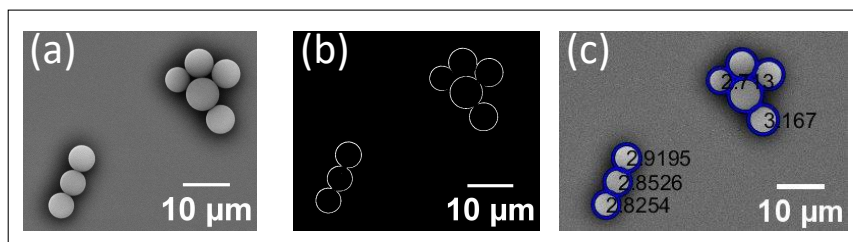


Figure 3.1: Bead size measurement protocol. (a) SEM images of the bead suspension. (b) Pre-processed images: contrast was adjusted and the images were converted to binary format. (c) Contour-detected images: the MATLAB code identified circular contours for each particle and calculated their radii in both pixels and micrometers. The detected contours were overlapped on the original SEM images to confirm the accuracy of the detection.

3.1.3 Spore size measurement and nuclei quantification

To validate that size-based sorting can be used to separate *B. cinerea* spores with a single nucleus, it was essential to establish a correlation between spore size and the number of nuclei. This was necessary to confirm the hypothesis presented previously that single spores can be isolated based on their size. The process began with the preparation of a suspension of *B. cinerea* spores whose nuclei were labeled with green fluorescent protein (GFP). After collection, the spores were stored in water at 4 °C in a refrigerator and used within 24 hours. Brightfield and fluorescence images were then captured using an epifluorescence microscope (DMI8, Leica, Germany) and saved as TIFF files. These images were analyzed using a purpose-built Python script, run within a Jupyter Notebook interface. The process in detail is outlined below:

3.1.3.1 Size measurement protocol

1. **Sample preparation** A 10 μL spore solution is introduced into a cell counting chamber (Quick-Read™ Precision Cell, Globe Scientific Inc.). To prevent sample evaporation, the chamber is sealed with dental resin. The sealed chamber is then observed under the microscope.
2. **Imaging.** Images were acquired in both brightfield and fluorescence modes at 63x magnification for the same field of view. A high number of spores was captured using the TileScan option and subsequently merged into a single image.
3. **Contour and size detection.** The contours of the spores were detected from the brightfield images. A custom Python script processed the images, by reading them and converting them first to a grayscale. Then, three filters were applied. First, the **Laplacian filter** helped to highlight areas with a rapid variation in pixel

values that generally correspond to the borders of the objects. Second, the **high-pass filter** was useful to enhance fine details by emphasizing high-frequency components. Third, the **Gauss filter** was employed to remove the background noise from the image.

After this treatment, a **thresholding step** was applied to separate the spores from the background. The resulting image displayed the spores in black while the background was white. To facilitate edge detection, the pixel value was inverted, resulting in white contours on a black background. This inversion facilitated edge detection.

The surface area enclosed by the detected edge was computed, and the equivalent diameter (D_{eq}) was calculated as the diameter of a circle with the same surface area. Additionally, the coordinates of the **bounding rectangle** for each spore were used as a unique identifier in the next step to correlate the same spore detected in both brightfield and fluorescence images.

4. **Nuclei detection.** The coordinates of the bounding rectangle from the previous step were used to crop the corresponding region in the fluorescence image, where the contour was also drawn. Each crop of the fluorescence image was displayed, hence allowing the manual counting of nuclei, which were then stored in a list. Importantly, by linking the unique identifiers assigned to each spore in the brightfield images, the spore surface area was correlated with the nuclei count at the end of the process.

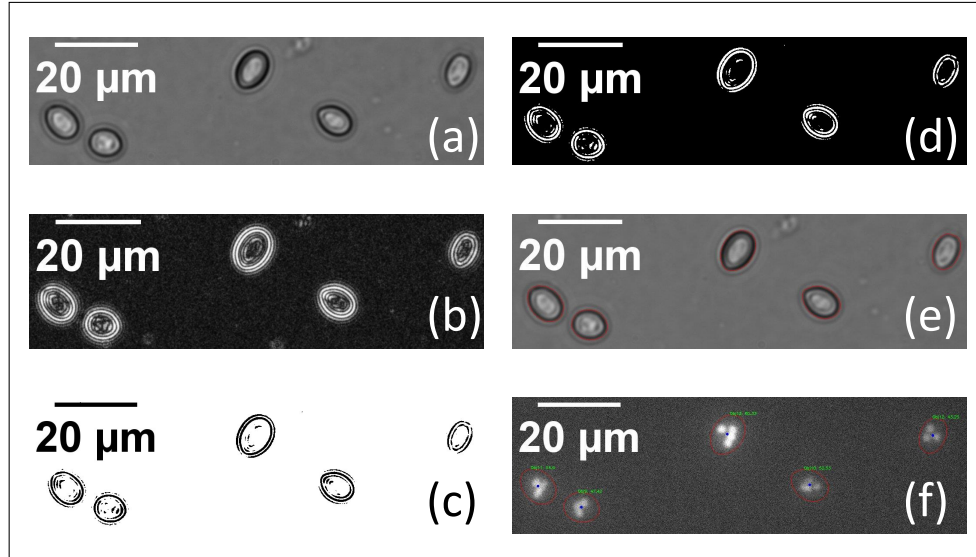


Figure 3.2: Spore size measurement protocol. (a) Brightfield image of the spore suspension. (b) Pre-processed image: first converted to grayscale, then Laplacian, high-pass, and Gaussian filters are applied. (c) Thresholded image: used to separate spores in black while the background appears white. (d) Inverted image: to improve visualization, pixel values were inverted so that contours appear white against a black background. (e) Contour detection image: a custom Python script detected the edges of the spores in the inverted image; the area and equivalent diameter D_{eq} were computed. The contours were then overlaid onto the original brightfield image to verify the accuracy of the detection. (f) Nuclei detection image: the coordinates of the bounding rectangle were used to match spores in the brightfield with those in the fluorescence field. Nuclei were manually counted.

3.1.3.2 Statistical analysis

The statistical analysis followed a series of steps to determine the appropriate tests for the data set. The analysis was performed for each nuclei group, where spores with 1 nucleus, 2 nuclei, and n nuclei were treated as different groups.

First, the non-normality of each group was assessed using the **D'Agostino-Pearson test** for groups with more than 30 spores, while the **Shapiro-Wilk test** was used for groups with less than 30 spores. Outlier detection was performed next, using **Grubbs' test** for normally distributed data and the **interquartile range (IQR) method** for non-normally distributed data.

Because all the data did not have a normal distribution, non-parametric tests were used: the **Mann-Whitney U test** for comparisons between two groups and the **Kruskal-Wallis test** with a **Dunn post hoc test** for comparisons of more than two groups.

All analyses were conducted using RStudio, and statistical significance was set at $p < 0.05$. Results are therefore presented in terms of median and quartiles, as required for non-parametric statistics.

3.1.4 Device fabrication

The PDMS microfluidic devices were fabricated using a standard soft-lithography technique [117]. The process begins with the fabrication of the master mold. To achieve this, a negative photoresist, SU8 2015 (MicroChem), was deposited at the center of a clean silicon wafer. The photoresist was then spin-coated at a low speed of 200 rpm for 10 s, followed by a high-speed of 2100 rpm for 30 s, resulting in a layer with a theoretical thickness of 20 μm . The following step was to perform a soft bake step (pre-bake) at 95 °C for 3 minutes 45 seconds. After this, the resist was exposed to direct writing using the μPG 101 laser lithographer (Heidelberg Instruments®). The laser beam was automatically focused using the pneumatic function. The power was set at 55 % of 55 mW with the filter off. A post-bake was performed after exposition at 95 °C for 4 minutes and 45 seconds. Next, the wafer was developed using a Propylene Glycol Methyl Ether Acetate (PGMEA) solution for 2 minutes, dissolving the unexposed areas. Therefore, it ensures precise pattern formation. Immediately after development, the wafer was rinsed with isopropyl alcohol to stop the developer's action. Finally, the master mold underwent a hard-bake process at 200°C for 30 minutes to enhance the mechanical strength of the resist.

The master mold was silanized, using Trichloro(1H,1H,2H,2H-perfluorooctyl)silane from Sigma-Aldrich, under a vacuum for 3 hours to facilitate demolding. The subsequent replicates were done with a mixture of a base and a curing agent (in a 9:1 w/w ratio). The mixture was then degassed for approximately 10 minutes and let to solidify at 60°C for at least 90 minutes. Once solidified, the PDMS was peeled off, and inlets and outlets were punched using a 0.75 mm puncher. After 3 minutes of oxygen plasma treatment using a O₂ plasma bonding machine (Harrick Plasma®), the PDMS was bonded to a glass slide and kept until use at 60°C to enhance the bonding and preserve hydrophilic properties.

3.1.5 Experimental set-up

All the microfluidic experiments described in this chapter were performed using a Leica DMI8 microscope and particles behavior was recorded with a high-speed camera (Miro Lab310 from Phantom). The recorded data was then analyzed using ImageJ software for the bead or spore trajectory.

Before injecting the spore or bead suspension into the inlet, a 0.22 μm filtered PBS solution was injected into the microfluidic device for 15 minutes at 600 mbar using a pressure-based flow controller (MFCS series, FluigentTM) to minimize particle-to-particle aggregation and particle-to-surface adhesion. Afterward, the sample and buffer solutions were connected and run for 15 minutes. This duration was chosen to ensure consistent sample collection, as the device tends to clog after 15 to 20 minutes, caused by the progressive formation of particle clusters or aggregates over time. The samples collected at the outlet corresponded to the big and small outlet fractions, which were analyzed according to the specific experimental requirements. Medical-grade polyethylene tubes (Scientific Commodities Inc.), 40 cm and 30 cm long, were used to connect the Eppendorf tubes to the inlets and outlets, respectively. This setup is shown in Fig. 3.3.

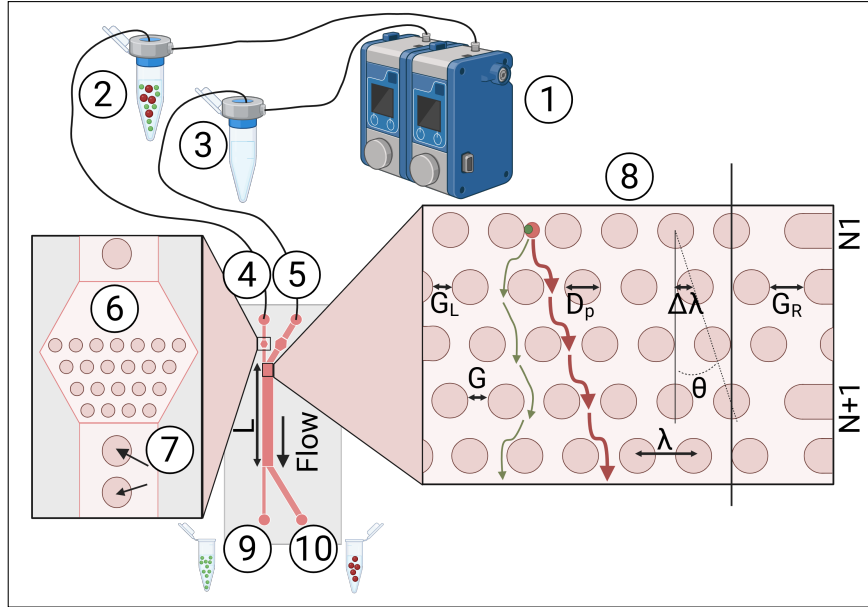


Figure 3.3: Schematic overview of a deterministic lateral displacement (DLD) microfluidic system setup. The system includes: (1) a pressure-driven pump, (2) sample reservoir, (3) buffer reservoir, (4) sample inlet, (5) buffer inlet, (6) filter structure, (7) support pillars, (8) deviation array, (9) small outlet, and (10) large outlet.

The following section presents the first step of this study: establishing a correlation between the equivalent diameter of the spores and the number of nuclei they contain. This analysis aims to determine whether a size-based cell sorting approach is suitable for isolating spores with a single nucleus, under the assumption that their size may differ from that of spores containing two or more nuclei.

3.2 Correlation between spore equivalent diameter and nuclei number

The spores' equivalent diameter (D_{eq}) was measured on bright field images (see Fig. 3.4a), while the number of nuclei is determined on fluorescent images (as illustrated in Fig. 3.4b) for each spore, using the protocol described in section 3.1.3. Their D_{eq} distribution according to the number of nuclei was then analyzed using the statistical tests described previously (section 3.1.3.2); the data are presented in a violin plot (Fig. 3.5).

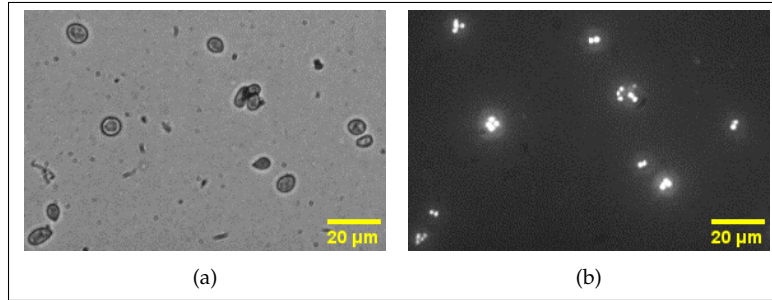


Figure 3.4: Video-microscopic observations of a suspension of *B. cinerea* expressing GFP nuclei. (a) Equivalent diameter measurement in brightfield mode. (b) Nuclei quantification on the fluorescence mode.

In general, spores with higher nuclei numbers tended to have larger equivalent diameters. Spores with 1 nucleus, which are the main focus of this study, presented a statistical difference regarding spores with 2 or more nuclei ($p < 0.05$). However, the equivalent diameter distributions exhibited overlapping between groups, particularly spores with 1, 2, and 3 nuclei. The spores equivalent diameter in the N1 group (1 nucleus) and the N2 group (2 nuclei) ranged from 5 to 10 μm , while those in the group N3 (3 nuclei) ranged from 6 to 12 μm .

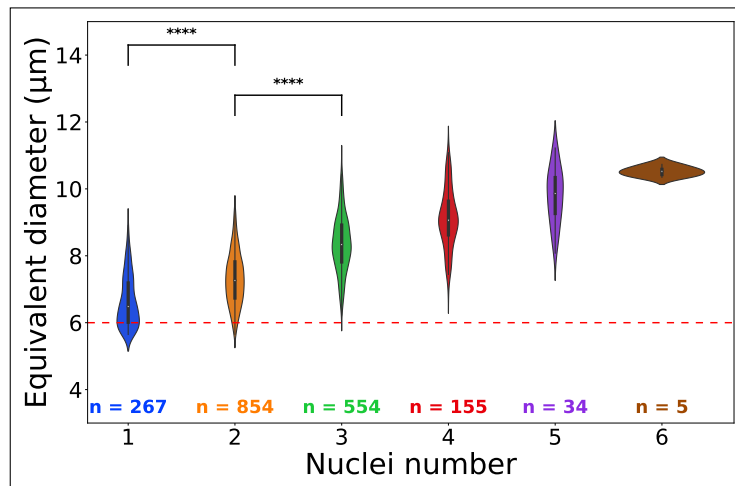


Figure 3.5: Distribution of spore equivalent diameters according to the nuclei number. The red line at 6 μm represents the theoretical critical diameter (D_c) which is the threshold for distinguishing populations composed mostly of N1 spores from those that mainly contain multiple-nuclei spores.

It can be observed in Figure 3.5 that a D_{eq} of 6 μm (marked by a red dashed line)

could be used as the theoretical critical diameter (D_c) to separate the spore population into two subpopulations. The first one consists of spores with an equivalent diameter below the theoretical critical diameter and corresponds mainly to groups N1 and N2, with almost no spores containing 3 nuclei. In contrast, the second one comprises spores with an equivalent diameter equal to or greater than the theoretical critical diameter. While some spores in this group have a single nucleus, the presence of multinucleated spores becomes more significant.

It can also be observed that using a higher theoretical threshold does not lead to a clear separation, as multinucleated spores (e.g. N4) start to appear. This would reduce the enrichment of N1 spores, making the distinction between populations less effective. Since the number of spores below 6 μm is minimal, lowering the theoretical threshold would not be effective, as most spores would still fall into the population above D_c , making the separation unclear.

Table 3.1: Percentage distribution and total number of spores which have equivalent diameters below three different theoretical critical diameter thresholds ($D_c = 5.7, 6$, and $7 \mu\text{m}$).

D_c	N1	N2	N3	N4	N5	N6	Total number of spores
7 μm	34.9%	58.9%	6%	0.19%	-	-	533
6 μm	67.71%	31.25%	1.04%	-	-	-	96
5.7 μm	80%	20 %	-	-	-	-	15

To better illustrate the previous discussion on how the theoretical critical diameter (D_c) influences the enrichment of specific subpopulations, Table 3.1 summarizes the total number of spores falling below different D_c (5.7, 6, and $7 \mu\text{m}$), based on the observations from Figure 3.5. This table provides a visual and quantitative summary of the trends previously described, showing how small variations in D_c can significantly affect the enrichment of spores with a single nucleus, as well as the total number of spores collected.

For example, considering the total number of spores below each critical diameter, we observe that at $D_c = 7 \mu\text{m}$, the largest population is obtained (533 spores), compared to 96 and 15 spores at $D_c = 6$ and $5.7 \mu\text{m}$, respectively. However, analyzing the composition reveals that only 31% (186 spores) are N1 at $7 \mu\text{m}$, while this proportion increases to 67.7% (65 spores) at $6 \mu\text{m}$ and 80% (12 spores) at $5.7 \mu\text{m}$. Although $5.7 \mu\text{m}$ offers the highest N1 proportions, the total number of spores is too low. In contrast, $D_c = 6 \mu\text{m}$ provides a better compromise: it provides a reasonably large sample with a good N1 proportion. Moreover, no N4 spores and very few N3 spores are present at this threshold.

These findings support the use of a size-based criterion to sort spore solutions, with $6 \mu\text{m}$ as the theoretical threshold for separating a population enriched in a single nucleus from another that, while still containing N1 spores, also includes a significant proportion of multinucleated spores

Having established that single-nucleated spores can be enriched from multinucleated

ones based on size, the next step is to develop a microfluidic device capable of performing this size-based separation. The following section presents the design and optimization of the selected microfluidic approach—DLD—to address the specific objectives of this application.

3.3 Design optimization

The following paragraphs present the different versions of the DLD device and its optimization process. It is important to note that circular pillars were chosen because they are the most well-documented geometry, with established theoretical models and equations. In addition, the reference particles used in our DLD studies are typically spherical beads and spores, which, in a first approximation, can be assumed to be nearly spherical, as shown in Appendix A.2. This pillar's geometry is therefore suitable, although it could be interesting to see if another pillar shape may improve the sorting performances of non-spherical particles.

The discussion is structured into two sections: the first, **Device Design**, covers the system's design and optimization, including parameters such as channel length, inlets, and outlets. The second, **Array Design**, explores the different deviation array configurations implemented.

3.3.1 Device Design

The 2-10 μm beads suspension was used in the experiments carried out in this section. The large size difference between the 2 μm and 10 μm particles facilitated an initial assessment of the device design because it was possible, using a high-speed camera, to observe whether 2 μm particles were directed to the **small outlet** (designed for particles below D_c) and the 10 μm particles to the **big outlet** (designed for particles larger than D_c).

The first version, **DLD v1**, included a sample inlet and two outlets. It was observed that when the particles entered the device, they dispersed in all directions, without a clear deviation (Fig. 3.6a). This behavior was likely due to the absence of a buffer inlet, which is essential for flow focusing. Without it, the system failed to properly focus the sample flow, preventing the particles from following their expected trajectories. As a result, both particle sizes could be recovered in the small outlet (Fig. 3.6b). Lastly, a ceiling collapse was observed in both the inlet and outlet channels, which could be caused by the high aspect ratio of the channels.

In the new version, **DLD v2**, a buffer inlet (Fig. 3.3 item 5) was included in the design to focus the flow against the upper wall, which was important based on observations from the previous version. The width and length were also modified. The length was chosen so that, even if a particle enters at the first pillar (close to the wall) it should still follow the expected deviation along the pillar array and reach the beginning of the big outlet. When the 2-10 μm beads suspension was injected into the DLD v2, it was observed that particles were flow-focused toward the sample inlet walls due to the buffer flow, and both expected trajectories were visible: 2 μm beads followed a straight path, while 10 μm beads deviated along the channel. However, some 10 μm

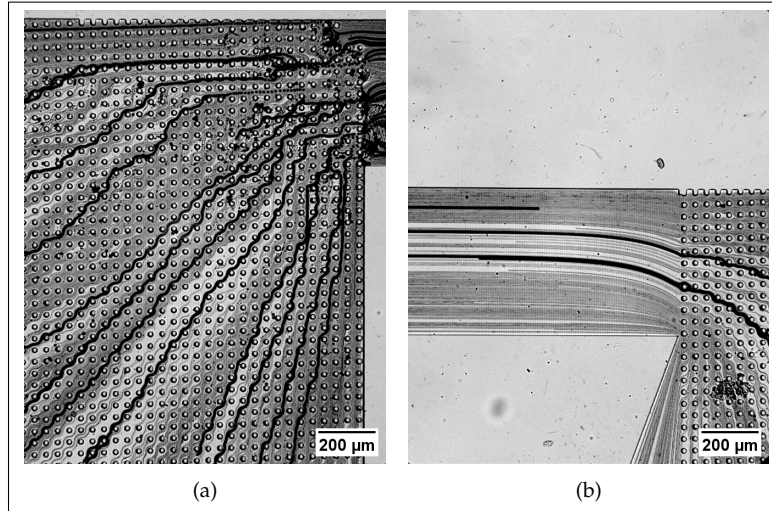


Figure 3.6: Video-microscopic observations of a suspension. (a) Inlet. (b) Outlet. Trajectories were extracted using a Z-projection (MIN) in ImageJ. Dark lines represent large beads (10 μm), light gray lines represent small beads (2.2 μm).

did not deviate enough and ended up in the small outlet, likely because the device length was designed with no safety margin. Support pillars (Fig. 3.3 item 7) were added to the buffer inlet and outlet channels design to address the issue of ceiling collapse. The support pillars have a diameter of 130 μm and are spaced 270 μm apart. The dimensions were chosen arbitrarily.

Based on the previous observations, in **DLD v3** the array's length was designed to ensure that particles on a bumping trajectory reach halfway into the big outlet, minimizing the risk of contamination (i.e., particles ending up in the wrong outlet). This required length depends on the deviation angle, ensuring the system operates effectively. As a result, a complete separation of 2 and 10 μm was visible. Additionally, following improvements made in DLD v2, where support pillars were added to the buffer inlet and outlet channels to address ceiling collapse, support pillars were also added to the sample inlet in this version. The size of the support pillar was increased from 130 μm in version 2 to 160 μm in version 3 to find the optimal dimensions. During the experiments, it was observed that the device tends to clog easily, which reduces its operational lifetime.

To prevent clogging in the deviation array, a filter structure (Fig. 3.3, item 6) was added to the inlet channel, upstream of the array, in **DLD v4**. This structure, consisting of an array of posts, is designed to break down particle aggregates that could potentially clog the device. If clogging occurs, it takes place in the filter, thereby protecting the deviation array from disrupted flow lines or complete blockage. Similar filter structures have already been implemented in microfluidic devices (e.g., Lee et al. [118]). In DLD v4 version, the support pillars have a diameter of 200 μm with 300 μm of spacing, which was not optimal as recirculation was observed at the small outlet, causing 2 μm beads to flow back into the big outlet (Fig. 3.7).

In the next version, **DLD v5**, recirculation had to be avoided. Since increasing the support pillar size was not effective, the pillar size was instead reduced to 50 μm , with a pitch of 50 μm . However, recirculation did not completely disappear. Additionally,

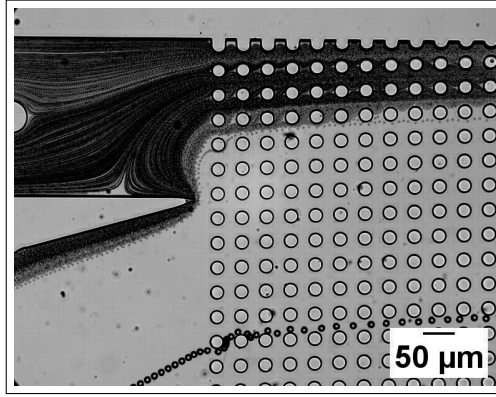


Figure 3.7: Recirculation observed in the small outlet of DLD v5, causing $2.2\ \mu\text{m}$ particles to be redirected to the large outlet instead of being collected as expected.

some debris was observed entering the buffer inlet. To avoid it, a filter structure was also added to this inlet.

In the latest version, **DLD v6**, the only modification was the addition of pillars ($20\ \mu\text{m}$ in diameter) in the small outlet to disrupt the flow and successfully eliminate recirculation. This final design, illustrated in Fig. 3.3, will be used in all future experiments, with only the deviation array (Fig. 3.3 item 8) being modified.

3.3.2 Array design

DLD devices are characterized by arrays of pillars that shift laterally in each successive row. The key design parameters of a DLD array are the gap (G) and the row shift fraction (ϵ), both of which are discussed in detail in Chapter 2 (Section 2.1.3) and illustrated again in Figure 3.3, item 8.

To achieve spore separation around a threshold of $6\ \mu\text{m}$, three deviation arrays were designed with theoretical critical diameters of 6 , 5 , and $4\ \mu\text{m}$, using the empirical formula proposed by Davis [63] for circular pillars. All the parameters mentioned here are better explained in Chapter 2 (Section 2.1.3).

This approach allows testing different DLD configurations to identify the most effective one for spore separation at $6\ \mu\text{m}$. As the DLD devices analyzed here consist of narrow arrays (< 25 columns), boundary corrections introduced by Inglis [71] were applied on both the left and right edges to minimize sidewall effects that could compromise separation performance. A summary of the geometrical parameters of the three devices is presented in Table 3.2.

Table 3.2: Deviation array parameters of the DLD devices

Parameters	Symbol	DLD 6	DLD 5	DLD 4
Theoretical critical diameter	D_c (μm)	6	5	4
Row shift fraction	ε	0.04	0.031	0.031
Period	N	25	32	32
Array tilt angle	θ	2.31°	1.76°	1.81°
Gap width	G (μm)	20	19	15
Channel depth	h (μm)	17.1	16.3	16.3
Post diameter	D_p (μm)	20	20	20
Inter-post distance	λ (μm)	40	39	35
Array length	L (mm)	11	17.4	16.8

After finalizing the DLD device design, the next step is to investigate how flow parameters affect the device's performance. The following section examines this influence and establishes the optimal flow parameters for the DLD system.

3.4 Flow parameters

This section presents the qualitative results of experiments carried out in a DLD device to examine how the inlet pressures could impact the macroscopic flow patterns, and therefore the deviation efficiency. This experiment aimed first, to establish the maximum pressure at which the system operates effectively and second, to determine the optimal pressure ratio (calculated as the ratio of sample inlet pressure P_s to buffer inlet pressure P_b) range for reliable performance.

As in the previous section 3.3.1, a 2-10 μm beads suspension was used for these trials. Experiments were conducted at three different buffer inlet pressure levels: low (300 mbar), medium (600 mbar), and high (900 mbar). These values were chosen based on preliminary tests, where lower pressures resulted in no sample recovery, and 900 mbar was close to the upper limit of the pressure-based flow controller. For each pressure, the particle flow pattern was observed. Additionally, the **pressure ratio** was evaluated to determine its effect on the system performance. P_b was fixed at a constant value during the experiments. The pictures were taken after 100 s to ensure steady-state conditions.

The sample width of 2 μm particles was measured, and a normalization process was applied. In the DLD system, normalization was based on the channel width of the small outlet. Particles with a normalized flow width greater than 1 exited through the big outlet, while those with a flow width of 1 or lower remained in the small outlet. Since 10 μm beads are significantly larger than the critical diameter of all devices, they were expected to always deviate towards the big outlet, serving as a reference that the system is working as expected. The flow was first characterized in DLD 6 and then in the other two devices.

3.4.1 Impact of different pressures at the inlets

Following the described protocol, four repetitions were performed for the 600 mbar condition. The small standard deviation observed indicated that the results were consistent and reproducible. Therefore, the experiment was conducted only once for the other conditions, without repetitions. The curves describing the flow behavior under different inlet pressure conditions are presented in Fig. 3.8.

It can be observed that at 300 mbar (red stars), the sample flow had a slightly greater width compared to 600 mbar (green diamonds) and 900 mbar (blue triangles) for the same pressure ratio values. For example, at a pressure ratio of 1, the sample width was slightly above 1, for a buffer pressure of 300 mbar. In contrast, at 900 mbar, the sample flow was narrower, measuring around 0.8. As expected, this confirms that at lower pressures, the sample flow expands more, whereas higher pressures limited its width. Since a clear separation was observed under all pressure conditions, **900 mbar** was selected as it allowed for a higher sample throughput.

3.4.2 Impact of the pressure ratio

In the previous section, a pressure of 900 mbar was selected for subsequent experiments. The next step is to identify the optimal parameters to ensure effective particle separation, avoid cross-contamination between outlets, and maximize the performance of the DLD device.

Figure 3.8i illustrates the effect of different pressure ratios on the sample width. Across all pressure ratio conditions, the behavior of larger particles (e.g., 10 μm) remained consistent: they followed a bumping trajectory and exited exclusively through the large outlet. Consequently, the measured sample width corresponds to the distribution of 2.2 μm particles. These beads, being significantly smaller than the critical diameter (D_c), are theoretically expected to follow a straight trajectory along the channel.

Overall, for all three buffer pressure levels tested, the sample width increased quasi-linearly with the pressure ratio. Three distinct regions can be identified in the graph, highlighted by colored bands:

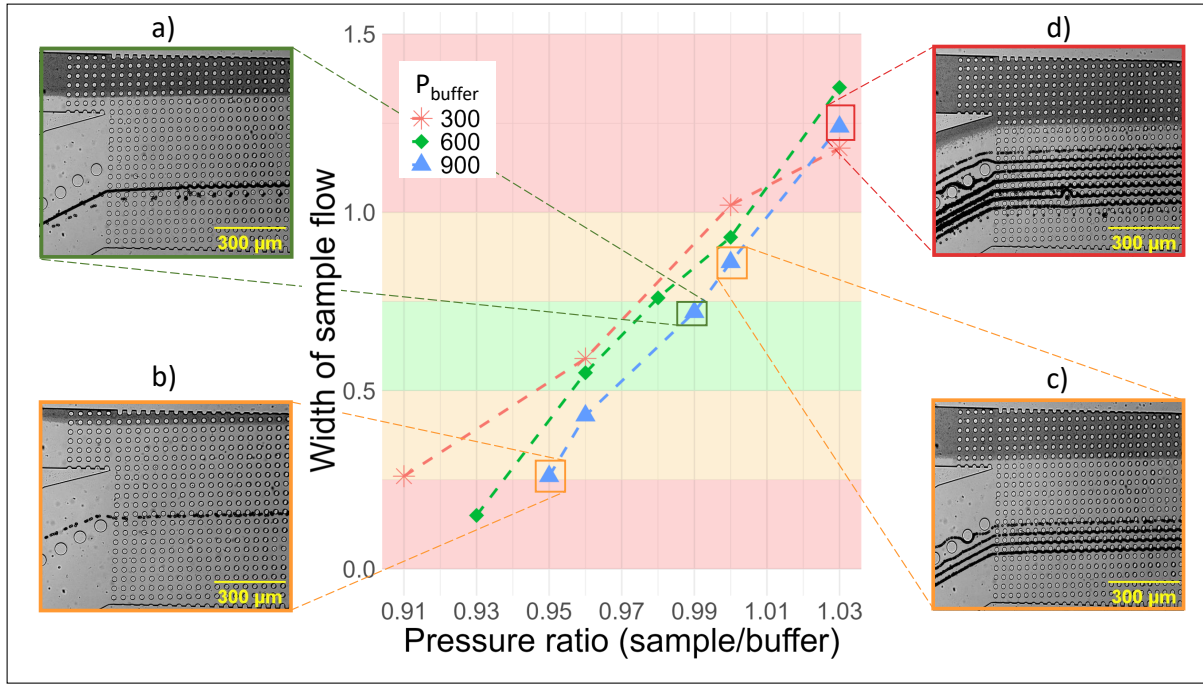


Figure 3.8: Sample width at the small outlet as a function of the pressure ratio (sample to buffer) for three applied buffer inlet pressures: 300, 600, and 900 mbar. The colored bands represent different regions: (a) green band – optimal parameters, (b) lower orange, and (c) upper orange are the safe margins, and (d) upper red - leads to contamination.

- **Green band (optimal parameters)** This range corresponds to the conditions under which the sample stream follows a straight trajectory and exits exclusively through the small outlet. Here, the sample width lies between 0.5 and 0.75 (green band). This behavior is illustrated in Fig. 3.8a. The pressure ratio for this optimal sample width range is found between 0.97 and 0.99.
- **Orange band (Safe margin)**
 - Upper orange band. This region (sample width between 0.75 and 1) defines the maximum allowable sample width before contamination occurs (Fig. 3.8c)). Contamination refers to particles going to the wrong outlet. For the P_b of 900 mbar, this band is presented at a pressure ratio above 0.99 and below 1.01.
 - Lower orange band. This region represents the conditions where there is no contamination, but the sample inlet flow is significantly reduced, meaning the device does not operate at full capacity (Fig. 3.8b)). This band corresponds to a pressure ratio between 0.965 and 0.95 for a P_b of 900 mbar.
- **Red band (operational limits)**
 - Upper red band. When the width of the sample exceeds 1, it splits into both outlets, leading to contamination of the big outlet and loss of particles of interest (Fig. 3.8d)). This occurs at a pressure ratio above 1.01.
 - Lower red band. The flow ceases almost entirely below a pressure ratio of 0.95, which is represented by the lower red band. As shown, there is

no sample width in this region because no flow occurs. An exception was observed at 600 mbar (pressure ratio of 0.93), where a very thin sample width appeared, though it was nearly imperceptible.

The experiment was repeated for DLD 5 and DLD 4, using an inlet pressure of 900 mbar. The results are presented in Appendix A.1. Similar behaviors were observed for the three DLD devices. However, the slope is slightly different. This indicates that the pressure ratio range required to stay within the green zone is narrower for DLD 4 than for DLD 5 and DLD 6, meaning that the slope for DLD 4 is steeper compared to the others. In other words, DLD 4 is more sensitive to the pressure ratio than the others.

The pressure applied at the inlets has a direct impact on separation efficiency, as previously discussed. A similar observation was reported by Aghajanloo et al., who qualitatively investigated this effect using a DLD device with one sample inlet and two buffer inlets. To visualize the flow behavior, they dyed the sample and buffer fluids with different food colorings and analyzed the interface under varying buffer-to-sample flow rate ratios.

Their study revealed distinct regimes, including an optimal zone that ensured complete separation. When the flow rate ratio was increased, the sample stream expanded, leading to cross-contamination between outlets. Conversely, when the ratio was too low, the sample stream became too narrow, preventing the device from operating at full capacity. These findings are consistent with our results and confirm the importance of fine-tuning flow conditions to optimize DLD performance.

The selected pressures, $P_b = 900$ mbar and $P_s = 882$ mbar, correspond to a sample-to-buffer pressure ratio of 0.98. These parameters fall within the green band, maximizing sample throughput while ensuring reliable separation.

Following the optimization of flow parameters, the next step consisted of finding the correct strategy to reduce particle-to-particle aggregation and particle-to-surface adhesion. The next section presents the approaches used to achieve this.

3.5 Buffer Optimization

In a DLD device, a high surface area-to-volume ratio results from the presence of numerous pillars. This increases the likelihood of particle non-specific adhesion to the pillars or the walls. The adhesion can cause particle losses, perturbation in the flow lines, and eventually clogging, thereby reducing both the function performance and the device's lifespan. This section discusses the different approaches tested to address these challenges.

3.5.1 Pre treatment

Previous studies suggest that a pre-treatment step before sample introduction may reduce particle-surface interactions and minimize bubble formation in the system. For instance, Loutherback et al. [119] used a solution of $1 \times$ PBS, 1% BSA (w/v), and 1

mM Ethylenediaminetetraacetic acid (EDTA) to wet the tubing and devices. Similarly, Inglis et al. [72] flushed the device with 0.22 μm filtered PBS before injecting a suspension of *Aspergillus* spores. Since fungal spores were also used in this study, the same solution, 0.22 μm filtered PBS, was selected to wet the DLD devices. A reduction in particle-surface adhesion was observed when using this solution, and after 15 minutes of running this wetting solution, no bubbles remained in the device.

3.5.2 Buffer

The first buffer tested was 9% w/v Dextran dissolved in 2% w/v BSA. This solution had been previously used by the team for human cells to prevent sedimentation and reduce particle-to-particle aggregation [120]. When the spore suspension was introduced into this buffer, the particles flowed through the device, allowing the sample to be processed for approximately 15 to 20 minutes. However, due to its high viscosity, sample recovery was minimal after the run.

Subsequently, three additional Dextran concentrations (0.9% w/v, 2% w/v, and 5% w/v) dissolved in 2% w/v BSA were tested. As shown in Fig. 3.9 for 0.9% w/v, a low Dextran concentration led to spore agglomeration, which rapidly clogged the device. Similar clogging issues were observed with the 2% w/v and 5% w/v Dextran solutions.

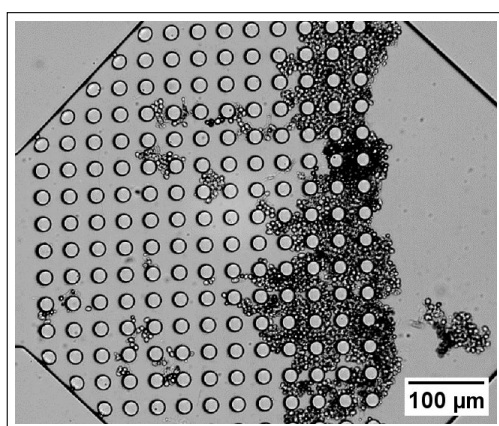


Figure 3.9: Spore agglomeration and clogging observed in the filter structure when using a Dextran solution at concentrations of 0.9% in 2% BSA. Lower Dextran concentrations promoted faster clogging due to increased spore aggregation.

The second buffer solution tested consisted of 0.1% w/w polyethylene oxide (PEO) and 0.01% v/v Tween 20 dissolved in PBS. This buffer was previously used by Park et al. [54] to prevent microparticle aggregation. Although it slightly reduced aggregate formation, the system operated for a maximum of only 10 minutes before clogging occurred.

Finally, the last buffer tested was PBS containing 1% v/v Tween 20. This formulation was based on a previously reported buffer by Inglis et al. [72], who used 0.1% v/v Tween 20. In our case, the concentration was increased tenfold as part of an empirical adjustment aimed at further reducing surface tension and improving spore dispersion. This modification effectively eliminated spore agglomeration, allowing continuous flow through the system for approximately 15–20 minutes, during which sample recovery was possible.

If the three buffer solutions tested are compared, the 9% w/v Dextran in 2% w/v BSA and PBS with 1% v/v Tween 20 were the most effective, as they reduced particle-to-

particle interactions and allowed spores to flow through the device for 15 to 20 minutes. However, the PBS-Tween 20 solution enabled a higher sample recovery volume under the same injection pressure and collection time. Additionally, Tween 20 is routinely used in the MAP laboratory to prepare spore solutions, ensuring cell viability and compatibility with spore cultures after passing through the DLD device. Therefore, PBS with 1% v/v Tween 20 was selected as the buffer for all future experiments.

As can be observed, the process parameters have been optimized, including the flow parameters and the buffer solution. The next step is to characterize the device by determining its empirical cutoff and quantifying its separation efficiency, which will be done in the following section.

3.6 Cut-off and separation efficiency determination

In this section, we aimed to determine the empirical cut-off size and evaluate the separation efficiency of our three DLD devices. Together, these two parameters provide a comprehensive assessment of the system's performance. The empirical cut-off defines the actual threshold diameter above which particles are laterally displaced, while the efficiency reflects how precisely the system performs this separation.

The spores of *B. cinerea* exhibit a continuous size distribution and a non-spherical, tear-shaped morphology. To enable more controlled and reproducible characterization, we used 6 μm polystyrene beads as model particles. These beads are spherical and possess a continuous size distribution comparable to that of the spores, making them a suitable reference for evaluating device performance.

Characterization using these beads was carried out across the three DLD designs. SEM was employed to image particles collected from the inlet and both outlets, as described in Section 3.1.2. The resulting images were processed using MATLAB to extract the surface area and calculate the diameter of individual particles, enabling quantification of the separation outcomes.

First, the surface area and diameter of each bead were measured. Then, using RStudio, the diameter values were divided into regular intervals (from 3 to 8. μm , in 0.1 μm steps), and the number of particles within each interval was counted. This allowed us to build a frequency distribution of particle diameters. The choice of interval width was arbitrary. Although the frequency distribution was calculated using intervals of 0.1 μm , for clarity in the graph interpretation, data points were plotted every 0.5 μm . The continuous line connecting these points, however, reflects the full resolution of the 0.1 μm interval analysis.

Next, the deviation efficiency (DE %) was calculated, indicating the percentage of particles in each interval that deviated towards the Big Outlet. Therefore, to calculate DE %, only the particles in the Big Outlet were considered and compared to the total number of particles on both outlets. Consequently, it can be defined as

$$DE(\%) = \frac{I_{nBig}}{I_{nOutlets}} * 100 \quad (1)$$

where $I_n Big$ is the number of particles in the n th interval of the big outlet and $I_n Outlets$ is the total number of particles in both outlets in the n th interval. This deviation efficiency then fits a sigmoidal model of the form

$$DE(\%) = \frac{A}{1 + e^{-k(x-x_0)}} \quad (2)$$

Where A is the maximum value of efficiency (normally it should be 100%), k is the slope, which indicates how fast efficiency grows, and X_0 is the point where the efficiency reaches 50%. In Fig. 3.10, the sigmoidal curve for the deviation efficiency for the DLD 4, DLD 5, and DLD 6 can be seen. Three areas can be observed:

- **No deviation (gray band):** This region represents the range of diameters in which no deviation occurs, meaning the particles follow a straight trajectory without deviating. Here, the deviation efficiency is 0%. This occurs because the particles are smaller than the critical diameter of the system. In this case, none of these particles are expected to be collected in the Big Outlet.
- **Mixed (blue band)** This region corresponds to the transition zone, where the deviation efficiency progressively increases. Here, the deviation efficiency goes up to 95 %. Some particles begin to deviate, while others continue in a straight trajectory. And particles with diameters close to the critical diameter may exhibit both behaviors. The larger the particle diameter, the higher the probability of deviation.
- **Full Deviation (yellow band):** In this region, all particles are completely deviated. This occurs because their diameter is larger than the device threshold.

For DLD 4 (Fig. 3.10a), the no-deviation zone includes particles up to 3.8 μm in diameter, while deviation begins around 5.3 μm . This indicates that DLD 4 displaces smaller particles compared to DLD 5 and DLD 6. This behavior is expected, as DLD 4 was designed with a lower theoretical cut-off than the other two devices.

In DLD 5 (Fig. 3.10b), particles smaller than approximately 5.25 μm are not displaced, while those larger than 6.6 μm are fully deviated toward the large-particle outlet.

In DLD 6 (Fig. 3.10c), the no-deviation zone extends up to approximately 6 μm , suggesting that smaller particles are not displaced. However, a clear deviation zone is not observed. This may be due to the size distribution of the 6 μm bead suspension not being broad enough to fully characterize the system and identify the onset of deviation. As a result, DLD 6 does not meet our expectations, since the expected theoretical diameter of deviation around 6 μm is not evident.

Comparing the mixed zones of the three devices, DLD 6 shows the largest mixed zone. This suggests lower separation accuracy near the theoretical cut-off (D_c), reducing the overall efficiency of the device. It may also result in cross-contamination between particle populations, with some particles ending up in the incorrect outlet and decreasing the purity of the collected samples.

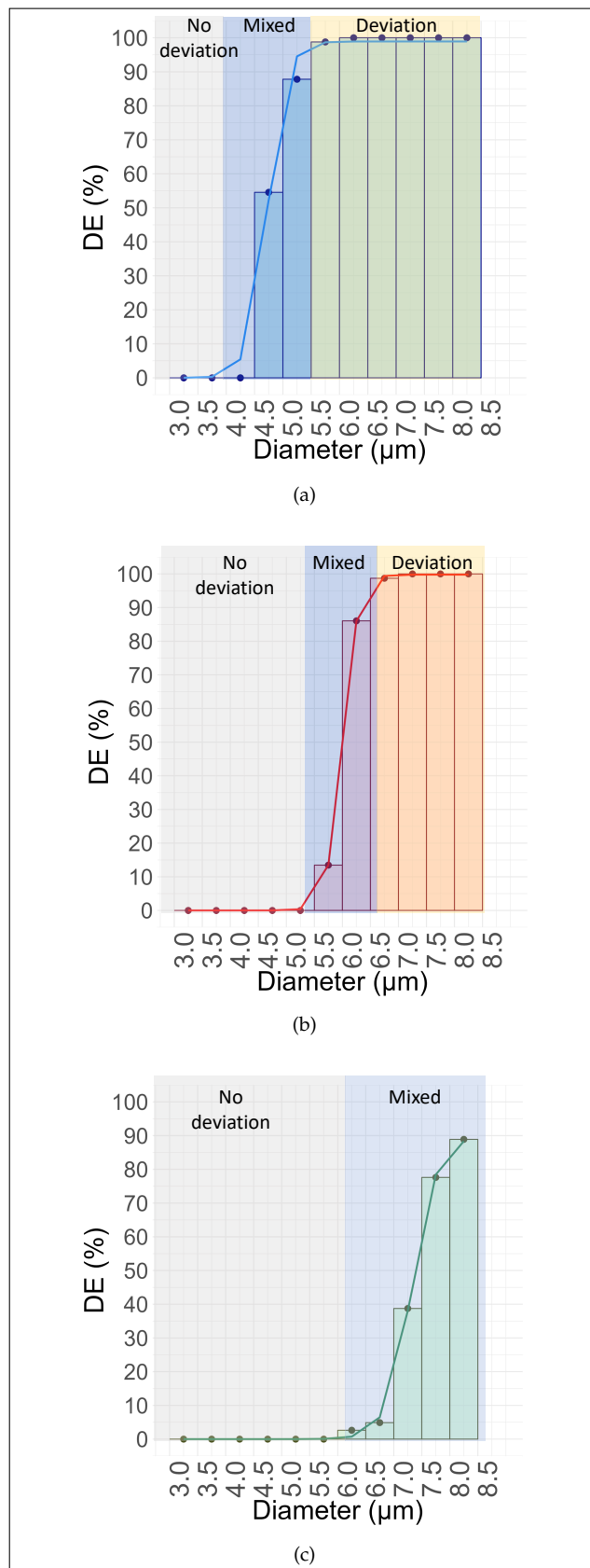


Figure 3.10: Deviation efficiency (%) as a function of particle diameter for (a) DLD 4, (b) DLD 5, and (c) DLD 6. The gray, blue, and yellow bands indicate regions of no deviation (0%), mixed deviation (up to 95%), and full deviation (95-100%), respectively.

The region referred to here as the mixed zone has been previously studied. For example, Pariset et al. [121] identified an intermediate displacement mode in which particles near the theoretical critical diameter exhibit both zigzag and bumping behaviors

due to wall effects. These effects generate unexpected particle trajectories that compromise separation efficiency. They suggested that this phenomenon occurs primarily in narrow DLD devices—such as those presented here, which have approximately 20-25 posts across the channel width—and may explain the presence of this mixed mode in all three DLDs. They proposed calibration curves that incorporate the number of columns across the channel width to better predict particle trajectories. Although edge correction was applied in our devices, it appears that further refinement is needed.

Since no sharp critical diameter is observed, the mixed area will be considered as the empirical critical diameter, D_{Emp} , in this study. Table 3.3 summarizes all the discussed regions for each device.

Table 3.3: Summary of deviation regimes for DLD devices 4, 5, and 6. The table shows the diameter ranges corresponding to no deviation, mixed zone (empirical critical diameter, D_{Emp}), and full deviation. *For DLD 6, the upper limit of the mixed zone is an estimate, as the full deviation region was not observed.

Device	No deviation	Empirical Critical Diameter (D_{Emp})	Full deviation
DLD 4	< 3.8 μm	3.8 - 5.3 μm	> 5.3 μm
DLD 5	< 5.25 μm	5.25 - 6.6 μm	> 6.6 μm
DLD 6	< 6 μm	6 - 8.5* μm	Not visible

In conclusion, when evaluating the devices concerning the objective of enriching the N1 spores population by separating spores at 6 μm , DLD 5 appears to be the most suitable option. In this case, particles between 5.25 and 6.6 μm exhibit an intermediary mode, alternating between zigzag and bumping behaviors, while particles larger than this threshold are fully deviated.

Based on the results obtained, **DLD 5** is the most suitable device for our purposes, as it ensures deviation efficiency beyond 6.6 μm for spherical particles.

After characterizing the DLD devices and identifying their empirical critical diameters, the next step is to evaluate their performance with the target particles: *B. cinerea* spores. Unlike the spherical beads used in the initial tests, the spores have a teardrop-like shape, which may influence their behavior in the system.

3.7 Evaluation of *B. cinerea* spores separation in DLD Devices

In Section 3.6, we analyzed the empirical cut-off and separation efficiency of the DLD devices using spherical particles. However, since *B. cinerea* spores have a teardrop-like shape; their separation behavior may differ. Therefore, the three DLD devices were tested with the spore suspension of *B. cinerea* to evaluate their performance with these non-spherical particles.

The analysis focused not only on the equivalent diameter of the spores collected at the small outlet, but also on the number of nuclei they contained. The main objective

is to enrich mononucleated spores to obtain a suspension with homogeneous DNA content.

A suspension of *B. cinerea* spores, whose nuclei were labeled with GFP, was injected into the three different DLD devices. Samples were collected from both outlets, and their size was measured following the protocol described in Section 3.1.3

Figure 3.11 presents the equivalent diameter distribution of spores grouped by their number of nuclei (N1, N2, etc.), comparing the populations collected at the inlet and both outlets of the three DLD devices. It is important to note that the inlet data correspond to a subsample of the inlet reservoir and do not represent all spores entering the DLD device. This was performed to characterize the inlet population properly. To complement this analysis, Table 3.4 reports the proportions of spores recovered at the small outlet—our main focus—classified according to their nuclear content.

In this context, **proportion** refers to the percentage of a specific subpopulation of spores within a given sample. It is calculated as the number of spores with a specific number of nuclei (Sample N_n , where N_n represents N1, N2, ..., N_n) divided by the total number of spores in that sample (Sample_{total}):

$$\text{Proportion (\%)} = \frac{\text{Sample } N_n}{\text{Sample}_{\text{total}}} \times 100 \quad (3)$$

Additionally, we define **enrichment** as the fold increase in the proportion of a given subpopulation (e.g., N1) in the small outlet relative to its original proportion in the inlet:

$$\text{Enrichment (\%)} = \frac{\text{Outlet Proportion } N_n}{\text{Inlet Proportion } N_n} \times 100 \quad (4)$$

Finally, **recovery** refers to the total number of spores collected at a specific outlet, regardless of their nuclear content. It represents the absolute yield of the device and is used to assess the amount of material recovered after the separation process.

For **DLD 4**, which has the lowest empirical cut-off, almost no particles were recovered at the small outlet (Fig. 3.11b). This was expected, as the empirical critical diameter lies between 3.8 and 5.3 μm , as described in Section 3.6. Moreover, the size distribution of spores at the inlet (Fig. 3.11a) indicates that only a few spores fall below this critical diameter.

As shown in Table 3.4, the small outlet of DLD 4 collected N1 and N2 spores in equal proportions (50% each). However, this corresponds to only a single spore of each group. Although the enrichment of N1 spores was high (683%) compared to the inlet—where N1 spores have a proportion of only 7.32%—the overall recovery was extremely low. This suggests that DLD 4 has very poor recovery, making it unsuitable for our practical application. Additionally, despite the low cut-off, N2 spores were still present at the small outlet, which confirms an overlap in diameter distributions, as discussed in Section 3.2.

In contrast, for **DLD 5**, spores were successfully recovered at the small outlet, as shown in Fig. 3.11d. Most of these spores had one nucleus (21.28%) or two nuclei (69.15%), with a smaller fraction presenting three nuclei (9.57%). The presence of N3 spores could be explained by the higher empirical cut-off of this device, which allows slightly larger particles than DLD 4 to enter the small outlet. In this case, the enrichment of N1 spores reached 260%, indicating a notable increase in their proportion in the collected suspension relative to the inlet. Furthermore, a larger number of spores (94 in total) were collected at the small outlet, highlighting an improvement in recovery compared to DLD 4.

For **DLD 6**, the empirical cut-off is higher than in the other devices, allowing larger spores—typically with more nuclei—to be collected at the small outlet. This supports the observed trend: as the empirical critical diameter increases, spores with a higher number of nuclei become more frequent at the small outlet. Compared to the previous devices, the proportion of N1 spores decreased to 11.5%, while N3 spores increased to 37.82%, and a small percentage of N4 spores was also observed (2.56%). As a result, the small outlet contained a smaller proportion of N1 spores and a bigger proportion of N3 and N4 spores than both DLD 4 and DLD 5. In terms of recovery, DLD 6 collected the highest number of spores, but it showed the lowest enrichment of N1 spores, with only a 157% increase.

Based on the performance of the three DLD devices, **DLD 5** was selected as the most suitable system for isolating *B. cinerea* spores. Although DLD 4 exhibited the highest enrichment of mononucleated (N1) spores (683%), its extremely low recovery—only two spores collected—renders it impractical for our application. On the other hand, DLD 6 achieved the highest recovery, but with a significant presence of multinucleated spores and the lowest enrichment of N1 spores (157%). DLD 5 offered the best compromise between recovery and enrichment: it collected a reasonable number of spores (94) and achieved a notable N1 enrichment of 260%. Furthermore, the small outlet of DLD 5 yielded a favorable spore composition, with a majority of N1 and N2 spores and minimal contamination by spores with three or more nuclei. These results indicate that DLD 5 provides an optimal balance for effectively enriching and recovering mononucleated spores, which is essential for obtaining homogeneous DNA material.

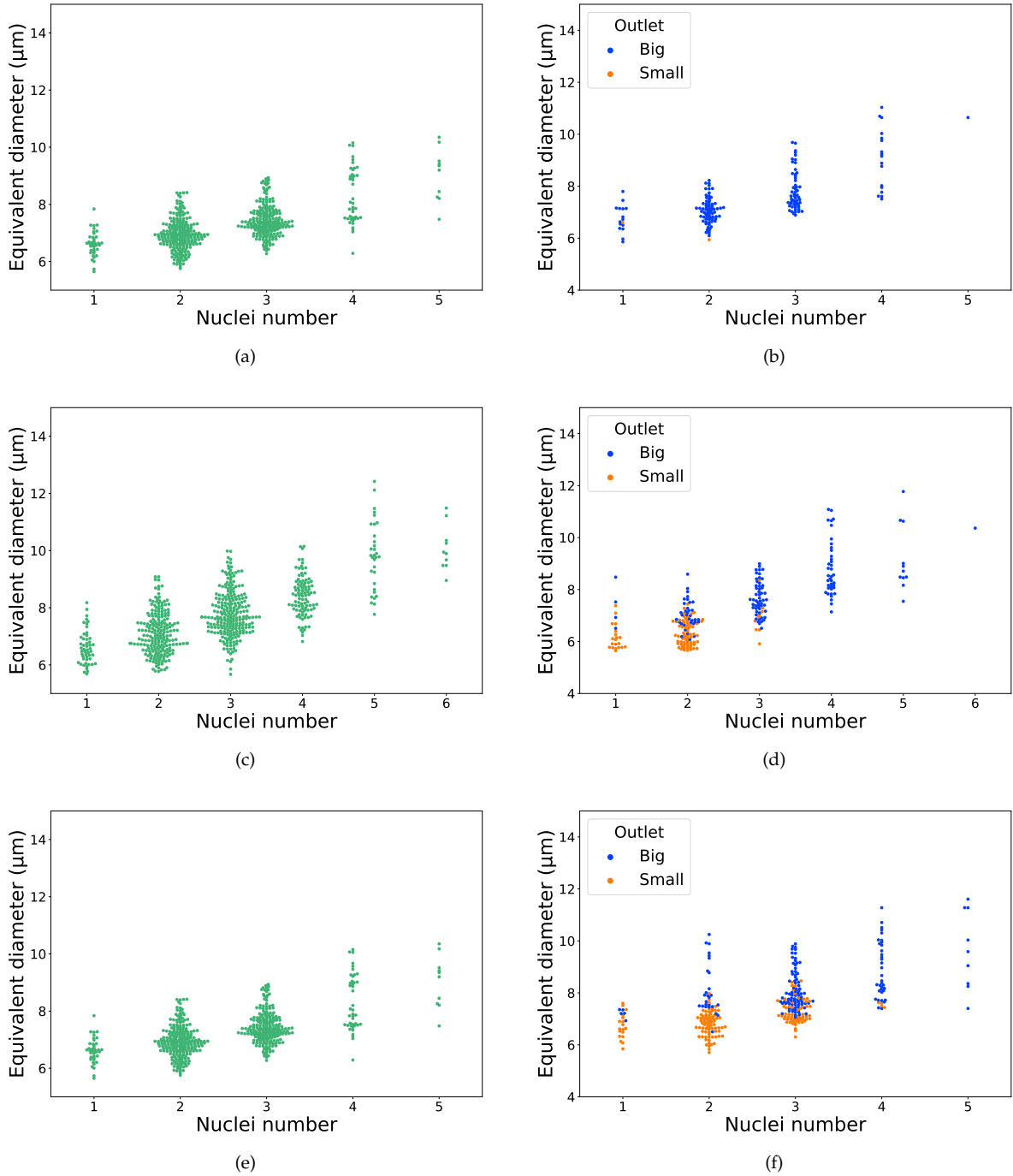


Figure 3.11: Distribution of spore equivalent diameters (μm) as a function of nuclei number, for DLD devices 4, 5, and 6. Inlet data are shown in (a), (c), and (e); corresponding outlet data are shown in (b), (d), and (f).

It was observed that the separation results obtained with spores were fairly similar to those obtained with spherical beads, for the three DLD devices. To evaluate how the shape of spores might influence their behavior in the DLD, we analyzed the same dataset used in Section 3.2—where the critical diameter was determined—and fitted each spore to an ellipse to calculate its axis ratio (major axis divided by minor axis).

As shown in Appendix A.2, most spores exhibited axis ratios between 1 and 1.9, regardless of their number of nuclei. An axis ratio of 1 corresponds to a perfect circle,

Table 3.4: Proportion of spores collected at the small outlet for each DLD device.

Nuclei number	DLD 6(%)	DLD 5 (%)	DLD 4 (%)
N1	11.54	21.28	50
N2	48.08	69.15	50
N3	37.82	9.57	-
N4	2.56	-	-
Recovery	156	94	2

so values in this range indicate that spores are only moderately elongated and remain relatively close to a spherical shape. This moderate ellipticity may help explain why their separation behavior is not drastically different from that of rigid spherical beads. Even though spores are biological and not perfectly round, their overall shape remains close enough to spherical for the DLD device to function effectively.

To validate the performance and reproducibility of **DLD 5**, four additional independent replicates were carried out. Figure 3.12 shows the equivalent diameter distribution of spores collected at the small outlet, grouped by nuclei number for each replicate. In parallel, Table 3.5 summarizes the proportion of each spore subtype, the enrichment of mononucleated spores (N1), and the total recovery (i.e., number of spores collected) at the small outlet across all replicates.

These replicate experiments confirm that DLD 5 consistently enriches for mononucleated (N1) spores while maintaining a comparable recovery level across runs. In the case of replicate DLD5_05, the lower enrichment may be attributed to limited statistics, as the total number of spores collected was low. The observed variability in spore proportions is expected given the biological nature of the samples, where the composition at the inlet can vary between experiments. Nevertheless, the overall distribution profiles remain consistent, reflecting the reproducibility of the separation performance. It is also worth noting that each experiment was conducted under the same conditions, with a constant sample running time of 15 minutes.

Table 3.5: Spores proportions, N1 enrichment, and recovery in the small outlet for each replicate of DLD 5.

Replicate	N1(%)	N2 (%)	N3 (%)	N4 (%)	N1 Enrichment (%)	Recovery
DLD5 01	21.28	69.15	9.57	-	206	94
DLD5 02	14.73	58.14	27.13	-	624	129
DLD5 03	19	66	14	1	805	100
DLD5 04	25.56	54.44	20	-	236	90
DLD5 05	10.87	67.39	21.74	-	101	46

Previous studies have demonstrated the high efficiency of DLD devices for separating fungal spores. Notably, Inglis et al. [72] reported nearly 100% purity and high recovery when isolating spherical *Aspergillus niger* spores from debris using a DLD chip. In their case, the size difference between the spores and residual particles was substantial, which facilitated the separation. In contrast, our goal was to enrich a specific subpopulation—mononucleated (N1) spores of *B. cinerea*—from a heterogeneous suspension of spores all belonging to the same species. These subgroups differ in their

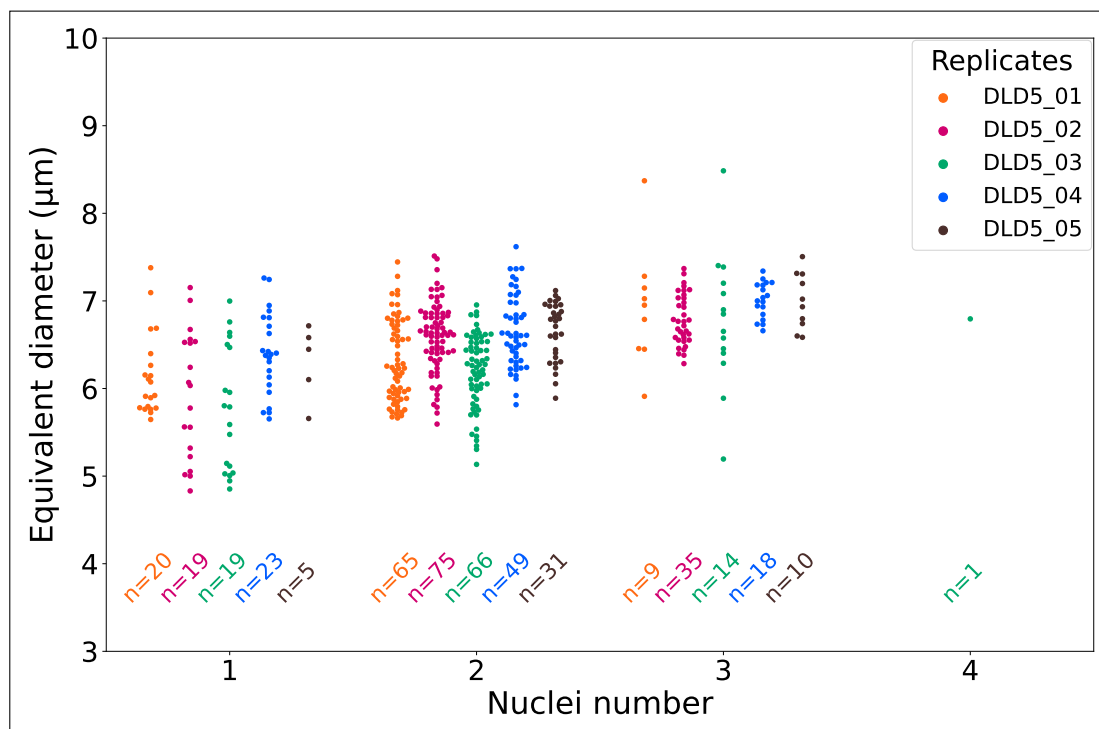


Figure 3.12: Equivalent diameter (μm) of spores grouped by nucleus number (N1 to N4) for all five replicates of DLD 5 (DLD5 01 to DLD5 05). Each point represents an individual spore measured at the small outlet.

nuclear content (N1–N6), but their sizes are only slightly different and partially overlapping. This makes the separation more demanding, as it requires distinguishing subtle variations within a continuous biological distribution, and more importantly, because our interest is in those with homogeneous DNA content.

Although we measured the DLD 5 device’s performance and reproducibility, showing an enrichment between 200%-800%, this does not fully demonstrate its efficiency for DNA homogeneity. While we target single-nucleus spores, some two-nuclei spores are also collected and can contribute to have homogenous DNA. To assess effectiveness, we require that at least one colony out of 50 shows homogeneous DNA, which is in line with the MAP protocol, where 50 colonies can be tested simultaneously. If this criterion is met, a single DLD run can accelerate the purification process.

Therefore, full characterization requires verifying that sorting preserves spore viability and performing DNA analysis on sorted spores/colonies. These experiments will be discussed in the following sections.

3.8 Spores viability

The viability of spores collected from the small outlet of the **DLD 5** device is a key aspect of this study. As previously mentioned, once DNA homogeneity is confirmed, these spores are cultured to investigate the effects of mutations on fungal pathogenesis. After the sorting process, we expected to obtain spores containing mostly one, two, or occasionally three nuclei. However, until now, the viability of mononucleated

spores had not been evaluated in the MAP laboratory, mainly due to the difficulty in isolating them. This experiment represents the first attempt to assess the survival and growth capacity of mononucleated spores, made possible by their physical separation using the DLD 5 device. Following the previous experiment, we continue to use a strain with green fluorescent nuclei to assess spore viability after sorting with DLD 5.

Spores collected from the small outlet were inoculated in liquid growth medium into a 96-well plate. The spore suspension was diluted to ensure that individual spores were sufficiently spaced and did not come into contact with each other. Time-lapse images were captured for 18 hours using the Epifluorescence Microscope Micro XL (Molecular Devices). Fig. 3.13 shows the progression of germination. After 18 hours, clear filamentous growth was observed, indicating successful germination and hyphal development.

These observations confirm — for the first time — the viability of mononucleated spores isolated using the DLD 5 device. This result demonstrates that spores with a single nucleus, once sorted, are capable of survival and growth under appropriate conditions.

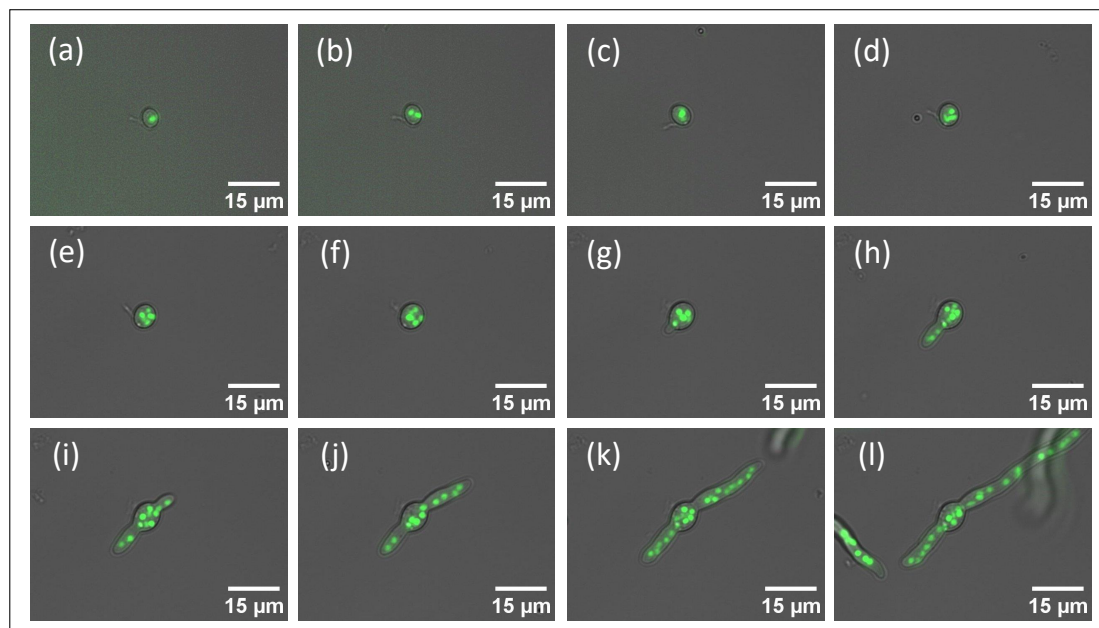


Figure 3.13: Time-lapse images of a mononucleated spore taken for 18 hours, showing the germination and subsequent growth over time.

Now that DLD 5 has been characterized and its performance quantified, showing good results and confirming the survival of mononucleated spores, we can proceed to assess the increase in purity (homogeneous DNA material) from the inlet compared to the small outlet of DLD 5

3.9 Microfluidics Genetic Purification

The final step is to assess the effectiveness of the genetic purification of *B. cinerea* spores using a DLD device. Specifically, we aim to evaluate the genetic homogeneity of the fungal colonies grown from spores collected at the small outlet of the device, following

a single DLD sorting run. As stated in previous sections, to demonstrate the efficiency of the DLD device, we require that at least one out of the 50 tested from the small outlet shows homogeneous DNA content. This result would indicate that a single DLD run is sufficient to obtain a sample with homogeneous DNA, thereby accelerating the purification process.

This section presents the results of PCR analyses performed on *B. cinerea* filaments derived from spores individually isolated at the small outlet. We compare the genetic profiles of the samples taken both at the inlet and at the small outlet to determine the level of purification achieved. The goal is not only to increase the percentage of purified spores relative to the initial population but also to significantly accelerate the purification process. Ideally, a single DLD run should yield a highly purified mutant population at the small outlet, thereby eliminating the need for further rounds of traditional genetic purification protocol.

As explained in Chapter 1, Section 1.1.1, the genetic modification consists of the insertion of a DNA cassette containing a marker gene into the fungal genome. This insertion can occur either randomly or at a specific location (genetic locus).

The previous sections, which focused on evaluating the performance of the DLD devices, used spores whose nuclei were stained with GFP. However, this type of modification is introduced randomly into the genome. To perform PCR analysis reliably, it is crucial to know the exact location of the mutation.

Therefore, in this section, we worked with *B. cinerea* spores carrying a targeted genetic modification consisting of the insertion of a DNA cassette—containing a nuclear localization signal (NLS), the mCherry red fluorescent protein, and an antibiotic resistance gene—into the known *niaD* locus of the genome (Fig. 3.14). This specific insertion allows a precise molecular characterization using PCR. In contrast to previous experiments involving randomly modified GFP-expressing spores, this well-defined mutation enables a reliable comparison between wild-type strain (WT) and mutant strains. Moreover, this color-based mutation is inserted similarly to the mutation affecting spore structural properties, which are of primary interest to MAP biologists. So they can serve as relevant models.

For molecular identification, PCR products are measured in terms of fragment length. These lengths are commonly expressed in base pairs (bp), which correspond to the number of nucleotides (nt) in double-stranded DNA [122]. In this study, we use both terms interchangeably, with "bp" being the standard notation in gel electrophoresis results, and "nt" used to match the terminology in our figures.

Two PCR reactions were designed to target distinct regions:

- **PCR 1** amplifies a segment spanning the *niaD* locus. The wild-type allele produces a band of 2316 bp, while the mutated locus (with the insertion) yields a band of 6308 bp.
- **PCR 2** targets the antibiotic resistance cassette inserted in the mutant strains, producing an 818 bp band present only in these mutants.

These molecular signatures enable a clear distinction between WT and mutant geno-

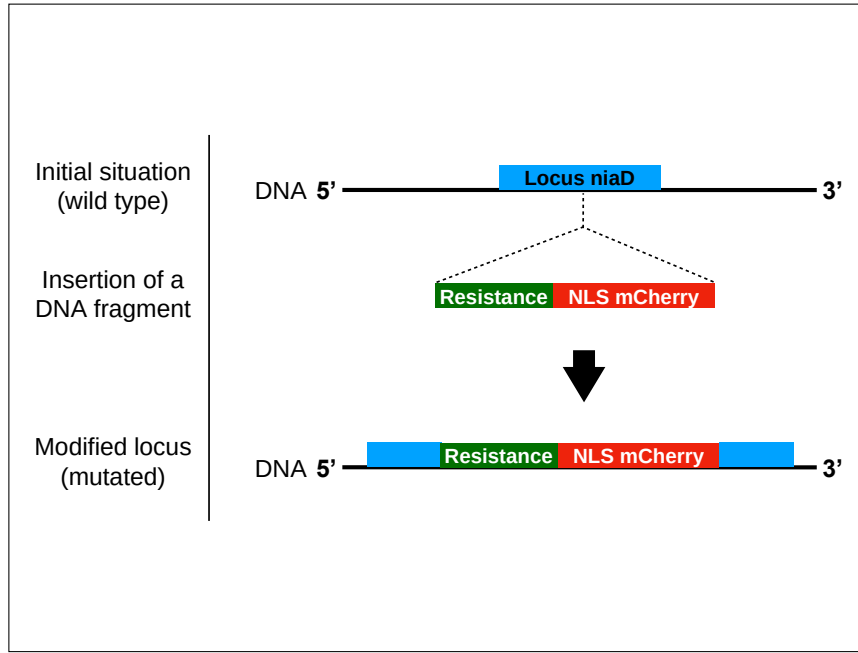


Figure 3.14: Schematic representation of the targeted insertion of a DNA cassette into the *niaD* locus of *B. cinerea*. The cassette contains a nuclear localization signal (NLS) fused to the mCherry red fluorescent protein, and an antibiotic resistance gene for selection.

types (DNA content). To facilitate visualization of this concept, Fig. 3.15 illustrates that distinction. It shows spores containing multiple nuclei: the red-colored ones indicate the presence of the *mCherry* marker (mutant genotype), whereas the blue ones correspond to the wild type genotype. The expected PCR1 & 2 banding patterns are displayed for the different DNA contents: homogeneous (wild type or mutant type) or heterogeneous (wild type and mutant combined), depending on whether the insertion is present in all or only some of the nuclei

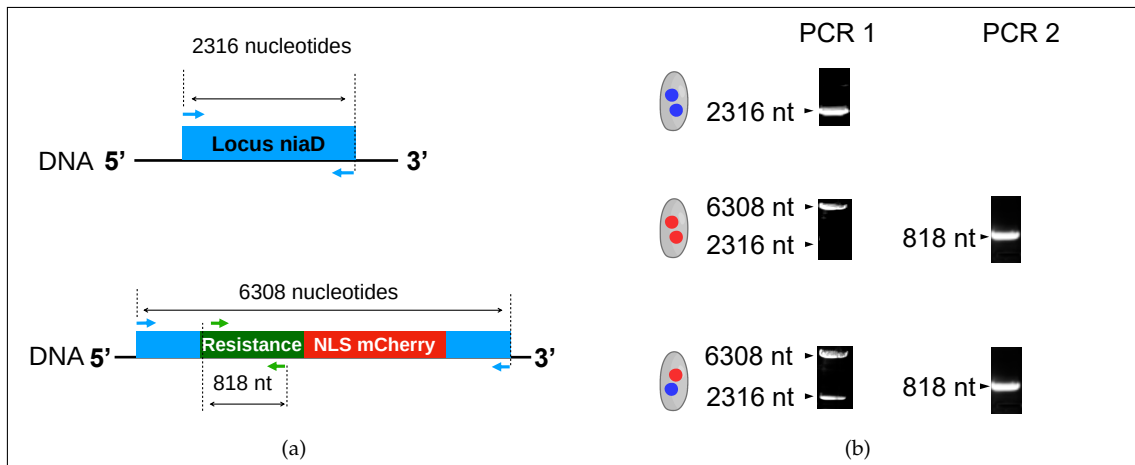


Figure 3.15: Schematic illustration of the two PCR reactions designed to distinguish wild-type (WT) and mutant *B. cinerea* spores. (a) PCR 1 targets the *niaD* locus, producing a 2316 bp band for WT and a 6308 bp band for mutants carrying the insertion. PCR 2 amplifies the antibiotic resistance cassette (818 bp), present only in mutants. (b) Spores with nuclei color-coded by genotype: red nuclei express the *mCherry* marker (mutant), and blue nuclei correspond to WT. Expected PCR banding patterns are shown for spores with homogeneous or heterogeneous nuclear genotypes.

We performed PCR analysis on colonies by spores from the initial solution (inlet) and from the solution collected at the small outlet. First, PCR 2 was used to detect the presence of the antibiotic resistance gene. This result indicates the percentage of spores that would potentially survive when grown on nutrient agar containing antibiotics. The results show that 90% (19/21) of the colonies tested from the inlet carry the resistance gene in at least one nucleus, while 73% (8/11) of the colonies formed by spores collected in the DLD 5 small outlet tested positive. This would indicate that 27% of the spores collected in the small outlet did not contain the resistance gene in their genome. Such spores are expected, as the mono-nucleated spores sorted by the DLD 5 device can contain either the genetically modified nucleus or the wild type nucleus. In the 73% of colonies that tested positive, the presence of the resistance gene was hence confirmed, but this did not provide information about genetic homogeneity.

To assess homogeneity, we used PCR 1. Among the spores carrying the antibiotic resistance gene (resistant strains), only 26% (5/19) of the colonies from the inlet were fully mutant (i.e., all nuclei contained the insertion). In contrast, after DLD 5 sorting, the proportion of fully mutant colonies increased to 75% (6/8). This result demonstrates that DLD sorting not only enriches for spores with fewer nuclei (as discussed in Section 3.7), but also significantly increases the proportion of genetically pure colonies in a single run, representing approximately a threefold improvement compared to the inlet.

Table 3.6: DNA analysis of isolated spores performed at the inlet and small outlet of the DLD

Repetition	Sample	Isolated spores	Antibiotic resistance gene	Pure strain	
				over total	over resistant strains
01	Inlet	21	19/21 (90%)	5/21 (24%)	5/19 (26%)
01	Small outlet	11	8/11 (73%)	6/11 (56%)	6/8 (75%)
02	Small outlet	50	42/50 (84%)	18/50 (36%)	18/42 (43%)
03	Inlet	48	46/48 (96%)	0/48 (0%)	0/46 (0%)
03	Small outlet	48	48/46 (96%)	1/48 (2%)	1/46 (2%)

Interestingly, in the present experiment, the inlet already showed a higher percentage of pure colonies than typically observed by MAP biologists (usually below 2%). Due to the low number of colonies initially studied, two additional replicates were performed to obtain more robust statistics. Table 3.6 summarizes the results.

In the second replicate, we observed that in the small outlet, 84% (42/50) of the colonies tested positive for the antibiotic resistance gene. Among these resistant strains, 43% (18/42) were fully mutant. Unfortunately, we were unable to obtain PCR results for the inlet sample in this replicate, so a direct comparison of purity between the inlet and the outlet was not possible.

In the third replicate, we isolated 48 spores from both the inlet and the outlet. In each case, 96% (46/48) tested positive with PCR 2. Among these resistant strains, 0% of the inlet colonies were fully mutant, whereas in the small outlet, 2% (1/46) exhibited homogeneous mutant DNA.

These results suggest that the increase in purity strongly depends on the inlet sample. Since spores are biological particles, their characteristics can vary significantly between replicates. For example, in the first repetition for the resistant strains, it is

possible that the sample contained a higher proportion of spores with a single nucleus (which are inherently homogeneous) or spores with two or three nuclei where all nuclei happened to have the mutation, leading to homogeneous DNA material. In contrast, the third repetition, for the resistant strains, may have included more spores with multiple nuclei exhibiting different DNA contents, thus resulting in heterogeneity and reducing the likelihood of obtaining homogeneous DNA material.

These variations highlight the biological variability of the samples and suggest that more replicates are needed to draw robust statistical conclusions. Given this variability, it may not be appropriate to define a fixed value for the purity increase. However, a consistent trend was observed: after processing through the DLD 5 device, the number of colonies with homogeneous DNA material increased.

Although in the third repetition only 2% (1/46) of the colonies were fully mutant, this still indicates that even a single run of DLD can yield at least one pure colony. This represents a potential advantage over the current MAP protocol by reducing the time required for strain purification.

While these findings are encouraging, further testing with a larger sample size is necessary to better assess the improvement in purity between inlet and outlet samples.

3.10 Conclusion and perspectives

During this chapter, we presented the implementation and characterization of the Deterministic Lateral Displacement microfluidic device designed to enrich a suspension of spores. The resulting sample is enriched with mononucleated spores, thereby increasing the likelihood of obtaining homogeneous DNA material.

The first step was to corroborate the hypothesis that there is a correlation between the number of nuclei and the size of the spores. We observed that, indeed, a correlation exists: as the number of nuclei increases, spores tend to be larger. However, there is an overlap in the distribution of equivalent diameters. Therefore, it is important to define a relevant critical diameter to divide the spore population into two groups.

After analysis, a critical diameter of 6 μm was chosen, as it offers a good compromise between the enrichment of mononucleated (N1) spores and the recovery of a sufficient number of cells. Particles above this threshold—mostly multinucleated spores—are discarded, while those below—mainly mononucleated, and spores with 2 nuclei, although some may contain three nuclei—are collected. This strategy enables the selection of a subpopulation with a higher probability of containing homogeneous genetic material.

We tested three different DLD devices with theoretical critical diameters of 4 μm , 5 μm , and 6 μm . After a series of characterization experiments using a bead suspension—including flow parameter optimization, buffer selection, and empirical validation—we identified that the DLD 5 device (5 μm theoretical threshold) achieved the best performance. Although it did not produce a sharp cut-off at 5 μm , it exhibited a gradual separation profile (empirical threshold around 6 μm), which aligned well with our target. The device demonstrated good separation efficiency and consistent

reproducibility.

Regarding the spore suspension, which is the main focus of this project, the results show that the small outlet of the DLD 5 device successfully collects the smaller spores, which predominantly contain one or two nuclei, and a few with three nuclei. When we assessed the purity of the sorted samples, we observed variability among replicates, largely dependent on the composition of the initial suspension. Although further repetitions could provide a more robust evaluation of the device's performance, the three replicates conducted here demonstrate that a single round of DLD sorting can yield a sample with homogeneous DNA content. This significantly reduces the time required for the purification process compared to the existing protocol at the MAP laboratory. This time reduction represents one of the key contributions of this thesis, as it accelerates the isolation of purified material, thereby facilitating the genetic characterization of mutant strains and improving our understanding of the genes involved in the infection process.

Furthermore, prior to this work, the MAP laboratory could not isolate and study mononucleated spores due to the technical difficulty of manual separation. The implementation of the DLD 5 device now makes this feasible. In addition to enabling a more efficient purification process, this tool opens new opportunities to study spores with a single nucleus and compare them with multinucleated spores. This is especially relevant for genomic studies aiming to understand the role of nuclear content in fungal development. Recent studies have highlighted the growing interest in understanding the biology and genetic dynamics of multinucleated fungi [123], and with our DLD-based sorting system, the MAP laboratory is now equipped with a powerful tool to contribute to this field.

Résumé du Chapitre 3

Dans ce chapitre, nous avons mis en œuvre une technique de tri microfluidique basée sur la taille, connue sous le nom de **Déplacement Latéral Déterministe (DLD)**.

Nous avons fait l'hypothèse que la taille des spores fongiques était corrélée avec le nombre de noyaux contenus à l'intérieur. Si cette hypothèse est correcte, alors une méthode de tri basée sur la taille peut être employée pour enrichir l'échantillon en spores mononuclées, qui sont la source la plus simple pour obtenir du matériel ADN homogène.

Ce chapitre est organisé en trois sections: "Materials and Methods", "Experimental results", et "Conclusions and perspectives".

La section "Materials and Methods" inclue les protocoles de préparation des suspensions de microparticules, qui ont été utilisées comme particules modèles dans la caractérisation du dispositif. Elle décrit ensuite la préparation des suspensions de spores fongiques, qui diffèrent des billes en raison de leur non sphéricité. Enfin, on y retrouve les protocoles développés pour la mesure des distributions de taille des billes et des spores.

Ces deux objets ont fait l'objets de protocole de caractérisation distincts. En effet, la taille des billes doit être mesurée avec une grande précision, puisqu'elle est essentielle pour déterminer les performances du dispositif de DLD, tel que la taille seuil à laquelle s'effectue le tri. Or une petite erreur, soit dans la mise au point de la particule au moment de l'imager ou bien dans le calcul de son aire projetée peut entraîner une erreur non négligeable sur la détermination de sa taille. Par opposition, la mesure de la taille des spores implique non seulement la mesure de la surface projetée de la particule et son diamètre équivalent (c'est-à-dire le diamètre du cercle de même surface), mais également de compter manuellement le nombre de noyaux dans chaque spore. Finalement, la fabrication des puces microfluidiques et le dispositif expérimental sont décrits.

Nous avons débuté par confirmer notre hypothèse de départ, à savoir la corrélation entre la taille des spores fongiques et le nombre de noyaux qu'elles contiennent. Après avoir analysé plus de 1800 spores, nous avons observé une tendance : les spores avec un grand nombre de noyaux étaient associées à un diamètre plus important. Toutefois, il existe un recouvrement dans la distribution du diamètre équivalent. Afin de séparer la population d'intérêt dans le dispositif, un diamètre critique théorique de 6 μm est choisi. ce seuil permet un bon compromis : il permet de récupérer un nombre correcte de spores de taille inférieure à 6 μm ; lesquelles ont plus de chance de présenter un ou

deux noyaux, tout en minimisant la proportion de spores avec trois noyaux, et en ne collectant quasiment pas de spores avec quatre noyaux ou plus.

Ensuite, nous avons réalisé plusieurs optimisations du design du dispositif, tel que l'ajustement de la longueur du canal, l'ajout de filtres et de piliers de soutènement pour éviter l'effondrement du plafond de la puce. Après les optimisations, trois dispositifs de DLD ont été produits avec des diamètres critiques théoriques (D_c) de 4, 5 et 6 μm .

Les paramètres d'écoulement sont essentiels pour garantir une performance optimale du DLD. Une série d'expériences a donc été conduite pour identifier les meilleurs pressions d'injection qui équilibrent performances et débit.

Finalement, le choix d'un tampon approprié a été crucial. Différents tampons ont été testés et nous avons sélectionné le PBS avec 1% v/v de Tween 20, car il permettait d'éviter la formation d'aggrégats.

Une fois tous les paramètres fixés, nous avons déterminé le seuil et l'efficacité de séparation de chacun des trois dispositifs de DLD, en utilisant une suspension de billes. Ces indicateurs sont importants car ils permettent de comprendre de manière exhaustive les performances réelles de l'appareil.

La puce DLD avec un diamètre critique théorique de $5 \mu\text{m}$ (DLD 5) a été sélectionnée pour les expériences avec les spores fongiques car son seuil de séparation empirique est plus proche de l'objectif de 6 μm .

Nous avons évalué la séparation de spores de *B. cinerea* en utilisant la puce DLD 5, en analysant la distribution des diamètres équivalents des spores récupérées dans la sortie dédiée aux petits objets, où les particules d'intérêt sont collectées. Nous avons montré que des spores de la taille attendue avaient été collectées avec succès à cet endroit. Plus important encore, ces spores contenaient pour la plupart un ou deux noyaux, seules quelques-unes en contenant trois.

Enfin, nous avons évalué l'augmentation de la pureté génétique obtenue grâce au dispositif DLD. En général, après séparation avec le DLD 5, la population de spores présentant un matériel ADN homogène a augmenté dans la sortie dédiée aux petits objets. D'autres expériences sont toutefois nécessaires pour obtenir des résultats statistiquement plus robustes.

En conclusion, dans ce chapitre, nous avons présenté la mise en œuvre et la caractérisation d'un microsystème fluidique de DLD designé afin d'enrichir une suspension de spores fongiques, en spores contenant un unique noyau. L'échantillon obtenu est effectivement enrichi en spores mono-nucléées, augmentant ainsi la probabilité d'obtenir du matériel génétique homogène.

De plus, avant ces travaux, le laboratoire MAP ne pouvait pas facilement isoler et étudier les spores mononucléées en raison de la difficulté technique que représentait leur séparation manuelle. La mise en œuvre du dispositif DLD 5 rend désormais cela possible. Outre le fait qu'il permette un processus de purification plus efficace, cet outil ouvre de nouvelles perspectives pour étudier les spores à noyau unique et les comparer aux spores multinucléées. Cela est particulièrement pertinent

pour les études génomiques visant à comprendre le rôle du contenu nucléaire dans le développement fongique. Des études récentes ont mis en évidence l'intérêt croissant pour la compréhension de la biologie et de la dynamique génétique des champignons multinucléés. [123]. Grâce à notre système de tri basé sur la technologie DLD, le laboratoire MAP dispose désormais d'un outil puissant pour contribuer à ce domaine.

U-shaped Traps for Single-Spore Isolation

In the previous chapter, we introduced a method for speeding up the genetic purification process of *B. cinerea*, using a microfluidic system based on Deterministic Lateral Displacement. This device isolates mainly spores with one or two nuclei, with a smaller number of spores containing three, and discards spores with 4 or more nuclei (up to 8 nuclei). By focusing on spores with fewer nuclei, the likelihood of obtaining genetically homogeneous material increased. To confirm which colonies contain genetically homogeneous material, PCR analysis needs to be performed after collecting the sample from the small outlet of the DLD device.

As stated in Chapter 1, part of the protocol of genetic purification consists of isolating individual spore colonies before performing the PCR analysis. This step is crucial because if two or more spores grow too close to each other and the colonies overlap, the risk of cross-contamination or mixing of genetic material increases, affecting the reliability of the PCR results. By ensuring that each spore grows separately, the genetic analysis will accurately reflect the genotype of a single, isolated colony.

At MAP, the individual spore colonies isolation can be carried out in two ways, which are described in Chapter 1. The first method (standard method) consists of sufficiently diluting the spore suspension, so that each spore grows into a distinct colony without touching the others. Then, the diluted suspension containing spores are added into a Petri dish filled with nutrient-rich gel (Fig. 4.1 Ia). The Petri dish is kept in the dark at room temperature, and visible colonies begin to appear after approximately three days. At this point, the colonies are transferred manually to a square Petri dish with a 5x5 grid, which also contains nutrient agar (Fig. 4.1 Ib). Each colony occupies a separate well. The spores continue to grow for another two to three days. After this period, the surface of each colony is gently scraped to collect a small amount of filaments, which will be used for PCR analysis.

The second method involves injecting the spore suspension into a Smart Aliquoter from iBioChips (Fig. 4.1 Ib), which helps to isolate individual spores. The suspension is injected into the inlet channel, and the device automatically distributes it into 100

individual wells. On average, the Smart Aliquoter device isolates about 30 single cells across these wells. Because the distribution is random, it follows a Poisson distribution: most wells remain empty, some contain a single cell, and a few receive two or more. Then, the aliquots corresponding to single spores are collected from the wells. As in the previous method, the spores are transferred to a 5x5 Petri dish and cultured individually for about three days. After this period, the surface of each well is scraped, and the filaments collected are subjected to a PCR assay.

The first method is easier to perform but requires more preparation time because the spores are first cultured in a Petri dish for 2 to 3 days and then transferred to another dish. In contrast, the Smart Aliquoter is faster; the suspension is simply injected into the device, and then each single spore is collected from the wells and transferred. When comparing their efficiency, the first method does not guarantee that the spores will not overlap, even if the suspension is diluted. Moreover, the second method presents one major disadvantage: it is expensive, while designed for single use only.

Hence, a microfluidic system based on U-shaped traps is proposed for isolating single spores. This approach offers several advantages, including low cost, spore viability, and direct integration with the DLD output, enabling spore separation and individual isolation in a single step. The following sections describe the materials and methods used, the preliminary results obtained, and outline future perspectives.

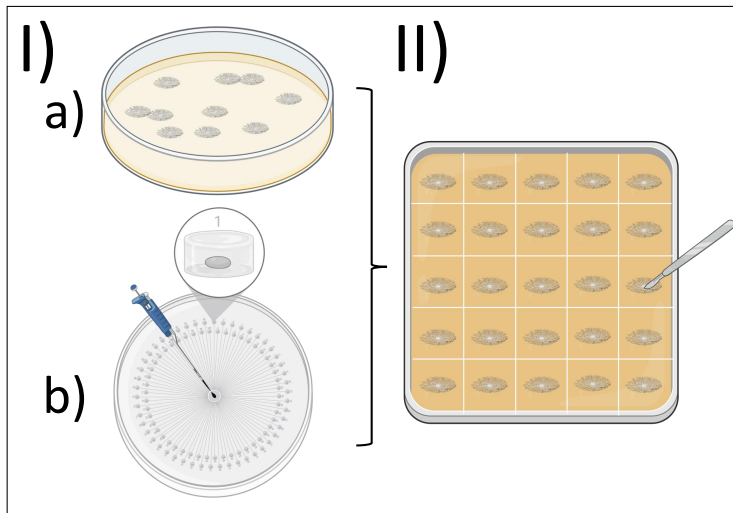


Figure 4.1: Isolation and Culture of Colonies following of a spore suspension. I) single colonies isolation a) Using a nutrient-rich Petri dish b) Using a Smart Aliquoter device II) Colonies culture in a 5x5 grid Petri dish

4.1 Material and methods

In this section, the sample preparation procedures used throughout this chapter are presented, along with the experimental setup.

4.1.1 Spores preparation

The spore suspension used in this study was prepared in the same way as in the previous Chapter (Section 3.1.1.2). First, the transformants were grown for 3-5 days, in the dark at 21°, on solid medium containing the necessary nutrients. Then, to trigger the reproduction, they were exposed for 3-5 days to white blue light. After this period, the spores were collected by gently scraping the gel's surface, resuspending

in water, and stored at 4°C until use, typically within 48 hours. This suspension was used to demonstrate the proof of principle of our approach. Once the system is fully characterized and optimized, the spore suspension expected to be used will be the one coming directly from the small outlet of the DLD 5.

Before use, the spore suspension was passed through a 10 µm filter to simulate the expected diameter distribution at the outlet of DLD 5. Finally, it was diluted in a 1% (v/v) Tween 20 solution in PBS to reach a final concentration of approximately 3.125×10^4 spores/mL.

4.1.2 Gel Handling

The nutrient-rich gel is composed of 2.4% Potato Dextrose, which provides nutrients for fungal cultivation, and 0.7% oxid agar, which is responsible for forming the gel. The oxid agar concentration can be adjusted depending on the desired consistency of the solution. Once prepared, the gel can be stored in a fridge at 4 °C until needed.

Because of the gel's thermal hysteresis, it must be heated to a temperature between 95–100 °C before use to ensure that the oxid agar is fully dissolved and the mixture reaches a liquid state. After dissolution, the preparation is maintained at 50 °C — a maximum temperature compatible with the survival of *B. cinerea* spores — until it is injected into the microsystem. Since agar is transparent, fluorescein was added to improve visualization as it entered the device. The agar then solidifies as it cools down to a temperature between 35–40 °C.

4.1.3 Device design and fabrication

The U-shaped traps consist of two microstructures arranged in a U-shape [124] A diagram of this U-shaped structure is shown in Fig. 4.2 item 4. It has one larger opening (10 µm) and one smaller opening (5 µm). These dimensions are adapted to match the size of the spores expected after DLD 5 sorting. Only one spore can fit into each trap, so the next spore will continue flowing through the device until it finds an available trap. In this initial version, the traps are arranged symmetrically, with a spacing of 30 µm between adjacent traps and 20 µm between rows.

In addition, to facilitate the transfer of spores from the U-shaped traps device to the 5x5 grid Petri dish, it was important that the bonding between the PDMS and the glass slide was reversible to facilitate access to the spores. Therefore, a peripheral channel—Aspiration channel— was added to the system (Fig. 4.2 item 3). By applying negative pressure to this empty channel, its ceiling channel collapsed, causing the system to adhere firmly to the surface. When the pressure was released, the channel returned to its original shape, allowing the PDMS stamp to detach from the glass slide.

The microfluidic devices in PDMS were fabricated using standard soft lithography techniques [117], following the same procedure described in Chapter 4, Section 3.1.4. The only modification was that here the target channel height is 10 µm. To achieve this, the negative photoresist SU-8 2010TM (MicroChem) was spin-coated at 200 rpm for 10 s, followed by 2000 rpm for 30 s.

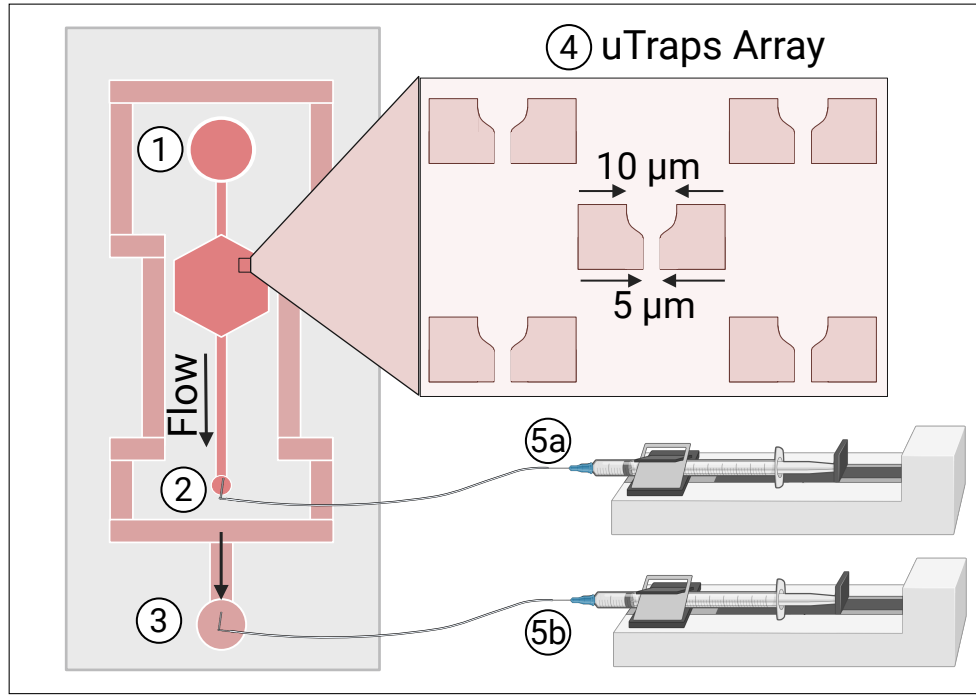


Figure 4.2: U-shaped traps experimental setup. Spores are directly injected into the inlet chamber (1). The outlet (2) is connected to a syringe (5a) that creates negative pressure to aspirate the liquid through the device. The aspiration channel (3) is connected to another syringe (5b), which generates vacuum and seals reversibly the PDMS stamp onto the glass slide [125]. Spores are individually captured in the array of U-shaped traps (4) [124].

4.1.4 Experimental set-up

The experimental setup begins by connecting the empty aspiration channel (Fig. 4.2, item 3) to an empty syringe (Fig. 4.2, item 5b), which is used to create a vacuum that helps adhere the PDMS stamp to the glass slide [125]. Once the device is sealed, a second empty syringe (Fig. 4.2, item 5a) is connected to the outlet of the microfluidic device (0.75 mm diameter) (Fig. 4.2 item 2) to generate negative pressure and draw fluid from the inlet chamber (Fig. 4.2 item 1).

Both syringes (15 mL) are manually pulled to create and maintain negative pressure. To hold them in the extended position, a 3D-printed support was designed and fabricated. Each syringe is fitted with a needle connected to an 80 cm-long medical-grade polyethylene tube (Scientific Commodities Inc.). The opposite ends of the tubes are attached to stainless steel connectors, which are inserted into the PDMS device.

First, a 0.22 μm-filtered **PBS solution** was introduced into the inlet reservoir (1.2 mm diameter) using a pipette. This pretreatment helps prevent particle adhesion to the channel walls and prevent the apparition of air bubbles in the system.

After this step, the **spore suspension** was added to the inlet reservoir. Once the spores entered the microfluidic system and were individually trapped, images were taken to calculate the percentage of successfully occupied traps. Then, the **nutrient-rich gel solution (in liquid state)** was injected into the inlet reservoir. Upon cooling down, the gel solidified, encapsulating the spores and providing nutrients that help their development. After 72 hours, spore development was monitored. This last step is

crucial. Injecting the nutrient-rich gel in its liquid state and allowing it to solidify not only provides nutrients for the spores to grow, but also immobilizes them in the U-traps when the device is opened for subsequent collection.

For the experiments conducted in this Chapter, images were captured under both brightfield and fluorescence modes using an epifluorescence microscope (DMI8, Leica, Germany).

4.2 Evaluation of single-spore capture efficiency in U-shaped traps

First, we evaluated the performance of the U-shaped traps for single spore capture. Three experimental replicates were conducted, and for each replicate, images were taken to manually quantify the number of traps occupied by zero spores, single spores, and multiple spores (Fig. 4.3). In this context, efficiency is defined as the proportion of traps that successfully capture a single spore compared to the total number of traps—i.e., the single-spore occupancy rate. Across the three replicates, efficiency ranged from 34% to 47%. These values are higher than the maximum efficiency previously reported for the Smart Aliquoter device, suggesting an improvement in single spore isolation with our system.

However, the efficiency was still lower than expected. For instance, Deng et al. [126] compared various microfluidic trapping designs for single-cell capture and reported that U-shaped traps achieved the highest efficiency in terms of single-cell isolation. Based on these findings, higher performance was expected in our system. One possible explanation for the reduced efficiency is the overall low trap occupancy rate (considering traps with single spores and those with multiple spores), which may be caused by clogging in the first row of traps (Fig. 4.2, pink circles). Once a clog forms, it blocks fluid flow to downstream traps, resulting in many of the traps remaining empty due to the lack of spore delivery.

Despite this limitation, it is important to highlight that most of the occupied traps successfully captured a single spore. Only 3–6% of the filled traps contained more than one spore, indicating high specificity in single-spore capture. In those cases, the multiple spores were typically attached to each other (Fig. 4.3 green square). Therefore, while overall trap occupancy could be improved, the system performs well in terms of selective single-spore trapping. A potential solution to increase the occupancy rate and prevent early clogging would be to implement a filter at the inlet, similar to the one used in the DLD device.

Nevertheless, this is a point to address in future designs. For now, we aimed to assess whether the trapped spores could grow and develop within the U-shaped traps. This aspect is explored in the following section.

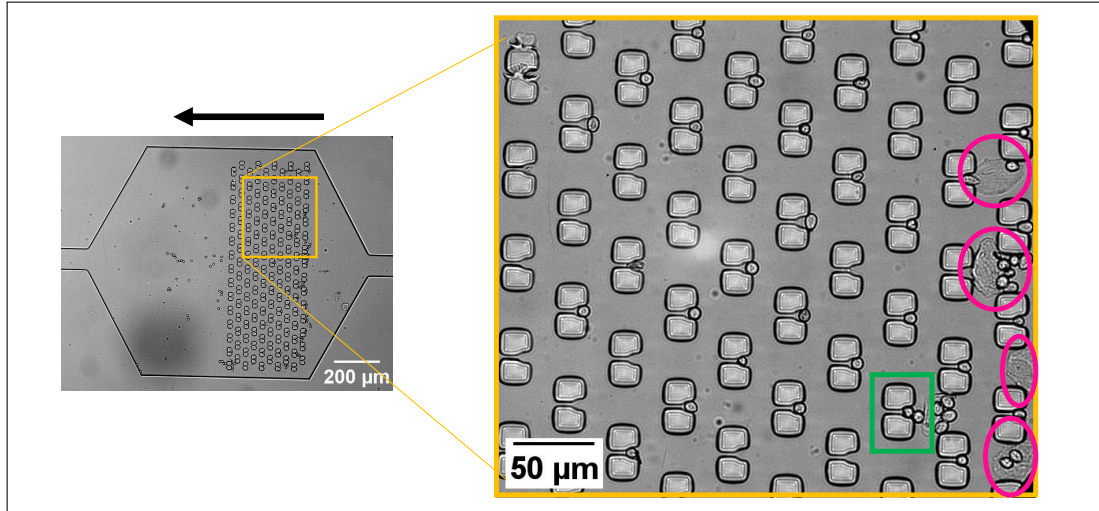


Figure 4.3: Individual spores captured in the U-shaped traps. Pink circles indicate clogging in the first row of traps. The green square highlights traps that contain more than one spore; this typically occurs when spores are attached to each other. Even though only one spore is inside the trap cavity, another spore remains attached just outside, so these cases cannot be considered as true single-spore captures.

4.3 On-Chip Spore Germination

Once it was confirmed that single spores could be successfully trapped in the U-shaped traps, the next step was to investigate whether they could germinate and withstand the temperature required for gel injection (50°C while liquid). Therefore, a germination assay was performed. Images were taken immediately after injecting the nutrient-rich gel (t_0), and then again after approximately 72 hours of incubation (t_{72}).

At t_0 (Fig. 4.4 I), the gel was observed to completely fill the microchannel, thus supplying nutrients to the trapped spores. By t_{72} (Fig. 4.4 II), partial evaporation of the liquid from the chip and the introduction of bubbles into the system had occurred, leading to dry, nutrient-depleted regions within the channel. In these areas, no spore germination was observed. However, in the regions where the gel remained intact, spores showed signs of germination, confirming that the spores were viable and capable of growing within the device in the presence of the nutrient-rich gel.

Compared to standard culture conditions implemented at MAP—where spores are grown on the surface of a nutrient-rich gel in a Petri dish—our system showed a reduced germination rate. Typically, under standard conditions, visible hyphal (filament) growth is observed after approximately 72 hours. In contrast, in our device, the spores exhibited only minimal filament growth (10-15 μm) after the same period.

One possible explanation for this limited growth is the reduced oxygen availability. On the surface of the gel, spores are in direct contact with air, whereas inside the microfluidic traps, even though PDMS is gas-permeable, oxygen diffusion may be limited. Additionally, the gel matrix itself introduces physical resistance, making it more difficult for the hyphae to grow and extend through the solidified medium. Another factor that may explain the limited spore development in our system is the relatively low nutrient concentration in the injected gel. Increasing the nutrient content could

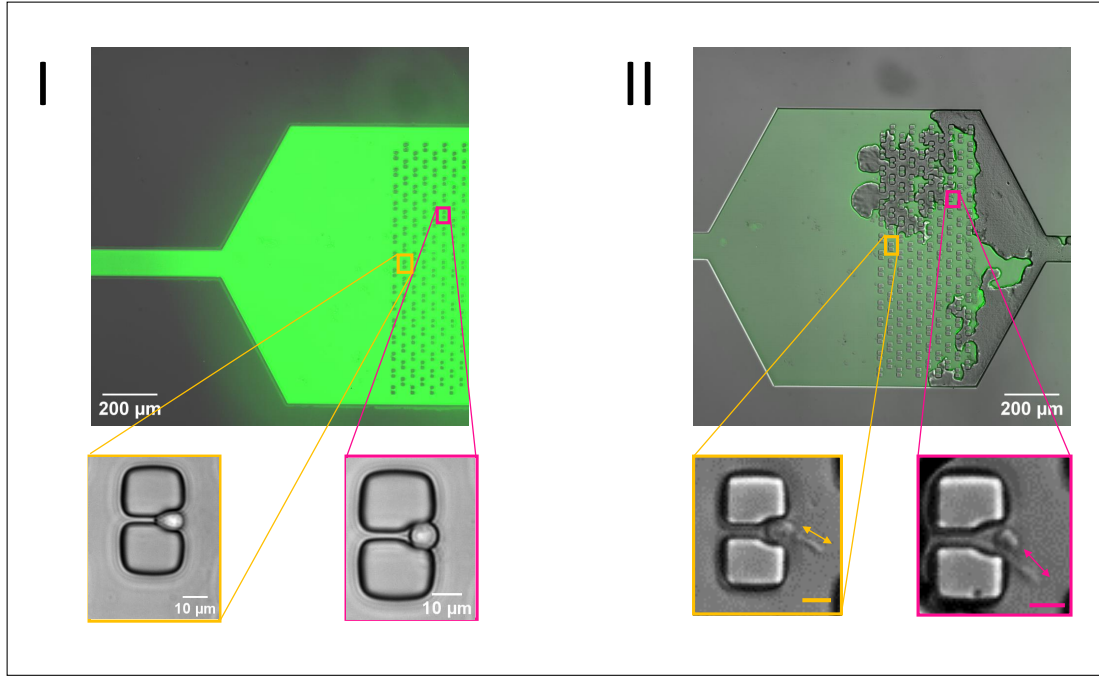


Figure 4.4: Spore culture. (I) Image taken at t_0 , where the gel fills the channel and provides nutrients to the spores. Close-ups are performed on two traps, each capturing a single spore. The bars represent 10 μm . (II) Image taken at t_{72} , showing spore germination in areas where the liquid remained in the gel and did not evaporate. However, the hyphal growth rate was low, as indicated by the pink and yellow arrows.

potentially enhance both germination and hyphal growth.

The potential of culturing spores in microfluidic devices has already been demonstrated in the literature. For instance, Richter et al. [127] developed a microfluidic platform—named the AMF-SporeChip—to study the germination and hyphal behavior of *Rhizophagus irregularis* and *Rhizophagus intraradices* under controlled physical obstacles. In their study, the spores exhibited hyphal elongation of a few millimeters, which is greater than what we observed in our system. However, they also reported a lower germination rate compared to standard Petri dish cultures. It is important to note that their system maintained a liquid culture medium throughout, whereas in our setup, the medium solidifies after injection. As mentioned, the solid gel matrix introduces physical resistance, which may limit hyphal extension and slow down their development.

Another example is the work by Hopke et al. [128], who conducted the "Fungus Olympics," engaging the international fungal research community to compare diverse filamentous fungal species in terms of their ability to grow in microfluidic channels and navigate through microfluidic mazes. Among all the competing species, the fastest-growing were *Trametes versicolor*, *Aspergillus nidulans*, *Rhizopus stolonifer*, and *Neurospora crassa*. Once again, this highlights the potential of microfluidic platforms for culturing and studying fungal spores. Unfortunately, *B. cinerea* was not among the species evaluated in this study, preventing any direct comparison of its hyphal growth rate with the growth rate observed in our system.

As mentioned earlier, partial evaporation of the nutrient-rich gel was observed at t_{72} .

To address this issue, we repeated the germination assay using a humidity chamber to maintain high ambient humidity and minimize gel evaporation. We monitored the evolution of the gel using time-lapse imaging. However, at t_{16} , we found that the gel had still partially evaporated. This could be due to the use of a reversibly bonded system, which may not be hermetically sealed and may allow slow water loss over time. Additionally, PDMS is known to be permeable, so some water vapor diffusion may have occurred, which could have further contributed to gel drying. These aspects require further investigation. Future improvements may include sealing the device edges, inlets, and outlets with resin to enhance moisture retention.

4.4 Conclusion and perspectives

In this chapter, we implemented a U-shaped trap microfluidic device to isolate single spores and culture them. The aim was to isolate the spores, promote their growth, and enable individual transfer to a 5×5 grid Petri dish. After two to three days, DNA analysis via PCR can be performed on these isolated colonies.

Currently, we have two options: the standard process takes 6–7 days to complete before PCR can be done, or alternatively, the Smart Aliquoter device offers faster processing time, but the isolation efficiency is low. Our goal is to speed up the isolation process while improving the efficiency of single-spore isolation at a low cost. Furthermore, the ultimate objective is to integrate the DLD sorting inline with the U-shaped traps (connecting the small outlet of the DLD 5 device to the U-shaped traps device) so that the entire sorting and single-spore isolation process can be performed in one step.

First, we analyzed the efficiency of the system in capturing single spores. Although the overall occupancy rate was low, we observed that among the occupied traps, less than 7% contained more than one spore. In most of these cases, the spores appeared attached to each other. Even though only one spore was actually located inside the trap cavity, another spore remained attached to it, partially outside the trap. To improve the occupancy rate, a possible strategy would be to add a filter at the inlet, similar to the one used in the DLD device. This could help prevent clogging in the first row of traps and break up spore aggregates, ensuring that spores flow individually and enter the traps one by one.

Second, we analyzed spore germination within the microfluidic device. We observed that the spores survived the 50°C temperature at which the gel is injected. Although hyphal elongation showed a low growth rate, strategies such as increasing the nutrient concentration in the gel could potentially improve germination and growth.

Third, evaporation of the gel inside the device was observed. A humidity chamber was added to reduce evaporation; however, it did not fully prevent gel drying. Nevertheless, in the regions where the gel remained, spore germination occurred, confirming the viability in the system.

Once the three issues mentioned earlier are fully addressed, several perspectives open up for further development. For instance, the spacing between traps needs to be carefully studied. It is important to ensure that there is enough room for manual manipu-

lation—when the PDMS stamp is separated from the glass, biologists need sufficient space to use tweezers or punchers to transfer individually the spores from the traps to the 5×5 grid Petri dish. One possible design improvement could involve arranging individual chambers in parallel rows, with each chamber containing a single trap. This configuration would allow for greater spacing between traps, facilitating manual handling. A similar concept has been presented by Robinson et al. [129], who used parallel chambers to isolate individual giant unilamellar vesicles (GUVs).

In summary, in this chapter, it was demonstrated the feasibility of isolating and culturing single spores in a microfluidic U-shaped trap system which will provide a more efficient and scalable workflow. While several technical challenges remain, such as improving trap occupancy, enhancing germination conditions, and minimizing gel evaporation, the results presented here mark a promising first step. Future efforts will focus on refining the device design, optimizing spore handling, and integrating this platform with the DLD 5 device to achieve fully in-line spore sorting and single-spore isolation.

Résumé du Chapitre 4

Dans ce chapitre, nous avons présenté un dispositif microfluidique conçu pour isoler les spores individuelles en vue de leur croissance. Dans un premier temps, le chapitre introduit la pertinence de ce système microfluidique, ainsi que ses principaux objectifs. Cette étape, qui consiste à isoler chaque spore individuellement, est cruciale, car si deux spores ou plus se développent trop près l'une de l'autre, leurs colonies peuvent se chevaucher pendant leur croissance. Il en résulte un risque accru de contamination croisée, ce qui compromet la fiabilité des analyses telles que la PCR. Il est donc essentiel d'isoler chaque spore et de leur permettre de se développer indépendamment. Ce n'est que dans ces conditions que les résultats de la PCR peuvent refléter avec précision le génotype de chaque colonie individuelle.

Le microsystème de piégeage en U, présenté ici, vise à remplacer les méthodes d'isolation actuelles : (1) la culture de suspension hautement diluée, et (2) le système commercial Smart Aliquator. Les principaux avantages et inconvénients de ces deux approches ont été discutés. De plus, le système microfluidique de piégeage en U peut être intégré en série avec la sortie de la puce du DLD 5 dédiée aux petits objets. Ceci permettrait de proposer une solution combinant l'enrichissement en spores fongiques mono-nucléées et isolement des spores individuelles en une seule étape.

Ce chapitre continue en présentant les matériels et méthodes utilisées durant ces développements. Cela inclue la préparation des suspensions de spores et du gel marqué à la fluorescéine pour améliorer sa visualisation. Le design des pièges en U est aussi décrit. Il convient de noter que le dispositif comprend un canal d'aspiration qui permet un collage réversible de la puce microfluidique sur la lame de verre. Le montage expérimental est illustré et décrit en détail.

Les travaux expérimentaux comprenaient l'évaluation de l'efficacité de piégeage des spores individuelles dans les pièges en forme de U, l'observation de leur germination et l'évaluation de la manipulation du gel.

L'efficacité de capture a également été évaluée par la réalisation de 3 réplicats indépendants. Cette efficacité a été quantifiée par la proportion de pièges qui ont capturé des spores individuelles par rapport au nombre total de pièges disponibles. La seconde partie de l'étude s'est focalisée sur la germination des spores. Les spores ont été enregistré depuis leur piégeage (à $t=0$), et leur développement au cours du temps a été enregistré. La croissance des hyphes (filaments) a été observée et comparée aux conditions de culture standard sur des boîtes de Pétri. Enfin, l'injection du gel représentait une étape cruciale. Étant donné que le gel nécessite des températures élevées pour

atteindre son état liquide, il était essentiel d'évaluer si les spores pouvaient survivre à ce processus, ce qu'elles ont fait sans dommage.

La dernière partie du chapitre présentait les conclusions, les perspectives d'avenir et les améliorations possibles pour la puce microfluidique. Bien que plusieurs défis techniques subsistent, tels que l'amélioration de l'occupation des pièges, l'optimisation des conditions de germination et la minimisation de l'évaporation du gel, les résultats présentés ici constituent une première étape prometteuse. Les efforts futurs se concentreront sur le perfectionnement de la conception du dispositif, l'optimisation de la manipulation des spores et l'intégration de cette plateforme au dispositif DLD 5 afin de parvenir à un tri des spores et à un isolement des spores individuelles entièrement en une étape.



Part 2

Biophysical characterization

State of the Art: Biophysical Characterization

Biophysical properties, including the mechanical and electrical characteristics of cells, play a crucial role in regulating biological processes both at the molecular and cellular levels. Additionally, these properties can serve as label-free markers for identifying the physio-pathological state of cells [130].

The fungal cell wall (FCW) is a vital organelle that acts as a protective barrier, especially in terrestrial fungi. It is composed of a tightly regulated viscoelastic shell that counteracts internal hydrostatic pressure, known as turgor pressure, which is characteristic of walled cells. This pressure can range from as low as 0.1–0.4 MPa to values exceeding 2–3 MPa [131, 132].

Most studies in the literature focus on chemical analyses (*e.g.*, solid-state NMR) or electron microscopy imaging of the FCW [133]. For instance, Fontaine et al. [134] investigated the cell wall polymers of *Aspergillus fumigatus* using chemical methods to better understand the biological properties of this fungus. In another study on the same airborne species, Lamon et al. [135] observed that the composition of the FCW changes from the spores in the dormant stage to spores in the germination stage. Additionally, Fernando et al. [136] examined the chitin composition of the FCW in three different fungal strains—*Aspergillus*, *Rhizopus*, and *Candida*—using chemical approaches. They found species-specific chitin fingerprints among these fungi.

As demonstrated by the studies mentioned above, chemical approaches are effective for determining the composition of the FCW and revealing the heterogeneity of its polysaccharides [133]. However, biophysical analyses offer complementary insights into the FCW, providing additional layers of understanding regarding its structure and function. Therefore, in this chapter, we focus on various biophysical characterization techniques that could be applied to *B. cinerea*, beginning with mechanical methods and subsequently addressing electrical approaches.

5.1 Mechanical characterization

Mechanical characterization is essential for understanding cell morphology, structure, and behavior throughout the life cycle [137]. In fungi, the mechanical properties of the FCW play a key role in determining their shape and structural integrity [131]. In the literature [138], mechanical properties are commonly described in terms of a material's deformability or its resistance to deformation under an applied force. For solid materials, this behavior can be described using the stress–strain theory. Stress σ is defined as the force F applied per unit area A :

$$\sigma = \frac{F}{A} \quad (1)$$

Strain ε represents the deformation of a material in response to an applied stress and is defined as:

$$\varepsilon = \frac{L_f - L_0}{L_0} = \frac{\Delta L}{L_0} \quad (2)$$

where L_f is the final length and L_0 is the original length. Accordingly, the stress–strain relationship is described by Hooke's law, in which the slope of the linear region of the stress–strain curve corresponds to the Young's modulus (E):

$$E = \frac{\sigma}{\varepsilon} \quad (3)$$

In contrast, fluids deform continuously under applied stress and cannot store elastic energy. Dynamic viscosity μ quantifies a fluid's resistance to flow and is defined as:

$$\mu = \frac{\tau}{\dot{\gamma}} \quad (4)$$

where τ is the shear stress and $\dot{\gamma}$ is the shear rate [139]. Cells exhibit both elastic and viscous behavior and are therefore classified as viscoelastic. As such, elasticity and viscosity serve as key parameters for comparing mechanical properties in cells [140].

Most measurements of the elastic modulus in fungal spores have been conducted on yeast cells, primarily using Atomic Force Microscopy (AFM) [131]. This section presents various techniques developed to probe and assess mechanical properties, which can be broadly classified into three categories: collective, single-cell, and microfluidic approaches [141]. **Collective techniques** involve experiments on cell suspensions and yield averaged measurements across large populations. **Single-cell techniques**, as the name implies, analyze individual cells, although they typically suffer from low throughput. **Microfluidic techniques** integrate the benefits of both approaches, enabling high-throughput analysis at the single-cell level.

5.1.1 Collective techniques

As previously mentioned, collective techniques provide average measurements across large sample populations. Among these, the **rheometer** is a commonly used instrument to determine the flow and viscoelastic properties of materials. Rheometers allow for precise quantification of mechanical properties under well-defined geometries. The most frequently used type is the rotational rheometer. In this setup, the sample is placed between two plates with a controlled gap. The instrument can operate in either stress-controlled or rate-controlled modes—that is, one can apply a known torque and measure the resulting rotational speed, or conversely, impose a rotational speed and measure the torque required to maintain it [142].

Pollard et al. [143] used a rheometer to investigate the rheological properties of the filamentous fungus *Glarea lozoyensis*, which produces pneumocandins—compounds of pharmaceutical interest. In another study, Beckwith et al. [144] analyzed biofilms of *Candida albicans* using a rheometer. They found that the biofilms exhibited a relaxation time of 810 ± 19 s and a viscosity of 8.4 ± 0.2 MPa/s.

The second collective technique is the **viscometer** [145], an instrument used to determine the viscosity of a liquid. There are different types of viscometers, including:

- **Capillary**: measures the rate of fluid flow under pressure through narrow tubes.
- **Falling ball**: measures the rate at which a solid ball falls through a viscous medium.
- **Rotational**: measures the torque required to rotate a solid object within a viscous fluid.

These collective techniques are considered the gold standard for rheological measurements. However, they present several disadvantages: they require bulky experimental setups, provide only average measurements over entire cell populations, and therefore are unable to detect subpopulations. Moreover, the experiments are technically demanding and require proper training. Additionally, several milliliters of sample are typically needed to perform measurements [146, 147, 148]. Moreover, in literature, most of the works using these techniques are focused on deformable cells such as RCBs, leading to the question of whether they would be suitable for rigid particles such as the spores of *B. cinerea*.

To overcome these limitations and enable the detection of subpopulations, **single-cell techniques** can be employed. These approaches will be presented in the following section.

5.1.2 Single cell techniques

This section presents two well-established single-cell techniques, beginning with AFM, followed by micropipette aspiration.

5.1.2.1 Atomic Force Microscopy (AFM)

AFM is a type of scanning probe microscopy that enables high-resolution and accurate measurement of surface structures at the nanometer scale. It operates by using a flexible cantilever with a sharp tip that interacts with the sample surface [149]. AFM tips typically have a radius of up to 10 nm, while cantilevers generally range from tens to hundreds of micrometers in both length and width.

AFM has been widely used in biology because it allows imaging of biological samples in buffer solutions, at room temperature, and without the need for staining or labeling. The AFM system consists of three main components: piezoelectric transducers, a force transducer, and a feedback control system. The **piezoelectric transducers** precisely move the probe in three dimensions over the sample surface to reveal the topography of the biological object. With the help of the **force transducer**, the interaction force between the tip and the sample surface is measured. Finally, the **feedback control system** transmits the signal from the force transducer back to the piezoelectric system, maintaining a constant force between the tip and the sample [150, 151]. A typical AFM setup is illustrated in Fig. 5.1.

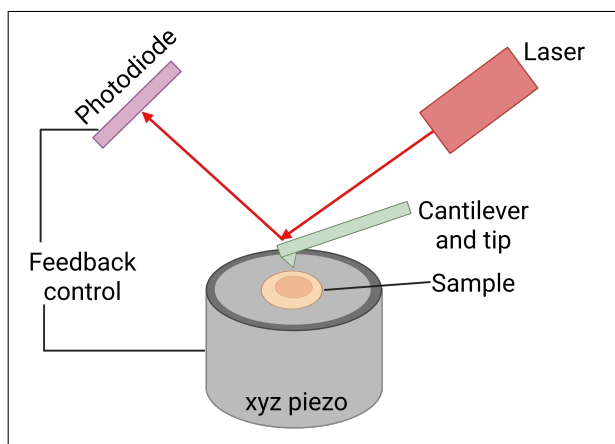


Figure 5.1: Schematic of an atomic force microscope experiment set up. Adapted from Butt et al. [149]

Typically, AFM operates in two main imaging modes: contact mode and dynamic mode. In contact mode, the probe tip remains in constant contact with the sample surface to obtain high-resolution images. However, this continuous contact generates lateral forces that can distort soft biological specimens. Dynamic mode includes both intermittent-contact (tapping) and non-contact modes. These modes significantly reduce lateral forces, thereby minimizing the risk of sample deformation. For this reason, dynamic modes are generally preferred when imaging biological samples [152].

AFM can be used not only to provide topographic imaging but also to measure the mechanical properties of samples. In these experiments, the cantilever functions as an indenter. The tip applies a force to the sample, and the resulting deformation is measured in response to this applied force. The bending of the cantilever (also called deflection), δ_c , is recorded as a function of the piezo actuator height (Z_p) as the tip approaches and retracts from the sample, generating a force–distance curve. This deflection is detected by a laser beam focused on the back of the cantilever and reflected into a photodetector [153]. The tip–sample force (F) can be expressed using Hooke’s law:

$$F = -\kappa_c \delta_c \quad (5)$$

where κ_c is the stiffness of the cantilever and δ_c its deflection. The resulting force–distance curve serves as the basis for analyzing mechanical properties such as the elastic modulus (Young’s modulus), stiffness, and adhesive forces [154].

To extract the elastic modulus, the cantilever deflection δ_c must be converted into the sample indentation δ , defined as the relative displacement of the tip into the surface after accounting for cantilever bending ($\delta = Z - \delta_c$, Z is the total displacement of the piezo). The Hertz model [155] is the most widely used approach for extracting the elastic modulus. This approach treats the AFM tip as a rigid, non-deformable spherical probe in contact with an elastic half-space (the sample). This elastic half-space is assumed to be isotropic and homogeneous, extending infinitely in all directions and in depth, with the top surface as a boundary [156]. And therefore, the Force-indentation curve is given by

$$F = \frac{4\sqrt{R}}{3} \frac{E}{(1 - \nu^2)} \delta^{3/2} \quad (6)$$

Here, R is the radius of the probe, E is the elastic modulus, and ν is Poisson’s ratio.

The main advantages of AFM include its ability to acquire high-resolution 3D topographic images and its versatility in characterizing a wide range of materials. Measurements can be performed in air or in liquid environments (close to physiological conditions), and sample preparation is relatively simple [157]. However, AFM also has some limitations, such as a small scan area, low scanning speed, and the relatively high cost of cantilevers [154, 158].

Due to its numerous advantages, AFM has been widely used to analyze fungal structures. For instance, Dague et al. [159] employed AFM to study the spores of *Aspergillus fumigatus*, probing the structural and physicochemical dynamics of the spore surface in real time by monitoring a single dormant spore throughout its germination process. Similarly, Dufrène et al. [160] examined the spores of *Phanerochaete chrysosporium* and observed structural differences between dormant and germinating spores. They also performed force–distance measurements and found that dormant spores exhibited no adhesion forces, whereas germinating spores showed strong adhesion. Zhao et al. [161] investigated the filaments of *Aspergillus nidulans* and reported that, when grown in complete medium, the filaments had an elastic modulus of 110 ± 10 MPa. This value decreased to 64 ± 4 MPa when the medium was supplemented with KCl, suggesting that osmotic conditions affect cell wall elasticity.

More specifically, AFM has also been applied to the study of *Botrytis cinerea*. El Oirdi et al. [162] investigated the exopolysaccharide (EPS) produced by *B. cinerea* and were able to measure its end-to-end distance, revealing that the EPS is a high molecular weight polysaccharide. Doss et al. [163] also employed AFM to study the adhesion properties of *B. cinerea* during the early stages of culture, when young germ tubes adhere to surfaces. Using AFM, they characterized the fungal sheath responsible for strong adhesion, providing insights into the mechanisms of fungal attachment.

While Atomic Force Microscopy provides valuable insights into surface structure and mechanical properties at the nanoscale, it is not the only single-cell technique used to study the physical behavior of biological samples. Another widely employed method is micropipette aspiration, which will be discussed in the following section.

5.1.2.2 Micropipette Aspiration

Micropipette Aspiration (MA) is a widely used and powerful single-cell technique for assessing cell elasticity. It is considered one of the gold standards for studying cellular mechanical properties [164].

The principle behind micropipette aspiration is as follows: cells in suspension are placed in a Petri dish, and a suction pressure, ΔP , is applied by connecting the micropipette to a pump. The micropipette is then carefully positioned near a target cell. Due to the negative pressure, the cell is immobilized at the tip of the pipette. As the suction pressure increases, the cell is partially drawn into the pipette and deforms. This deformation is quantified by measuring the extent to which the cell enters the pipette and the applied pressure required to achieve this. From these measurements, the Young's modulus of the cell can be determined [165].

The choice of micropipette bore size (inner diameter) is crucial for accurately determining mechanical properties. Similarly, the applied suction pressure plays a key role in controlling how much of the cell is drawn into the micropipette. For instance, when a cell is only slightly aspirated (requiring approximately 1–20 Pa), the mechanical properties of the cell membrane can be measured. In contrast, higher pressures (around 1 kPa) cause a larger portion of the cell to enter the pipette, enabling the assessment of cytoskeletal and cytoplasmic properties [166].

Th  ret et al. [167] proposed the first elastic model in 1988 to determine the Young's modulus of a cell. This model is expressed as:

$$\Delta P = \frac{2\pi}{3} E \frac{L_p}{R_p} \phi \quad (7)$$

where E is the Young's modulus of the cell, and ϕ is a geometric factor defined as the ratio between the pipette wall thickness and the pipette radius R_p ; typically, ϕ is approximately 2.1. L_p denotes the length of the cell projection inside the pipette at the applied pressure ΔP .

Micropipette aspiration offers the advantage of applying precisely controlled high pressures. However, its main limitations are low throughput and the time-consuming nature of the technique, since cells are randomly distributed in the Petri dish and the operator must locate them individually over a large area [168].

This technique has been employed to study fungal spores. For instance, Gonz  lez-Berm  dez et al. [164] investigated *Aspergillus nidulans* and identified a correlation between the pressure applied at the tip of hyphal filaments and their growth rate. They also observed a specific ΔP threshold sufficient to halt filament growth. Additionally, Jiang et al. [94] used micropipette aspiration to examine the plasma membrane

of *Candida glabrata*, analyzing its mechanical properties and the effects of antifungal treatments on membrane behavior.

In summary, the single-cell techniques presented here — atomic force microscopy and micropipette aspiration — are powerful methods that enable precise mechanical characterization of individual cells. However, due to their low throughput, recent research has increasingly turned to microfluidic-based approaches, which will be discussed in the next section.

5.1.3 Microfluidic Techniques

Microfluidic techniques offer an alternative for characterizing the mechanical properties of single cells with high throughput. In this section, microfluidic approaches are categorized into two types: active and passive. **Active** techniques require the integration of external actuation mechanisms to probe the cells. In contrast, **passive** techniques rely on the geometry and fluid flow within the microchannel, without the use of external actuators.

5.1.3.1 Active Techniques

Three active microfluidic techniques are presented in this section: optical stretching, electro-deformation, and compression-based experiments. For each technique, the working principle will be briefly described.

Microfluidic Optical Stretcher

The Microfluidic Optical Stretcher (μ OS) was first introduced by Guck et al. [169], who demonstrated that optical forces could be used to induce cell deformation. In their work, they successfully stretched human erythrocytes. Later, Guck et al. implemented the μ OS to analyze mouse fibroblasts and human breast epithelial cells, ranging from normal to cancerous states, including metastatic stages. Their findings showed that the device is sensitive enough to detect subtle changes in cell mechanics and to characterize the optical deformability of different cell lines [170].

The μ OS operates using two counter-propagating fiber laser beams that trap and deform cells through optically induced forces. The optical fibers must be positioned at equal distances from the microfluidic channel and perfectly aligned with each other. Two main forces act on the cell. The first is the **gradient force**, which pulls the cell toward the region of highest laser intensity—located at the center of the optical field. The second is the **scattering or axial force**, which pushes the cell along the direction of beam propagation. As a result, the cell is trapped at an equilibrium position along the centerline between the two laser beams [171]. An alternative strategy to control the position of particles was presented by Hu et al. [172], by balancing the gradient force and a third force, the **drag force** induced by the fluid.

The stretching procedure is as follows: first, cells are injected into the microchannels. When an individual cell reaches the laser-irradiated region, it is captured by a low optical force. At this point, the flow is stopped to ensure that only one cell is present in the trap. The laser power is then gradually increased in steps to a higher “stretching power,” causing the cell to deform progressively within its elastic range and elongate

along the direction of light propagation. This deformation is recorded, and the mechanical properties of the cell can be determined by analyzing the contour of the cell during the stretching process. After the measurements are completed, the laser power is reduced to its initial low value to maintain the cell trapped and allow it to relax back to its original shape. Finally, the laser beams are switched off, and the flow is resumed to release the cell [173, 174].

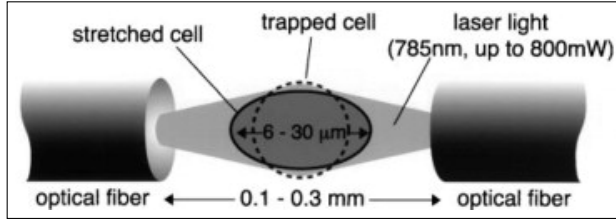


Figure 5.2: Schematic of the microfluidic optical stretcher set-up [169]

The main advantages of this single-cell technique are the wide range of cell types that can be analyzed and its ability to identify subpopulations. It provides quantitative information regarding cell deformation and can also monitor cell relaxation behavior [175]. The main disadvantages are that it does not offer high throughput compared to other microfluidic techniques, and there are heating issues that can damage cells since the laser wavelength must be chosen in the infrared range [110]. Additionally, cells must be spherical so that the force distribution can be precisely calculated [169].

Electro-deformation

The next active technique presented here is Electro-deformation. The idea of this technique is to study how cells respond to electric field forces, cells tend to stretch and contract in different directions, and therefore, the global mechanical characterization of the cell is obtained.

The device operates as follows: a droplet containing a cell suspension is placed onto the microdevice. When a single cell approaches the electrodes, a low-amplitude alternating current (AC) signal is applied to attract the cell toward the region of higher electric field (pDEP, which is explained and whose equations are presented in Chapter 2.2.1). For this to occur, the surrounding medium must be of low conductivity; only under this condition can pDEP be induced. Moreover, using a low-conductivity medium results in a stronger electrodynamic force, a CM factor close to 1 (maximum possible value), and a larger voltage drop across the medium rather than across the cell membrane, thereby decreasing the risk of electrolysis [176].

Subsequently, the voltage is highly increased, which results in a force exerted on the induced dipole strong enough to stretch the cell. This stretching is recorded by video microscopy. The captured images are processed using an image analysis algorithm to quantify cell deformation. From this deformation, the mechanical properties of the cell, such as Young's modulus, can be determined. For example, Chen et al. [177] used electrodeformation to calculate the Young's modulus of SiHa and ME180 cells. A schematic of the electro-deformation setup is shown in Fig. 5.3.

The main disadvantage of electro-deformation is that, being an active technique, it requires external equipment and a complex experimental setup.

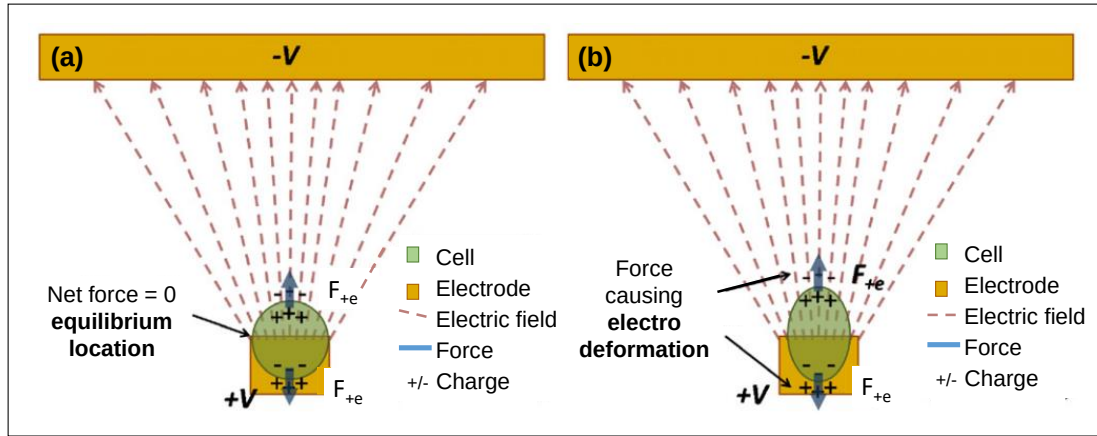


Figure 5.3: Electrodeformation: (a) Cell is at the equilibrium position. The DEP forces are balanced. (b) [177]

Compression experiments

Compression experiments are a technique used to determine the mechanical properties of various types of cells.

The microdevice typically consists of a cell culture chamber equipped with a flexible membrane layer (Fig. 5.4). A loading force is applied to this membrane, which in turn compresses the cells. The cells' mechanical properties can then be studied during the compression or relaxation cycle. This can be done under either static or dynamic conditions. In static conditions, the applied force remains constant throughout the experiment, with only the duration of the application being varied. In contrast, dynamic conditions involve applying a range of forces, often using an automated cycling mechanism [178].

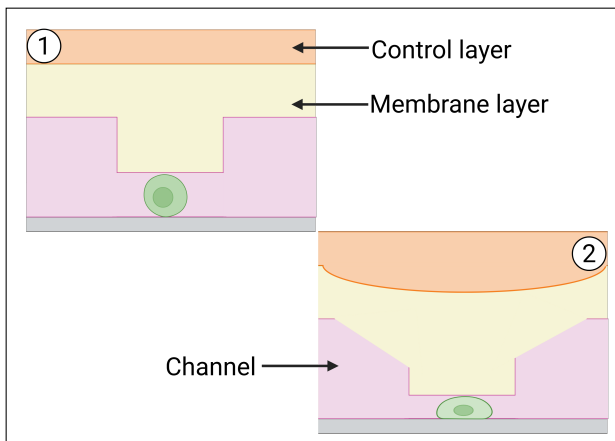


Figure 5.4: Schematic of the compression experiments device. (1) The compression control layer is uncompressed. (2) The compression control layer is compressed. Adapted from [176]

To calculate the Young's modulus of the cells, different numerical models can be employed. For example, Yokokura et al. [179] developed a microdevice that determined the Young's modulus of osteoblasts and demonstrated that their results were consistent with those obtained using AFM on the same cell type.

MEMS squeezers are also employed to compress single cells in aqueous environments, enabling the measurement of cellular mechanical properties both before and after membrane rupture. This approach provides valuable information about cell stiffness.

Barazani et al. [180] used MEMS squeezers to compress yeast cells and observed that the average force required to rupture the cell membrane was $0.47 \pm 0.1 \mu\text{N}$. Before rupture, the cells exhibited an average stiffness of $9.3 \pm 3.1 \text{ N/m}$, which significantly decreased to $0.94 \pm 0.57 \text{ N/m}$ following membrane rupture. The drawback is that it is highly complicated to fabricate and operate.

5.1.3.2 Passive techniques

Passive techniques, in contrast to active ones, do not require external actuators. Instead, they rely primarily on hydrodynamic forces and microchannel geometries to deform cells. In this section, two representative passive techniques are presented: the first and most popular, constriction channels, which include microfluidic micropipette aspiration and tapered aspirators. The second one is the hydrodynamic stretching technique.

Constriction channels

The first passive microfluidic technique presented here is the use of constriction channels. These channels are designed with a cross-sectional area smaller than that of biological cells. As a result, individual cells are driven by a pressure difference to pass through the narrow region, enabling the extraction of various quantitative parameters, such as the **entry time** (the time required for a cell to deform and completely enter the microchannel), **transit velocity** (the average velocity of a cell while passing through the channel), **elongation index** (the ratio of the length of the deformed cell to its initial diameter before entering the constriction) [181], and **pressure drop** (the change in hydrodynamic resistance of the constriction the cell is flowing through).

Leong et al. [182] observed in their experiments that the cell entry process can help distinguish between different cell types. They numerically modeled the entry of a breast cancer cell into a constricted channel and found that cell types could be differentiated based on their entry times. Additionally, they observed that the position of the nucleus depends on its stiffness and viscosity. Zhou et al. [183] extended these findings by investigating both the entry and exit of cells in microchannels, providing a more comprehensive characterization of the cell deformation and recovery process (Fig. 5.5).

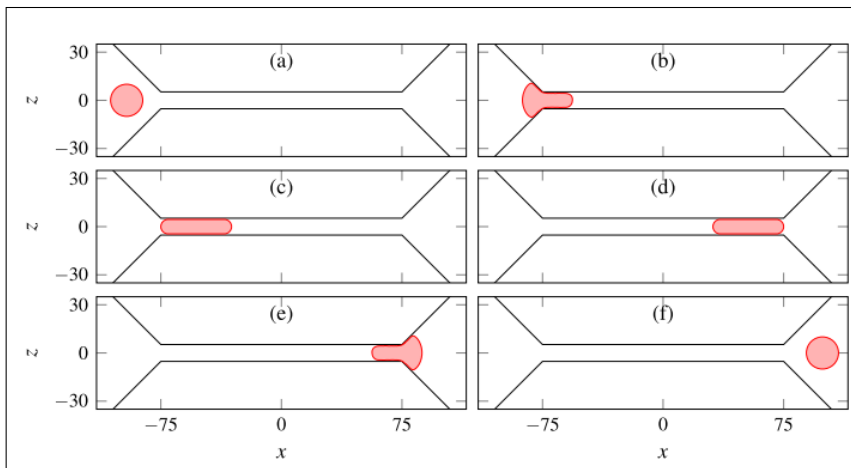


Figure 5.5: Snapshot of a simulation showing a cell crossing the constriction channel at different times: (a) $t = 0 \text{ s}$, (b) $t = 4.45 \text{ s}$, (c) $t = 16.41 \text{ s}$, (d) $t = 23.32 \text{ s}$, (e) $t = 24.92 \text{ s}$, and (f) $t = 29.79 \text{ s}$ [183].

Among the constriction channel approaches previously introduced **Microfluidic Micropipette Aspiration** is a microfluidic technique that operates in the low Reynolds number regime and is functionally equivalent to the traditional micropipette aspiration method described in the single-cell techniques section (see Section 5.1.2.2).

The technique involves deforming individual cells using a microfluidic device that incorporates a cup-shaped constriction designed to trap single cells [184] (Fig. 5.6). A suspension of cells is injected into the microchannel and flows toward the constriction. When a cell reaches the entrance of the constriction, it partially obstructs the flow, resulting in a localized pressure increase. As the applied pressure is incrementally increased, this pressure gradient forces the cell to deform and enter the constriction. The pressure at which deformation through the constriction is observed is recorded, as it allows the extraction of the intrinsic (size-independent) cell deformability [185, 186]. The model used to calculate this intrinsic deformability is the liquid-drop model, which assumes that the cell is a liquid-filled sac with constant volume. In this way, the intrinsic cell deformability is related to the persistent cortical tension in the cell membrane [187].

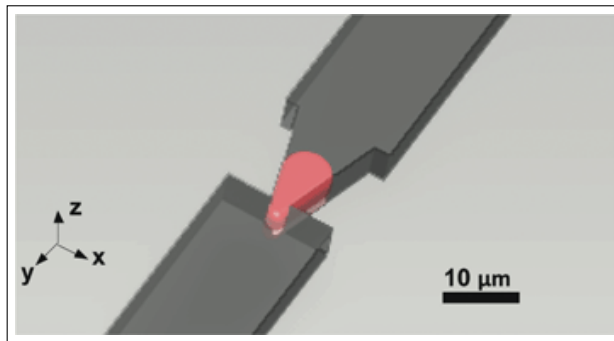


Figure 5.6: Schematic of a single cell being deformed through the MMA. [185]

Li et al. [188] demonstrated that microfluidic micropipette aspiration enables precise control over the pressure applied to individual cells, allowing for accurate measurements of cellular deformability. Moreover, they showed that the mechanical properties obtained with this technique are consistent with those measured using classical micropipette aspiration (MA), thereby validating its reliability. However, the high-throughput capacity of the device needs to be improved. In addition, a significant limitation is its lack of universality: the design of the constriction must be specifically tailored to the size and shape of the target cell type [185].

In addition, another representative constriction channel approach is **Tapered Aspirators**. Tapered channels have been used not only to determine the mechanical properties of particles but also as a size-based separation method [189]. In terms of mechanical characterization, Uddin et al. [190] employed this technique using rigid spherical microbeads. These beads were aspirated and individually trapped within the tapered channels, and their deformation was recorded at each increment of applied pressure. The deformed equilibrium shape of a bead at a given pressure was measured, and by applying a stress balance between the external stresses (e.g., wall pressure) and the internal elastic stresses, it was possible to determine the Young's modulus. Their results were compared with those obtained from mechanical compression. Although the absolute values were not the same, they exhibited the same tendency, for example, regarding the effect of polymer concentration in the microbeads.

This technique is similar to tapered micropipette aspiration, which employs tapered glass capillaries but has low throughput. In contrast, the use of microfluidic channels enables higher throughput by parallelizing tapered aspirators and automating the imaging system, as well as allowing the fabrication of precise geometries to better confine and deform the particles [191, 192].

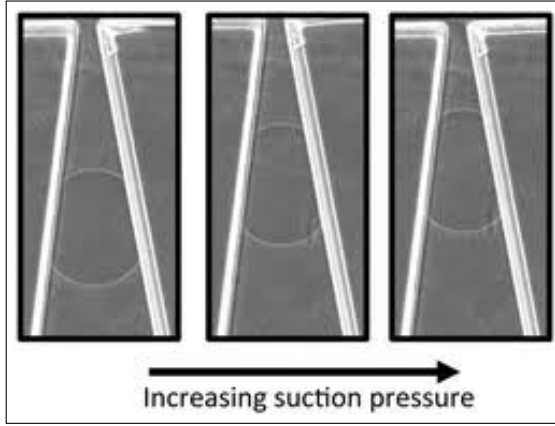


Figure 5.7: Snapshots at different aspiration pressures. The microbeads deform as the aspiration pressure increases (from left to right) [190].

Hydrodynamic Stretching

Hydrodynamic stretching is a microfluidic technique that enables high-throughput measurement of cell deformability and size by leveraging intrinsic fluid-dynamic stresses. This method allows for the tuning of hydrodynamic forces, thereby adjusting the sensitivity of measurements to cells with varying levels of deformability [193].

In this technique, single cells are injected into the microfluidic device at a high flow rate. The microchannel features asymmetric curvatures that focus the cells toward the center of the channel. The stretching region is located at a cross-junction, where perpendicular flows induce elongation of the cells as they pass through (Fig. 5.8). This deformation is captured through high-speed imaging and subsequently analyzed to quantify changes in cell shape, cell volume, and deformation index. After passing through the stretching region, cells continue toward the outlet and rapidly return to their original morphology [194].

Depending on the flow regime, two types of hydrodynamic stress are applied to cells for their deformation. In the inertial flow regime, where the Reynolds number (Re) is $\gg 1$, the compressive force is dominant, whereas in the shear regime, where $Re \ll 1$, the dominant force is the shear force. The inertial regime is particularly suitable for deforming the cytosol and nucleus, while the shear regime is more appropriate for deformation of the cell membrane and cytoskeleton. Thus, the choice of flow regime highly depends on the application [195].

A drawback of this method is that cell deformation might not be solely due to intrinsic properties, but also to asymmetries or non-uniformities in the flow and hydrodynamic pressures. Additionally, whereas in the previously discussed tapered channels, size does not affect the results, in hydrodynamic stretching, both the size and shape of the cell can influence the forces acting to deform it [193].

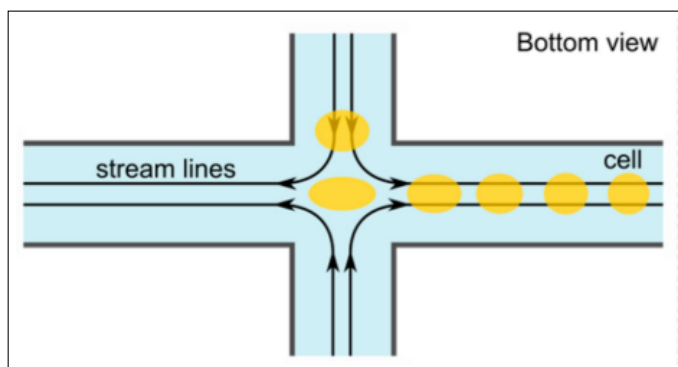


Figure 5.8: Schematic of a hydrodynamic device. [194].

In this section, different microfluidic approaches for mechanical characterization are presented. They offer the advantage of high-throughput single-cell mechanical characterization. The experimental setup is relatively simple, typically consisting of a microscope, a pump, and a high-speed camera, with the addition of an actuator for active techniques. Moreover, they require only small sample volumes [193, 196]. However, some limitations remain: the flow rate is constrained, as higher rates reduce the deformation time, which may make it difficult to capture the real deformation rate with current high-speed cameras. Additionally, elevated shear stresses may compromise cell membrane integrity [197].

It is also worth noting that there are relatively few studies on the mechanical characterization of fungal spores using microfluidic techniques. This may be due to the inherently high stiffness of their cell walls, which could limit their deformation under typical microfluidic conditions. For instance, Couttenier et al. [131] investigated *Candida albicans* spores' filaments using a microfluidic device and reported that their cell wall stiffness lies in the megapascal (MPa) range—considerably higher than that of many mammalian cells—highlighting the challenges of probing such rigid cells with conventional microfluidic setups, which are mostly made of PDMS. Kim et al. [198] determined the Young's modulus of PDMS at a 10:1 ratio (PDMS prepolymer to curing agent ratio) at different curing temperatures and obtained values around 1–3 MPa. This suggests that PDMS can be of the same order of rigidity, or even less rigid, than fungal spores. For example, in constriction channel experiments, PDMS could be more deformable than the spores. Therefore, there is a need to implement other materials to fabricate the microdevice, for example, poly(methyl methacrylate) (PMMA) which has a Young's modulus of 2–3 GPa [199, 200].

The next section provides a comparative summary of the advantages and disadvantages of the cell mechanical characterization techniques presented in this chapter.

5.1.4 Balance Sheet

This section presents an overview that highlights the key strengths and limitations of each method to help guide the selection of the most appropriate technique for characterizing the spores of *Botrytis cinerea* (see Table 5.1).

Given the limited information available regarding the mechanical properties of the fungal cell wall of *B. cinerea*, and considering the objective of comparing different strains (e.g., wild-type and mutant strains), the strategy will be to develop microflu-

idic approaches, as they offer higher throughput, miniaturization of the experimental setup, and improved user- friendliness, making them ideal for the subsequent large-scale screening and comparison of different *B. cinerea* strains. In parallel, AFM will be employed to obtain mechanical characterization of the spores. This will represent the first evaluation of the Young's modulus of *B. cinerea* spores, as there are almost no data in the literature.

Table 5.1: Mechanical characterization techniques comparison

Technique	Advantages	Disadvantages	References
Collective techniques	Gold standard in rheological characterization	Bulky experiments, require proper training, large sample volume	[146, 147, 148]
AFM	High resolution, easy sample preparation, single cell characterization	Limited scanning area	[154, 157, 158]
Micropipette Aspiration	Allow the application of precise controlled high pressure, single cell characterization	Time consuming, low throughput	[168]
Microfluidics	High throughput, single cell characterization, small sample volume	Not implemented with such stiff objects such as fungal spores	[193, 196, 197]

5.2 Electrical characterization

The electrical properties of cells are important and unique markers, as they are directly related to both their morphological and physiological states. Therefore, they can be used to monitor the status of cells.

The cell membrane, a thin phospholipid bilayer with very low conductivity, acts as a dielectric material and thus provides a capacitance (C_{mem}). The cytoplasm and extracellular environment can be considered as highly conductive ionic solutions, which behave as resistive elements characterized by conductivities σ_{cyt} and σ_{ext} , respectively [201, 202]. This electrical information can be used to monitor the physiological state of the cell.

The electrical response of the cell varies depending on the frequency of the applied signal. This frequency-dependent behavior provides insights into cell viability, apoptosis, or other biological processes [203, 204].

When an alternating electrical signal is applied, the cell exhibits impedance. Impedance, denoted as Z or $Z(\omega)$, is a complex quantity representing the electrical resistance and reactance of the cell. It strongly depends on the frequency (ω) of the excitation voltage (EV) [205].

Some commonly used methods to characterize cell electrical properties include electrical impedance spectroscopy, impedance flow cytometry, and electrorotation. In the

following paragraphs, these techniques will be discussed.

5.2.1 Impedance measurements

In this section, two techniques based on impedance measurements will be discussed, starting with Electrical Impedance Spectroscopy (EIS).

Electrical Impedance Spectroscopy is a technique that applies a sinusoidal signal to an electrical system in equilibrium or steady state. In our context, EIS is a technique in which a pair of electrodes is used to apply a voltage difference across the cell in order to characterize it from the point of view of its electrical parameters. The electrical current flowing through the cell can be considered as the probe. The ratio between the applied voltage amplitude and the measured current provides the complex electrical impedance of the cell. As this impedance depends on frequency, an impedance spectrum can be obtained by varying the frequency of the applied signal. An electrical model of the cell can be designed, typically involving resistors and capacitors, while inductors are generally less relevant [206]. The model is fitted to the measured spectrum to estimate its parameters. Models with different levels of refinement have been proposed to account for the various cellular structures, ranging from single- to multi-shell models [207, 208].

EIS has been implemented to study fungi. Favakeh et al. [209] measured single *Schizosaccharomyces pombe* (fission yeast) cells by sweeping the frequency from a few kHz to 6 GHz. They observed that the impedance of the cytoplasm in live yeast cells is clearly different from that in dead cells. This information can be used for viability assays.

Zhu et al. [210] used EIS in the frequency range of 10 kHz to 10 MHz to study budding yeast (*Saccharomyces cerevisiae*). Bud growth was monitored using EIS; although the increase in size is difficult to detect optically, it was clearly observed through impedance measurements. Furthermore, EIS enabled the distinction between bud growth and cell motion in real time.

Slouka et al. [211] applied EIS to monitor biomass production during aerobic and anaerobic batch cultivation of *Saccharomyces cerevisiae*. At frequencies below 1 kHz, the double-layer capacitance can be evaluated, and they observed that this capacitance is proportional to the increase in biomass. In other words, a strong correlation between double-layer capacitance and cell density was found.

EIS has also been used to evaluate the degradation of polymers by fungi, as shown in the study by Gu et al. [212].

Additionally, EIS has been used as a biosensor. D'Aponte et al. [213] employed this technique for the rapid detection of the abnormal presence of *Candida albicans* in urine, a condition known as candiduria.

Zhu et al. [210] studied single rod-shaped *Schizosaccharomyces pombe* using EIS to continuously measure cell growth. They were able to detect cellular and intracellular features with high spatial and temporal resolution.

Impedance flow cytometry (IFC), as explained in Chapter 2, is a technique that measures cell properties in real time while they flow through a microfluidic channel. The

measurements are generally performed at one or more frequencies (without obtaining an impedance spectrum) and can, for example, be used to classify cells based on their impedance.

These two techniques, EIS and IFC are complementary and can be combined within a single microfluidic chip (Fig. 5.9). For example, this was demonstrated by Feng et al. [97], who obtained the impedance magnitude of three different cancer cell types using IFC and the full impedance spectrum of individual cancer cells using EIS.

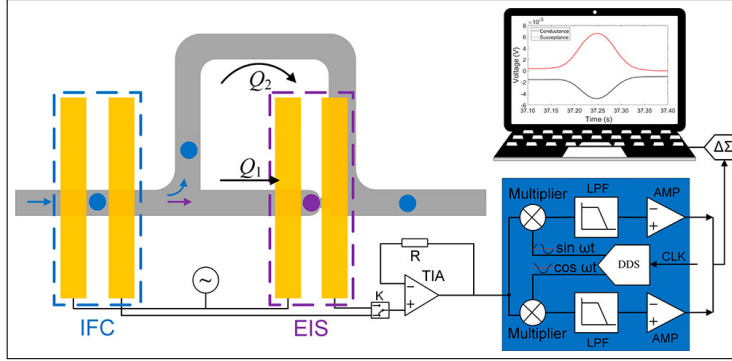


Figure 5.9: Schematic of the impedance measurement set-up implemented by Feng et al. [97]. The cell is passing or being trapped for IFC/EIS impedance measurements.

5.2.2 Electrorotation

Electrorotation refers to the rotation of dielectric bodies in rotating electric fields (Fig. 5.10). In this technique, a rotating electric field is applied to induce the rotation of cells. This occurs because the cell becomes polarized, inducing a dipole moment that, in the presence of the rotating electric field, generates a torque. This phenomenon typically occurs in aqueous cell suspensions within the frequency range of approximately 10^4 to 10^6 Hz. Electrorotation measures the rotation speed of the cells, which is highly dependent on the electric polarizability of both the cell and the surrounding medium, and varies with the frequency of the electric field. Using this method, it is possible to evaluate various cell properties, such as membrane capacitance, membrane conductivity, and internal conductivity of single cells. These properties are extracted by comparing the measured Electrorotation spectra with theoretical models, such as the single-shell or two-shell model [214, 215, 216].

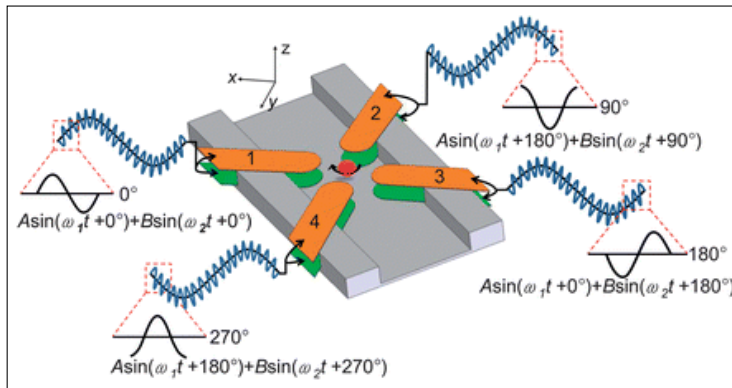


Figure 5.10: Schematic of the ROT-microchip [216].

Electrorotation has been applied to study yeast cells. For example, Hölzel et al. [217] analyzed the electrorotational spectra of yeast cells using a two-shell model, and they

were able to extract several cell parameters, such as the conductivity of the cytoplasm and membranes, as well as the capacitance of the cell membrane ($1.1 \mu\text{F cm}^{-2}$). It was also possible to measure the thickness of the cell wall ($0.11 \mu\text{m}$).

Later, Hölzel [218] used a more advanced model—a spherical four-shell model—to study yeast cells in greater detail. He examined the cell envelope, which includes the plasma membrane, the periplasmic space, and a thick inner and thin outer wall region. From this model, he was able to determine the specific capacitance of the plasma membrane ($0.76 \mu\text{F cm}^{-2}$), the periplasmic space ($0.5 \mu\text{F cm}^{-2}$), and the outer wall region ($0.1 \mu\text{F cm}^{-2}$). He also measured the conductivity of the cytoplasm, plasma membrane, and inner wall region.

Zhou et al. [219] characterized the Electrorotation of yeast cells under different concentrations of the biocide Cosmocil. Using the two-shell ellipsoid model, they obtained the electrical properties of yeast cells before and after treatment. The results showed that the conductivity of the cytoplasmic membrane increased as the concentration of the biocide increased. This quantitative analysis of chemical-induced damage can be applied to the development of new pharmacological agents.

Similarly, Kriegmaier et al. [220] studied rod-shaped fission yeast (*Schizosaccharomyces pombe*) by combining Electrorotation and electro-orientation (it orients anisotropic particles in an electric field). Their approach provided complementary information on how heat treatment affects the electrical properties of cells.

As observed, Electrorotation has been used for yeast characterization. The next section provides a comparison of the three electrical characterization methods presented in this Chapter.

5.2.3 Balance sheet

In this section, we compare the three electrical techniques discussed previously. Table 5.2 summarizes their main advantages and disadvantages.

Table 5.2: Electrical characterization techniques comparison

Technique	Advantages	Disadvantages	References
EIS	Full impedance spectrum; enables detailed electrical characterization	Low throughput; Calibration and high-frequency measurements are non-trivial and require careful implementation	[97, 210, 213]
Impedance cytometry	High throughput	Limited range of working frequencies	[97, 102]
Electrorotation	Provides insight into the internal structure of cells; non-contact method	Complex setup	[217, 218]

Our objective is to analyze the electrical properties of *B. cinerea* spores. Since this work focuses on a comprehensive electrical characterization, EIS was selected as the most suitable technique for this study. As mentioned earlier, EIS enables an extensive and detailed electrical analysis of fungal spores. Because it allows measurements over a wide frequency spectrum, it is possible to extract information about the cell wall, membrane, and internal organelles of *B. cinerea* spores. In particular, we expect to obtain an electrical signature related to the number of nuclei. This makes it possible not only to compare spores from different strains, but also to differentiate spores of the same strain based on their nuclear content. Electrorotation was discarded for this study, as it requires a more complex experimental setup than EIS.

Résumé du Chapitre 5

Dans ce chapitre, nous avons présenté différentes approches pour étudier les propriétés biophysiques des cellules.

Pour *B. cinerea*, la paroi cellulaire fongique (FCW) est un organite essentiel aux fonctions multiples. Elle sert de barrière protectrice et constitue l'interface principale entre la spore et son environnement [131, 132].

Il existe peu d'informations sur les propriétés biophysiques des spores de *B. cinerea*, notamment en ce qui concerne leurs caractéristiques mécaniques et électriques. La plupart des études existantes se concentrent sur des analyses chimiques ou des observations microscopiques [133, 134]. Les approches biophysiques sont particulièrement intéressantes car elles fournissent des informations complémentaires aux méthodes chimiques.

Nous avons d'abord présenté les approches mécaniques. Les propriétés mécaniques de la paroi cellulaire de la spore (FCW) jouent un rôle clé dans la détermination de sa forme et de son intégrité structurale [131]. Dans la littérature, les propriétés mécaniques sont généralement décrites en termes de déformabilité d'un matériau ou de sa résistance à la déformation sous l'effet d'une force appliquée.

Ici, nous avons présenté trois familles de techniques de caractérisation mécanique : les techniques collectives, les techniques à l'échelle de la cellule unique (par opposition aux techniques collectives) et les approches microfluidiques.

Parmi les techniques collectives, on peut citer le rhéomètre, un instrument couramment utilisé pour étudier les propriétés viscoélastiques des matériaux. Par exemple, Beckwith et al. [144] ont analysé des biofilms de *Candida albicans* à l'aide d'un rhéomètre afin de déterminer leur temps de relaxation et leur viscosité. Un autre exemple de technique collective est le viscosimètre [145], utilisé pour mesurer la viscosité des liquides. Il existe principalement trois types de viscosimètres : capillaire, à bille tombante et rotatif.

Le principal inconvénient des techniques collectives est qu'elles nécessitent des équipements encombrants, ne fournissent que des mesures moyennées sur l'ensemble des populations cellulaires et nécessitent généralement plusieurs millilitres d'échantillon [146, 147, 148]. Pour surmonter ces limites, des techniques à l'échelle de la cellule unique ont été développées.

Nous avons présenté deux techniques à l'échelle de la cellule unique : la Microscopie

à Force Atomique (AFM) et l'Aspiration par Micropipette.

La Microscopie à Force Atomique est un type de microscopie à sonde locale qui permet des mesures sur cellules individuelles avec une haute résolution spatiale [149]. L'AFM a été largement utilisée pour analyser les structures fongiques. Par exemple, Zhao et al. [161] ont étudié les filaments de *Aspergillus nidulans* et ont mesuré leur module élastique lorsqu'ils étaient cultivés dans différents milieux de culture.

L'aspiration par micropipette est une autre technique puissante utilisée pour évaluer l'élasticité cellulaire. Cette méthode a également été appliquée aux champignons. Par exemple, González Bermúdez et al. [164] ont étudié *Aspergillus nidulans* et ont identifié une corrélation entre la pression appliquée à l'extrémité des filaments hyphaux et leur vitesse de croissance.

Les deux techniques dites de cellules uniques sont des outils puissants permettant une caractérisation précise des cellules individuelles, bien qu'elles présentent un débit relativement faible.

La dernière approche mécanique présentée ici est l'utilisation des techniques microfluidiques. Ces méthodes peuvent être divisées en deux sous-catégories techniques actives et passives.

Les techniques actives nécessitent l'intégration de mécanismes d'action externe pour sonder la cellule. Parmi les approches actives, on peut citer l'étireur optique microfluidique, introduit par Guck et al. [169], qui ont démontré que des forces optiques pouvaient être utilisées pour induire la déformation cellulaire. Le principal inconvénient de cette technique est l'effet de chauffage généré par le laser, pouvant potentiellement endommager les cellules [110].

La technique active suivante est l'électrodéformation, qui étudie la réponse des cellules à un champ électrique appliqué. Dans de telles conditions, les cellules ont tendance à s'étirer et à se contracter, ce qui permet d'extraire leurs propriétés mécaniques globales.

La dernière méthode active présentée est l'expérience de compression. Cette approche consiste en un microdispositif avec une chambre de culture cellulaire équipée d'une couche de membrane flexible. Une force est appliquée sur la membrane afin de la déformer, comprimant ainsi les cellules, de manière à étudier leurs propriétés mécaniques pendant les cycles de compression et de relaxation [178].

Enfin, les techniques microfluidiques passives ne nécessitent pas d'actionneurs externes. Elles reposent principalement sur les forces hydrodynamiques et la géométrie des microcanaux pour déformer les cellules.

Les canaux de constriction sont conçus avec une section transversale plus petite que celle des cellules biologiques, obligeant les cellules à passer par une section étroite. Cette configuration permet d'extraire divers paramètres quantitatifs tels que le temps d'entrée, la vitesse de transit, l'indice d'allongement et la chute de pression [181]. Parmi les canaux de constriction, deux conceptions courantes sont l'aspiration microfluidique par micropipette et l'aspirateur conique.

L'étirement hydrodynamique est une technique microfluidique passive. Le microcanal présente des courbures asymétriques qui concentrent les cellules vers le centre du canal. La zone d'étirement se situe à une jonction croisée, où des flux perpendiculaires induisent l'allongement des cellules lorsqu'elles la traversent. Cette déformation est capturée à l'aide d'une imagerie à grande vitesse, puis analysée pour quantifier les changements de forme cellulaire, de volume cellulaire et l'indice de déformation [194].

Après avoir analysé les principaux avantages et limites de chaque méthode, la stratégie consistera à développer une approche microfluidique, car elle offre un débit plus élevé et permet la miniaturisation du dispositif. Parallèlement, l'AFM sera utilisée pour obtenir la caractérisation mécanique des spores. Cela représentera la première évaluation du module d'Young des spores de *B. cinerea*, puisqu'il existe presque aucune donnée disponible dans la littérature.

La partie suivante de ce chapitre présente les approches utilisées pour réaliser la caractérisation électrique. Les propriétés électriques des cellules sont des marqueurs importants et uniques, car elles sont directement liées à leurs états morphologiques et physiologiques. Nous présentons deux approches : les mesures d'impédance et l'électrorotation.

La Spectroscopie d'Impédance Électrique (EIS) est une technique dans laquelle une paire d'électrodes est utilisée pour appliquer une différence de tension à travers la cellule afin de la caractériser en termes de paramètres électriques. Comme l'impédance dépend de la fréquence, un spectre d'impédance peut être obtenu en faisant varier la fréquence du signal appliqué. Le spectre obtenu est ensuite ajusté à un modèle approprié pour estimer les paramètres électriques. Des modèles avec différents niveaux de précision ont été proposés pour prendre en compte les diverses structures cellulaires, allant des modèles à simple couche aux modèles multi-couches [207, 208].

La cytométrie de flux par impédance (IFC) permet également de mesurer les propriétés des cellules en temps réel lorsqu'elles traversent un canal microfluidique. Comme les mesures sont effectuées à une ou plusieurs fréquences fixes, cette méthode fournit une valeur d'impédance plutôt que le spectre complet. Les deux techniques sont donc complémentaires.

La deuxième approche électrique présentée est l'électrorotation. L'électrorotation mesure la vitesse de rotation des cellules, qui dépend fortement de la polarisabilité électrique de la cellule et du milieu environnant, et varie en fonction de la fréquence du champ électrique appliqué. Cette méthode permet d'évaluer diverses propriétés cellulaires, telles que la capacité membranaire, la conductivité membranaire et la conductivité interne des cellules individuelles [214, 215, 216].

Pour cette étude, nous avons sélectionné la Spectroscopie d'Impédance Électrique comme technique la plus adaptée, car elle fournit une analyse complète et détaillée des propriétés électriques des spores fongiques. Cette méthode permet d'extraire des informations concernant la paroi cellulaire, la membrane et les organites internes de *B. cinerea*. De plus, l'EIS présente l'avantage d'un montage expérimental moins complexe que l'électrorotation, ce qui facilite sa mise en œuvre. En particulier, elle permet d'étudier la signature électrique associée au nombre de noyaux, qui constitue un aspect clé de notre étude.

Mechanical characterization

The study of the fungal cell wall (FCW) is of great importance, as this dynamic organelle, designed to be both malleable and mechanically robust, is essential for cell viability, morphogenesis, and pathogenesis. It serves as the interface between the fungus and its environment, mediating interactions with other organisms. In plant - pathogen interactions, the fungal cell wall is recognized by plant receptors, triggering local and systemic defense responses, and thus the FCW could represent an important target for fungicides [221].

The fungal cell wall is mainly composed of polysaccharides, which account for approximately 90% of its dry weight. Among these, chitin is the most characteristic polymer of the FCW; however, it represents only a small fraction (1–3%) [222]. The most abundant polysaccharides are glucans, which make up about 50–60%. Glucans can be divided into two types, α - and β -glucans. While β -glucans are abundant and well characterized, α -glucans are less abundant and remain less studied [223, 224].

For this reason, the MAP laboratory is particularly interested in studying α -glucans. Their hypothesis is that these polysaccharides may play a role in the early stages of the infection process. To investigate this, they developed a mutant strain, Δ AGS, specifically designed to study the function of α -glucans. Preliminary results obtained from confocal microscopy experiments showed that this polysaccharide is located on the surface of the spores, outside the cell wall, forming microfibers that could facilitate the attachment process during spore germination.

Even though the FCW plays a crucial role in fungal biology, little information is available regarding its biophysical properties, and even less data exist for this specific mutant strain. This motivated our investigation into the biophysical properties of the fungal cell wall, aiming to provide biophysical insights into how the α -glucan contributes in the FCW properties.

In this chapter, we focus specifically on the mechanical approaches implemented in this thesis: microfluidics and atomic force microscopy (AFM). As a first test, the mi-

crofluidic approach has been tested with “soft” PDMS beads, while different types of measurements were conducted in AFM on actual wild type and mutant *B. cinerea* spores.

6.1 Microfluidics experiments

6.1.1 Sample preparation

“Soft” PDMS was prepared by mixing thoroughly the base with the curing agent at a ratio of 9:0.3 w/w (instead of 9:1 w/w for regular PDMS) and then degassing to remove most of the bubbles. 1 mL of this PDMS mixture is added to 10 mL of water with 10% w/v Sodium dodecyl sulfate (SDS), in a 15 mL flacon tube. The tube is then vortexed at 3000 rpm for 1 minute to form liquid PDMS droplets. Then, the droplets are cross-linked by heating the falcon tube in a water bath at 65°C for 2 hours. The production technique results in the fabrication of highly polydispersed “soft” PDMS beads.

In order to reduce size polydispersity, the bead suspension is filtered a first time through a 20 μm filter, where everything retained in the filter is discarded. It was then filtered again through a 10 μm filter, and this time, it was the particles retained in the filter that were collected and retrieved by rinsing the filter in the reverse direction.

“Soft” PDMS beads with a Young’s modulus of ~ 580 kPa (measured by AFM on equivalent bulk materials) were diluted in phosphate-buffered saline (PBS) containing 1% (v/v) Tween 20 at a concentration of 3×10^3 beads/mL. The suspension was then filtered again through a 10 μm filter to prevent clogging. This low concentration helps trap a single bead in the tapered channel and decreases the likelihood of multiple beads entering simultaneously.

6.1.2 Device design and fabrication

The microfluidic device was designed in Clewin software, inspired by the tapered channels presented by Uddin et al. [190], with dimensions adjusted to trap the spores effectively (Fig. 6.1).

The microfluidic tapered device was fabricated using standard soft lithography as described in Chapter 3.1.4. A negative resist, SU-8 2010 (MicroChem), was spin-coated to a theoretical height of 13 μm (200 rpm for 10 s, then 2000 rpm for 30 s), followed by silanization and curing to produce the PDMS stamp. The inlet was punched with a 4 mm puncher, while the outlet and the vacuum suction channel were punched with a 0.75 mm puncher. The device was then bonded to a glass slide using an O₂ plasma machine (Harrick Plasma®) and stored at 60°C to maintain hydrophilicity until use.

6.1.3 Experimental set-up

To begin the experiment, 10 μL of the PDMS bead suspension was introduced into the inlet reservoir, and the sample was injected into the device by applying negative pressure. To precisely control this pressure and regulate the increments, the outlet

was connected through medical-grade polyethylene tubing (Scientific Commodities Inc.) to a pressure-based flow controller (MFCS series, Fluigent™).

The sample was introduced into the device at an aspiration pressure of 50 mbar, allowing the beads to be individually trapped. The aspiration pressure was then increased in steps of 10 mbar up to 320 mbar. Images and particle behavior were recorded using a high-speed camera (Miro Lab310, Phantom).

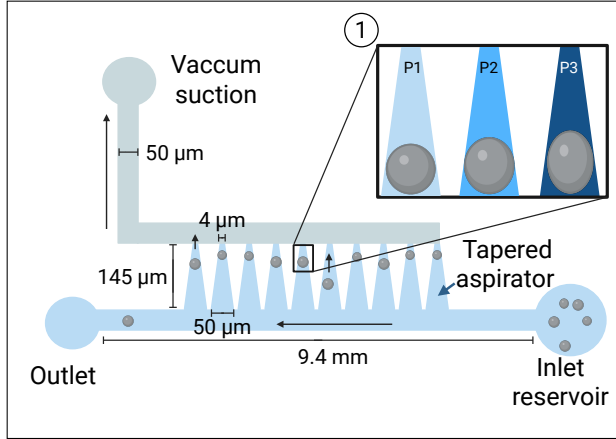


Figure 6.1: Schematic of the microfluidic tapered aspirator setup inspired by Uddin et al. [190]. The device consists of 10 tapered aspirators that trap individual particles. By increasing the aspiration pressure (item 1), the beads are deformed, and this deformation is recorded to calculate the mechanical properties of the spores.

6.1.4 Tapered Aspirators experiments

To demonstrate the proof of concept of the tapered aspirator microdevice, we initially tested it with a PDMS bead suspension, made of “soft PDMS”, rather than spores. These “soft” PDMS beads will serve as model particles, as their Young’s modulus has already been determined using AFM. We will compare these AFM results with measurements obtained using the tapered aspirators to assess the accuracy and reliability of the microfluidic method. Once the experiments and data analysis are validated, we will proceed with measurements on the spores. In this initial exploratory attempt, the pressure required to deform the PDMS beads was unknown. Therefore, we gradually increased the aspiration pressure, starting at 30 mbar, which was chosen arbitrarily. At 50 mbar, the beads were successfully injected and trapped individually in the tapered aspirators (Fig. 6.2a). It was observed that at pressures below 50 mbar, the injection of beads into the system was difficult.

We then continued to increase the aspiration pressure in steps of 10 mbar, up to 120 mbar. The behavior is illustrated in Fig. 6.2 at 90 mbar and 120 mbar; these two pressures were chosen arbitrarily. At both aspiration pressures, the beads did not appear to be deformed. This apparent lack of deformation could be due to two factors. First, the aspiration pressure applied may not have been sufficient to induce deformation. Second, the 40x magnification used may not have provided enough resolution to capture the behavior of individual beads accurately. Unfortunately, our current setup limits the maximum objective to 40x, preventing detailed observation of a single bead.

In the following experiments, we plan to further increase the aspiration pressure to investigate whether any deformation can be observed. Due to the 40x magnification limit of our current setup, we cannot improve the resolution, so our observations are restricted to detecting large changes induced solely by higher aspiration pressures.

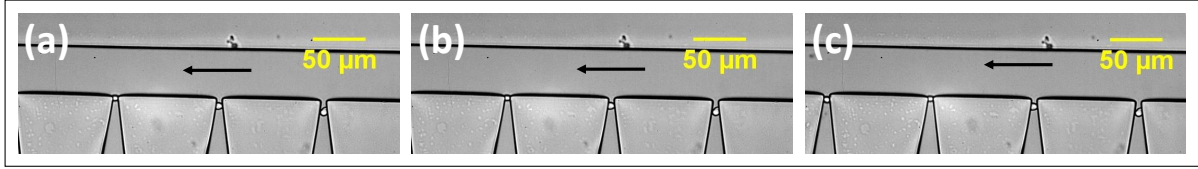


Figure 6.2: Images recorded from the tapered experiments: (a) At an aspiration pressure of 50 mbar, beads are injected and trapped in the narrowest opening of the tapered aspirators. The aspiration pressure is then increased in steps of 10 mbar. (b) At 90 mbar and (c) at 120 mbar, no noticeable differences are observed.

In a second attempt, higher aspiration pressures were applied. At around 200 mbar (Fig. 6.3a), more beads entered the tapered channels and formed clusters. Comparing the deformation of the bead located in the second tapered from left to right, at 200 mbar and 300 mbar (Fig. 6.3b), no significant differences were observed. However, at 320 mbar (Fig. 6.3c), the same bead appeared strongly deformed, with part of it crossing the narrowest section of the tapered channel and extending into the aspiration channel. Finally, at 350 mbar, the maximum pressure applied in these experiments, the tapered PDMS channels broke (Fig. 6.3d).

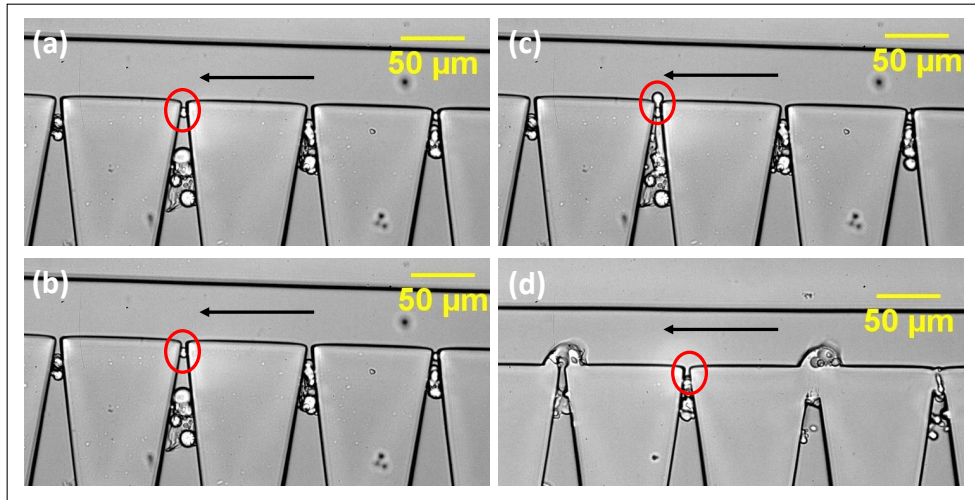


Figure 6.3: Images recorded from the tapered experiments: (a) At 200 mbar and (b) at 300 mbar, a high deformation of the beads is observed in the tapered channels. (c) At 320 mbar, the beads deform and move toward the aspiration channel, and (d) at 350 mbar, the aspiration pressure is so strong that it breaks down the tapered aspirators.

Comparing our results with the experiments performed by Uddin [190], they initially applied an aspiration pressure of 800 Pa to inject alginate beads into the device, which was then increased in steps of 100 Pa up to 1300 Pa (13 mbar) to induce deformation. They tested three polymer concentrations (1 wt%, 2 wt%, and 3 wt%), with the 3 wt% sample exhibiting the highest Young's modulus of approximately $26,532 \pm 4,522$ kPa. These particles are considerably less stiff than the "soft" PDMS beads used in our experiments, and thus, our PDMS beads need higher pressures to be deformed.

Based on our observations, we can conclude that an aspiration pressure of 50 mbar was sufficient to inject PDMS beads individually into the tapered channels, while deformation becomes noticeable at 300–320 mbar (using a 40x magnification). However,

the main limitation is that at these pressures, the channels become filled with multiple beads, preventing accurate deformation measurements. Moreover, these experiments suggest that PDMS may not be the most suitable material for this application, as its Young's modulus— ~ 2 MPa (measured in our laboratory with AFM), which is also in accordance with the literature (1.5–2.5 MPa) [225]—is not stiff enough compared to the "soft" PDMS beads (~ 580 kPa). It is important to remember that the ultimate goal is to characterize the deformation of *B. cinerea* spores. Although there is limited literature on this topic, the Young's modulus of *B. cinerea* spores is expected to be in the MPa range, similar to other cell-walled fungi such as yeast (around 2.9 MPa [226]), which is considerably stiffer than the "soft" PDMS beads and of the same order as the PDMS itself. Consequently, even higher aspiration pressures would likely be required to induce noticeable deformation in the spores.

Although PDMS was used to fabricate the microdevice for the proof-of-concept because of its availability in our laboratory, future experiments aiming for accurate mechanical characterization should employ a material with a much higher stiffness than that of the objects being tested. An alternative is poly(methyl methacrylate) (PMMA), which has been widely used to fabricate microfluidic devices and exhibits a Young's modulus of about 4–5 GPa [199, 200]. Another option is the thermoplastic cyclic olefin copolymer (COC), also suitable for microfluidic fabrication, with a Young's modulus of approximately 2.6–3 GPa [227, 228].

Due to the resolution and material limitations encountered in the microfluidics experiments, we completed our study with AFM measurements to characterize the mechanical properties of the spores and to provide a reference for future microfluidic experiments. These measurements are presented in the following section.

6.2 AFM experiments

6.2.1 Sample preparation

Two different strains of *B. cinerea* were used: the wild type and Δ AGS strain (genetically modified as described in Chapter 3.1.1.2. As stated earlier, in this mutant, α -glucan, one of the less abundant polysaccharides in the fungal cell wall, is not produced anymore. After 10 days of culture, the spores produced by both wild-type and mutant colonies were harvested by gently scraping the gel surface. The spores were then re-suspended in water, filtered through a 40 μ m filter, and stored at 4°C in the fridge. The suspension was used within 24 hours.

For the experiment, the spores were diluted in a nutrient-rich liquid solution to a concentration of up to 2.5×10^6 spores/mL. This concentration was chosen to ensure a sufficient density for AFM measurements while avoiding spore-to-spore contact.

Then, 10 μ L of this suspension was placed onto a glass slide in an area of approximately 2 x 2 cm, which was kept inside a Petri dish to minimize rapid evaporation. After 3 h 30 min, the glass slide was rinsed with deionized water. During this period, the nutrients in the medium promoted the initial adhesion of the spores to the glass surface. Once the nutrient-rich liquid was completely removed (to prevent further filament development), the glass slide was left under the hood for at least 1 hour to

ensure complete drying. This dried slide was then used as the substrate for subsequent AFM experiments.

6.2.2 Experimental set up

The AFM MFP-3D (Asylum Research, Oxford Instruments) was used to perform the measurements presented in this section. Here, we detail the calibration and the measurement protocol that we implemented.

Calibration

The experiment began by calibrating the AFM cantilevers to determine their actual spring constant (k_c), which often differs from the nominal value provided by the manufacturer [229]. First, to perform the calibration, the cantilever was placed in the AFM microscope, engaged with a hard surface (*e.g.* a clean glass slide), and switched to contact mode. A force curve is performed to record the cantilever deflection. The slope of the linear part of the curve gives the inverse optical lever sensitivity (InvOLS) (shown in Appendix B.1). This sensitivity is used in the next step to convert the photodiode voltage signal into an actual deflection in nanometers.

Next, we carried out the Thermal Tune method, which measures the thermal fluctuations of the cantilever when it is not in contact with the surface [230]. These fluctuations appear as a resonance peak in the power spectral density (PSD) of the photodiode signal, expressed in units of V^2/Hz . By integrating the PSD around the resonance peak, the mean-square voltage fluctuation $\langle V^2 \rangle$ is obtained. This value is then converted into a physical deflection by multiplying by the square of the inverse optical lever sensitivity (InvOLS, in nm/V):

$$\langle \Delta Z^2 \rangle = \langle V^2 \rangle \cdot (\text{InvOLS})^2 \quad (1)$$

From this mean-square deflection $\langle \Delta Z^2 \rangle$, the cantilever spring constant k_c is determined using the equipartition theorem. According to this theorem, each quadratic degree of freedom of a system in thermal equilibrium contributes an average energy E given by:

$$E = \frac{1}{2} k_B T, \quad (2)$$

where k_B is the Boltzmann Constant and T is the absolute temperature in Kelvin (K).

The AFM cantilever can be modeled as a spring, with its relevant degree of freedom being the deflection ΔZ , which fluctuates due to thermal motion. Its average elastic energy is therefore:

$$\frac{1}{2} k_c \langle \Delta Z^2 \rangle = \frac{1}{2} k_B T, \quad (3)$$

from which the calibrated cantilever spring constant can be calculated as:

$$k_c = \frac{k_B T}{\langle \Delta Z^2 \rangle}. \quad (4)$$

Measurements

Once the cantilever calibration was complete, it was positioned on the sample surface. Using the microscope, the surface of the spore was located and selected as the target area. The cantilever was then engaged and set to contact mode in order to record the force–distance curves of the different samples tested.

By entering the InvOLS and the calibrated spring constant values obtained during the calibration step, the force could be quantified using Hooke’s law:

$$F = k_c \cdot d = k_c \cdot (v_{\text{defl}} \cdot \text{InvOLS}) \quad (5)$$

where v_{defl} represents the deflection signal of the cantilever in volts, as detected by the photodiode, and d is the actual deflection of the cantilever in nanometers.

Force curves obtained on rigid surfaces, such as glass, show no indentation (see Appendix Fig. B.2). In contrast, those recorded on softer samples display an indentation region.

Therefore, the next step is calculating the indentation (δ). The total displacement of the piezo is related to the indentation and the deflection of the cantilever (d) as follows:

$$\delta = Z - Z_0 - d \quad (6)$$

where Z is the displacement of the piezzo, and Z_0 is the contact point between the cantilever and the sample. From this information, we obtained the force–indentation curves that were to be used for the analysis.

Post-processing includes calculating the Young’s modulus of the sample, from which the Young’s modulus of the spores was determined using the Hertz–Sneddon model:

$$F = \frac{3}{4} \frac{E}{(1 - \nu^2)} \tan(\alpha) \delta^2 \quad (7)$$

where F is the applied force, E is the Young’s modulus of the sample, ν is the Poisson’s ratio, α is the half-angle of the pyramidal tip, and δ is the indentation.

Alternatively, the indentation of the sample for a given force can be calculated and used to compare the mechanical response of different samples.

6.2.3 Force-indentation curves on *B. cinerea* spores

The first set of experiments was performed on a **wild-type** strain of *B. cinerea*. Fifty spores were tested using a pyramidal AFM tip (NanoWord®) with a calibrated spring constant $k_c = 0.14 \text{ N/m}$.

When analyzing the force-indentation curves obtained from these experiments, we observed that the curves show some "accidents", such as small bumps (Fig.6.4a). These accidents could be due to the AFM tip slipping over the spore surface rather than properly indenting it. One possible explanation is that the spring constant of the cantilever was not sufficiently stiff to indent the spore, resulting instead in the tip slipping along the surface. Therefore, the following experiments were conducted using a stiffer AFM cantilever to improve indentation accuracy and obtain more reliable data.

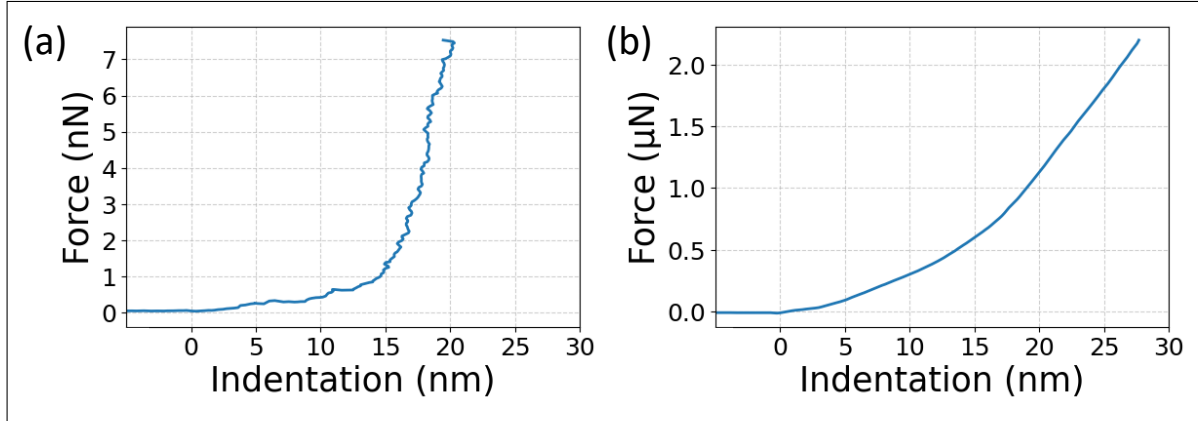


Figure 6.4: Force-indentation curves: (a) Using a soft cantilever, we observed irregularities/accidents in the curve; (b) Using a stiff cantilever, no accidents are observed in the curve.

In the second set of experiments, new force-indentation curves were acquired in air, using a stiffer AFM cantilever ($k_c = 52 \text{ N/m}$), to ensure proper contact between the tip and the spores' surface (fig. 6.4b). This time, both the **wild-type** and the ΔAGS strains were tested.

The spores are quite small objects (5-12 μm diameter), and it is not easy to be sure that we are actually probing the spores' surface. Therefore, we performed fast topographies of the spore surface (40 $\mu\text{m} \times 40\mu\text{m}$) in Tapping mode to locate the top of the spore. This approach allowed us to reduce the scanning area to approximately 5 $\mu\text{m} \times 5 \mu\text{m}$, focusing only on the top of the spore. By doing so, we ensured that measurements were performed directly on the spore surface and that the probing sites could be precisely selected before switching to contact mode. Within this area of 5 $\times$ 5 μm , about seven indentation measurements were performed at randomly chosen points on the same spore. A total of five spores per strain were tested, yielding approximately 35 force-indentation curves per strain.

The results are presented in Fig. 6.5. It can be seen that the Young's modulus of both spore strains lies within several hundreds of MPa. Although statistical analysis (explained in detail in Chapter 3.1.3.2) does not indicate a significant difference between the two strains, the distribution of Young's modulus values for the wild-type strain (ranging from approximately 100 to 1500 MPa) is broader than that of the ΔAGS strain (ranging from about 100 to 800 MPa). We hypothesize that the small area tested in the wild-type spores may present greater heterogeneity in mechanical properties compared to the small area tested in the ΔAGS strain. Moreover, the ΔAGS strain exhibits generally lower Young's modulus values than the wild type, suggesting that ΔAGS spores could be mechanically softer.

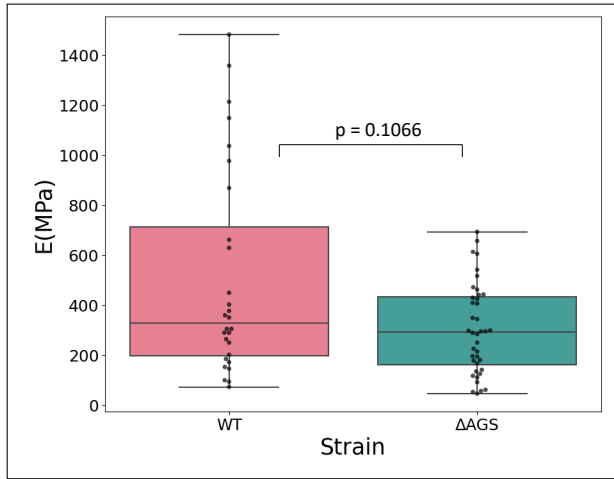


Figure 6.5: Boxplot showing the distribution of Young's modulus measured by AFM for the WT and Δ AGS strains.

At this point, it seemed interesting to further investigate the local surface properties of the spores. This will be addressed in the next section using a different approach.

6.2.4 NAP mode experiments

The next approach tested was the NAP mode experiments. This method consists of a two-pass technique [231]. During the first pass (Fig. 6.6, red dotted trajectory), the AFM probe scans the sample surface in standard tapping mode. The cantilever follows the sample surface, providing the topographical profile of the sample.

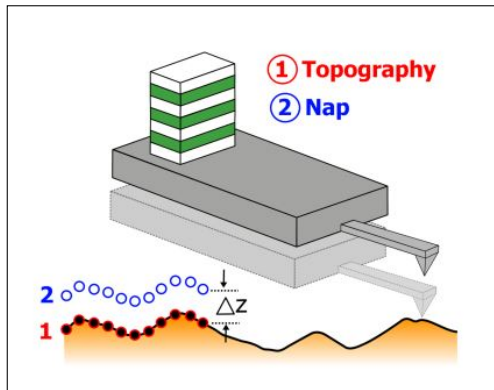


Figure 6.6: Working principle of NAP mode. First, the AFM probes the sample surface, providing the topographical profile. Second, the probe is raised by a certain delta height (Δz) and scans the sample surface again using the topographical information obtained previously [231].

This information on the topographic profile of both strains is shown in Fig. 6.7. The height profiles differ between the two strains, with the wild type (Fig. 6.7a) showing greater variations in height compared to the Δ AGS strain (Fig. 6.7b). Indeed, the grey level contrast is higher on the wild-type topography than on the mutant one. Moreover, it seems like the features are smaller in the wild type. Nevertheless, the topography should be repeated with a smaller scanning velocity to increase the resolution to confirm it. Indeed, the optimum parameters were not used here as the main purpose of the experiment was to obtain information on the mechanics of the spores.

In the second pass, called the "NAP" pass (Fig. 6.6, blue dotted trajectory), the AFM probe is lifted to a selected delta height (Δz) and retraces the sample surface based on the topography data from the first scan. During this pass, the height feedback is turned off, and the probe moves at a constant height above the surface. At the

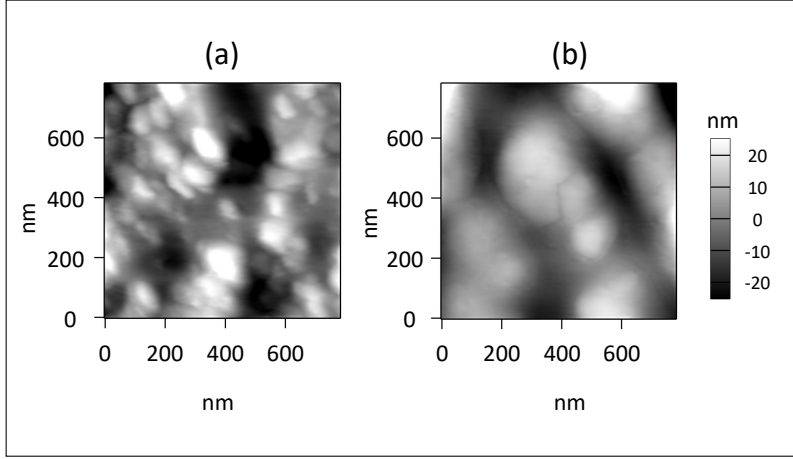


Figure 6.7: Topographical profiles of the strains obtained during the first pass in NAP mode: (a) wild-type and (b) Δ AGS strain.

same time, the drive amplitude is increased to maintain intermittent contact with the surface.

There is no strict rule for selecting the height, but it is typically in the range of 30–100 nm. The optimal value is the smallest height that effectively removes topographical noise while preserving a good signal-to-noise ratio. In our experiments, we chose a Δz of 50 nm.

During the second pass, any changes in the cantilever oscillation caused by dissipative forces are recorded. These variations are detected as phase shifts, expressed in degrees.

The phase shift reflects modifications in the energy dissipated during each oscillation cycle due to cantilever–sample interactions, and the resulting dissipation map provides a qualitative representation of the mechanical variations of the sample. In this dissipation map, regions of different colour (*e.g.*, red and blue) correspond to areas of higher or lower energy dissipation, respectively, reflecting local differences in the sample’s stiffness or adhesion properties.

We analyzed an area of approximately 800×800 nm on each spore, examining two spores per strain (**wild type** and Δ AGS) using a Tapping Mode AFM probe (Arrow-NCR, NanoWorld) with a calibrated spring constant of $k_c = 36$ N/m. The Δz was set to 50 nm. The results are presented in Fig. 6.8.

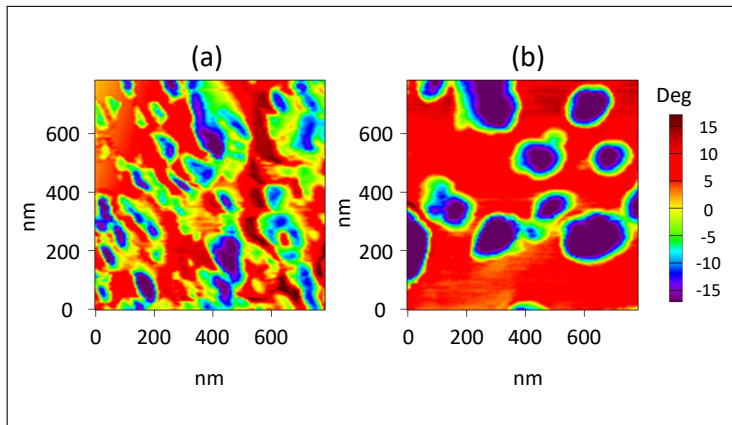


Figure 6.8: NAP mapping of the surface of (a) the wild-type and (b) Δ AGS strain ($\Delta z = 50$ nm).

Comparison of dissipation maps reveals differences in the distribution of energy dis-

sipation across the spore surface on the local scale ($800\text{ nm} \times 800\text{ nm}$). Generally, the wild-type strain (Fig. 6.8 (a)) exhibits a more heterogeneous pattern, consisting mostly of intermediate dissipation zones (green) and low dissipation zones (red), which correspond to stiffer or less adhesive regions, along with a few high dissipation zones (blue/purple) representing softer or more adhesive areas. In contrast, the Δ AGS strain (Fig. 6.8 (b)) displays more homogeneous regions with better-defined zones of low and high dissipation.

Comparing these results with those obtained previously (Fig. 6.5), we can suggest that the broad distribution of Young's modulus values in the wild-type spores could be attributed to the high heterogeneity of their surface present at the local scale. This heterogeneity likely explains the wider range of measured values. In contrast, the Δ AGS strain exhibits a more homogeneous surface at the local scale, which could explain the narrower distribution of Young's modulus values in this strain.

These results are encouraging; however, it would be valuable to investigate the spore at a global scale, as the local measurements presented here may not fully represent the properties of the entire spore.

Global measurements provide a complementary approach to characterizing the spore. For example, Chang et al. [232] employed a flat-tip cantilever to measure the Young's modulus of whole yeast cells by compressing them and generating a force–deformation curve. This approach proved to be effective to obtain reliable measurements of the Young's modulus at the whole-cell level.

6.2.5 Force-deformation curves on WT and Δ AGS spores using an AFM tipless cantilever

These experiments aimed to measure spores on a global scale rather than on a local one, in order to obtain a more complete characterization. The setup was the same as in the previous experiments, with the main difference being the use of a tipless cantilever (Fig. 6.9a) to apply a force on the spores and record their force–deformation curves (Fig. 6.9b). The deformation at a given applied force was measured from the contact point (red dot) to the corresponding point on the curve (green arrow), and these values were compared for each strain, allowing for a direct comparison between wild type and Δ AGS. The model used in the previous experiments is not applicable here due to the different contact geometry without a tip; however, this approach, while not providing a quantitative measurement of spore stiffness, allows for a comparative assessment of their mechanical response.

The experiments were carried out using a tipless AFM probe (AIOAI-TL, BudgetSensors) with a spring constant of $k_c = 46\text{ N/m}$. Deformations were recorded at applied forces of $3\text{ }\mu\text{N}$ (Fig. 6.10b) and $6\text{ }\mu\text{N}$ (Fig. 6.10c), with approximately 100 curves acquired for each strain.

The results showed statistically significant differences in deformation between the wild-type and Δ AGS mutant strain. Both strains displayed a broad range of deformation; however, the Δ AGS mutant exhibited higher deformation values than the wild type. This suggests that the Δ AGS mutant strain is softer than the wild type.

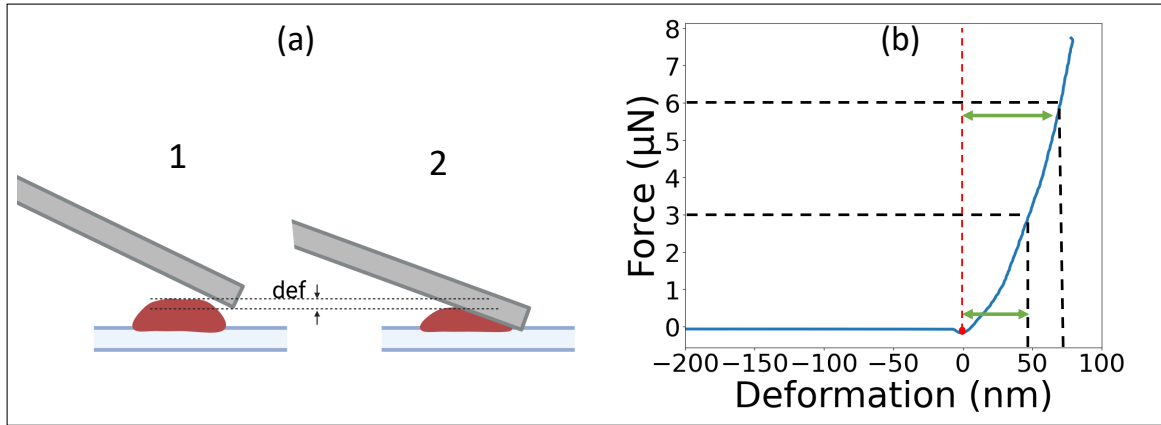


Figure 6.9: Force–deformation curves experiments. (a) Deformation of spores using a tipless cantilever: (1) spore before applying force, showing no deformation; (2) spore after the cantilever applied force, with the resulting deformation recorded. (b) Force–deformation curve: the deformation at a given applied force (e.g., 3 μN) is measured from the contact point (red dot) to the corresponding point on the curve, as indicated by the green arrow.

These initial results provide valuable insight into the global mechanical properties of wild-type and ΔAGS spores, and the approach appears promising due to its high throughput (compared to the local measurement approaches tested here).

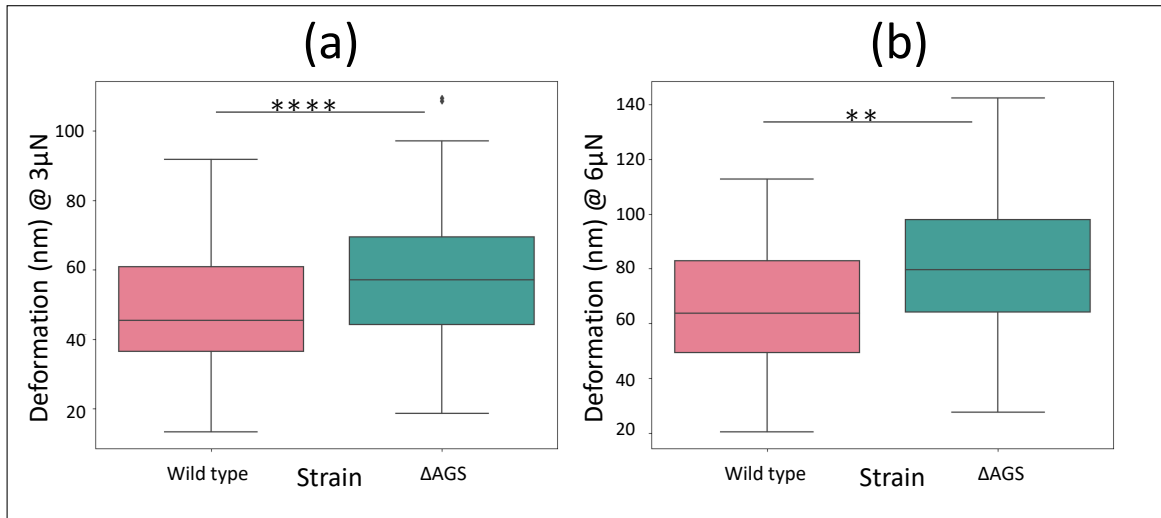


Figure 6.10: Force–deformation curves of wild-type and ΔAGS strains obtained using a tipless probe with $k_c = 46 \text{ N/m}$. (a) Deformation under an applied force of 3 μN ; (b) deformation under an applied force of 6 μN .

6.3 Conclusion

In this chapter, we conducted two main types of experiments: AFM experiments and microfluidics experiments.

Regarding the microfluidic experiments, we implemented tapered aspirators. We observed that at low aspiration pressures, the "soft" PDMS beads were not deformed, requiring higher pressures to achieve deformation. However, the PDMS device was not compatible with aspiration pressures above approximately 320 mbar, as it broke

down under these conditions. Spores of *B. cinerea*, as observed in this chapter by AFM, are expected to have a stiffness on the order of MPa, considerably higher than the "soft" PDMS beads tested. Therefore, even higher aspiration pressures would be required to deform them in the microfluidic device.

Since the rigidity of PDMS is similar to or even lower than that of the spores, it is not a suitable material to develop a microfluidic chip for deforming objects with stiffness in the range of MPa. This highlights the importance of using alternative materials to fabricate microfluidic devices, such as PMMA [199, 200], or COC [227, 228]. However, this device can still be implemented in our laboratory for cells with a lower Young's modulus.

Regarding the AFM experiments, we tested two different strains: the wild type and Δ AGS. These experiments can be divided into local and global approaches.

The local experiments included force-indentation measurements with a pyramidal tip using different spring constants, as well as NAP mode experiments. These experiments were useful to observe the spore surface at the local scale, showing its heterogeneous nature. With this type of experiment, we were able to obtain values of the Young's modulus. Nevertheless, they are highly localized, do not provide a full characterization of the spores, and are extremely time-consuming to perform. As a result, the number of measurements was limited, reducing the statistical robustness of the analysis.

On the other hand, global experiments offer the advantage of being considerably faster, and therefore, the throughput is higher than that of local experiments. However, they were performed with a tipless probe, and the resulting contact geometry is not compatible with our current models, making it difficult to extract reliable Young's modulus values. Nevertheless, this approach allowed for a comparative assessment of deformation under applied force between the two strains, revealing a statistically significant difference. To our knowledge, this is the first time that the mechanical properties of the Δ AGS mutant strain have been assessed. These results could provide new insights into the role of α -glucan in the cell wall and its influence on the interaction of the fungal cell wall of the spore with a plant.

Résumé du Chapitre 6

Dans ce chapitre, nous avons analysé les propriétés mécaniques de la paroi cellulaire de spores de *B. cinerea*, en comparant la souche sauvage et une souche mutante Δ AGS. Puisque très peu d'informations sont disponibles sur la déformabilité de ces spores, nous avons exploré deux approches complémentaires: des mesures en microfluidiques et des mesures en AFM.

Tout d'abord, nous avons réalisé des expériences microfluidiques à l'aide d'aspirateurs coniques issus de la littérature [190]. Afin de caractériser le dispositif, nous l'avons fabriqué en PDMS, dont le module d'Young est d'environ 2 MPa. Les premiers tests ont été réalisés avec des billes de PDMS "mou" (Module d'Young estimé à ~ 580 kPa). Ils ont montré que nous avons réussi à piéger une seule bille de PDMS à l'intérieur de chaque aspirateur conique, une étape essentielle pour l'analyse mécanique subse- quente. Cependant, avec notre configuration actuelle, il était difficile d'observer la déformation de la bille sans endommager le dispositif. Malgré cette limitation, ces résultats préliminaires constituent une première tentative précieuse. Les développe- ments futurs se concentreront sur la mise en œuvre de matériaux plus rigides que le PDMS pour fabriquer le dispositif microfluidique (PMMA ou COC), permettant ainsi la déformation contrôlée des billes et, à terme, des spores qui constituent le sujet prin- cipal de cette étude.

Deuxièmement, nous avons réalisé des mesures AFM à l'aide de trois approches dif- férentes. Nous avons d'abord enregistré des courbes de force-indentation afin de com- parer la rigidité des deux souches, en utilisant un cantilever souple ($k_c = 0,14$ N/m) puis un cantilever plus rigide ($k_c = 52$ N/m). Les résultats ont indiqué que le can- tilever souple n'était pas adapté à l'analyse des spores. En effet, les courbes de force- indentation présentaient de multiples « accidents ». Ces accidents suggèrent que le cantilever glissait sur la surface des spores plutôt que d'effectuer une indentation cor- recte. Le cantilever plus rigide s'est avéré plus approprié, nous permettant d'estimer un module d'Young des spores de *B. cinerea* de l'ordre de quelques centaines de MPa. Les spores de type sauvage présentaient une distribution plus large et des valeurs plus élevées du module d'Young par rapport à la souche Δ AGS. Ces mesures locales suggèrent qu'à l'échelle nanométrique, la surface des spores de type sauvage présente une plus grande hétérogénéité mécanique que celle de la souche Δ AGS.

Afin d'étudier plus en détail ces observations à l'échelle locale, nous avons effectué des mesures AMF en mode NAP. Ce mode donne accès à deux types d'informations: la topographie des spores et l'adhésivité et/ou de la déformabilité des spores. Ce dernier observable est mesuré *via* l'énergie dissipée à cause d'interactions à longue

portée. Les résultats topographiques ont montré que la surface des spores de la souche sauvage présentait de petites structures dont la différence d'altitude était supérieure à ~ 40 nm. Par opposition, les structures à la surface de la souche sauvage semble être plus larges avec une différence d'altitude moindre. Les cartes de dissipation obtenues ont montré qu'à l'échelle nanométrique, les spores de type sauvage présentaient une plus grande hétérogénéité de surface que la souche Δ AGS. Cependant, la technique NAP ne permet pas de discriminer de manière sûre l'adhésivité de la surface de la déformabilité de celle-ci.

Cette hétérogénéité locale pourrait ne pas refléter entièrement le comportement mécanique des spores à l'échelle globale. Nous avons donc réalisé des expériences à l'échelle de la spore à l'aide d'une sonde AFM sans pointe. À cette échelle, les spores Δ AGS avaient tendance à se déformer davantage que les spores de type sauvage, pour une même force appliquée. Cela suggère que la souche mutante était plus déformable que la souche sauvage; cette différence étant statistiquement significative.

Nous avons ici réalisé les premières étapes préliminaires pour étudier les propriétés mécaniques des spores de *B. cinerea*. Les mesures AFM ont fourni des informations précieuses sur les différences mécaniques entre les deux souches. La prochaine étape consistera à adapter le système microfluidique pour réaliser des expériences, ce qui augmentera le débit. Bien que des travaux supplémentaires soient encore nécessaires, ces premiers résultats sont prometteurs.

Electrical characterization

In this chapter, we present the setup developed to measure the electrical properties of fungal spores. As discussed in Chapter 5, we selected Electrical Impedance Spectroscopy (EIS) to characterize the electrical properties of *B. cinerea* spores. Having the impedance spectrum, models with different levels of refinement (from single- to multi-shell) can be used to extract the electrical properties corresponding to the cellular structures. The FCW can be probed at low frequencies (tens of kHz), whereas estimating nuclear content requires high-frequency measurements (>10 MHz) to overcome the capacitive effect of the FCW. Therefore, EIS is well-suited to expect obtaining information about the FCW as well as a signature related to the nuclear content.

7.1 Microfluidic system for EIS architecture

Since the measurements are performed at the single-cell scale, a microfluidic EIS system must be developed [233]. This microfluidic EIS system consists of two main layers, as shown in Fig. 7.1. The top layer corresponds to the trapping system. To obtain the complete impedance spectrum of a spore, it must remain stationary. Our microfluidic trapping system hydrodynamically captures single cells and holds them in place during the EIS measurements.

The bottom layer consists of an array of micro-electrodes patterned on a glass substrate. These gold electrodes are aligned with the microfluidic traps to enable electrical impedance spectroscopy measurements of single cells.

The design and fabrication process of each layer—the microfluidic and the electrode layers—is described in the following section.

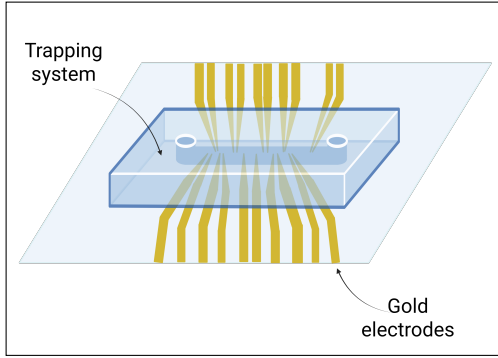


Figure 7.1: Microfluidic EIS system. It consists of a top layer (hydrodynamic single-cell traps) and a bottom layer (micro-electrodes).

7.2 Design and fabrication

7.2.1 Trapping system design & fabrication

The microfluidic channel was designed using CleWin software, inspired by the design reported in Tan et al. [234]. The microfluidic trapping system device (Fig. 7.2) consists of a symmetric layout with ten hydrodynamic traps in total, all connected through a main channel. Each trap (Fig. 7.2 item 4) has two symmetric funnel-like constrictions, called trapping sites, designed to capture only one spore at its narrowest point (5 μm). Once a trapping site is occupied, no more flow goes through it. The subsequent spores are then diverted through a bifurcation of the main channel, here referred to as the bypass, and continue toward the next trapping site. It is also worth noting that, as mentioned before, each trap has two trapping sites. Depending on the flow direction, one site will be occupied while the other remains empty, which allows for the implementation of a differential EIS measurement strategy.

In addition, an aspiration channel (Fig. 7.2 item 3) is included, similar to the U-shaped trap system (Chapter 4.1.3), allowing the glass-PDMS bonding to be reversible [125].

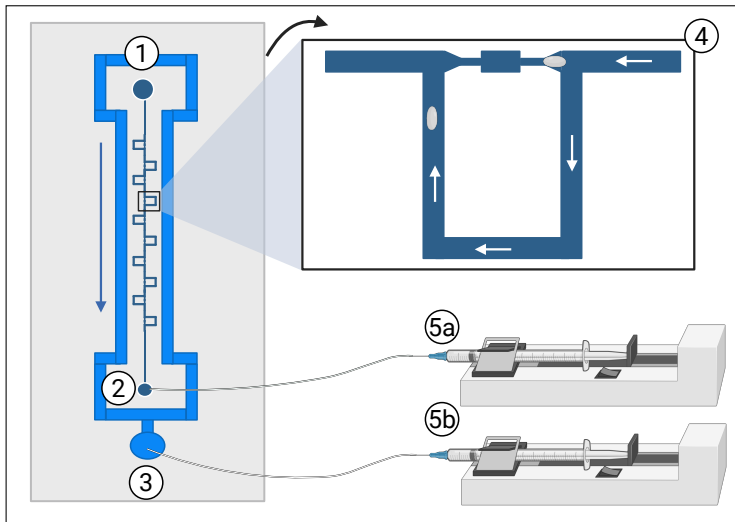


Figure 7.2: Trapping system device. (1) The sample is directly placed into the inlet chamber. It is then aspirated through the device and into the traps (4) by the negative pressure generated when withdrawing the plunger of an empty syringe (5a) connected to the outlet (2). The PDMS stamp is reversibly sealed by negative pressure generated in the same way, using another empty syringe (5b) connected to the aspiration channel (3).

A standard soft-lithography technique was used to fabricate the PDMS stamp [117]. The process is described in more detail in Chapter 3.1.4. The photoresist employed was SU-8 2010 (MicroChem), targeting a channel height of 13 μm . As with the U-shaped trap device, the negative photoresist was spin-coated at 200 rpm for 10 s, followed by 2000 rpm for 30 s. The PDMS replica is illustrated in Fig. 7.3.

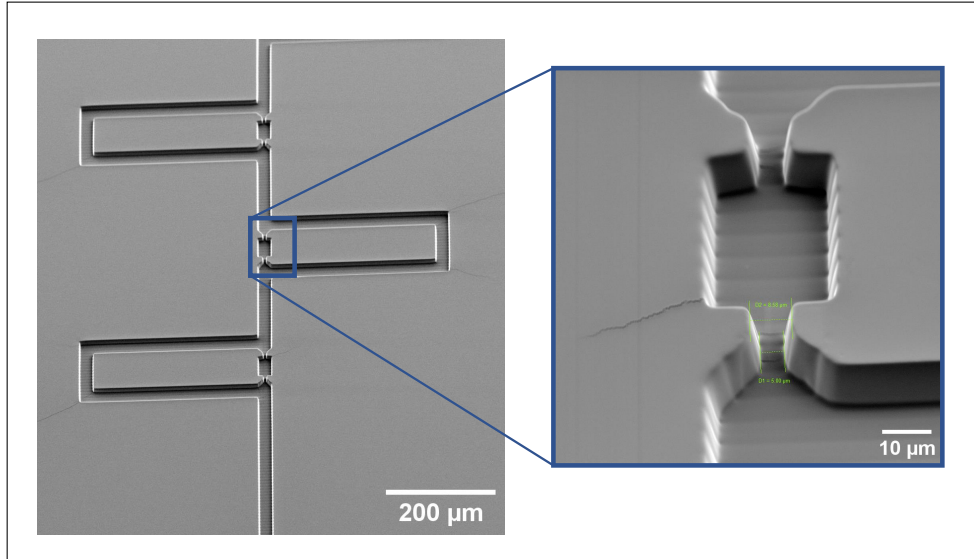


Figure 7.3: SEM view of trapping systems after replication in PDMS

7.2.2 Electrodes design & fabrication

A chrome mask (Nanofilm) patterned with the electrode design is made by direct writing using the μ PG 101 laser lithographer (Heidelberg Instruments[®]). The laser power was set to 50% of 7 mW with the filter on. The mask was then developed in a single puddle for 1 min using AZ400k developer diluted 4:1 with deionized water, and dried with nitrogen. Chrome etching was performed in a single puddle for 1 min, followed by rinsing with deionized water and drying.

The gold electrodes were fabricated through a lift-off process. First, negative photoresist AZ[®] nLOF[™] 2020 (Merck) was spin-coated at 3000 rpm for 30 s on a clean large glass slide (75×50 mm). A soft bake of 60 s at 110°C was performed on a direct-contact hot plate, followed by UV exposure at $66 \text{ mJ}/\text{cm}^2$ using the UV-KUB 2 (Kloe). The post-bake was done at 110°C for 60 s, also in direct contact with a hot plate. Subsequently, the development step was performed using AZ 326MIF developer for 20 s in a single puddle, then rinsed with deionized water.

The next step was the gold deposition using the EVA 300 e-beam evaporator (Alliance Concept). The glass substrates were placed inside the machine under vacuum conditions. First, a thin adhesion layer of chromium (5 nm) was deposited to improve the adhesion of the gold, which was then evaporated onto the substrate, forming a 50 nm conductive film. After evaporation, the unwanted gold was removed to obtain the desired pattern by immersing the glass slides into acetone, followed by a sonication bath for 60 s. Finally, the samples were rinsed with isopropanol and dried with a nitrogen gun.

7.3 Experimental setup Integration

In this section, the integration of the experimental setup is presented.

First, it is necessary to connect the electrodes to the measurement device (HF2LI from

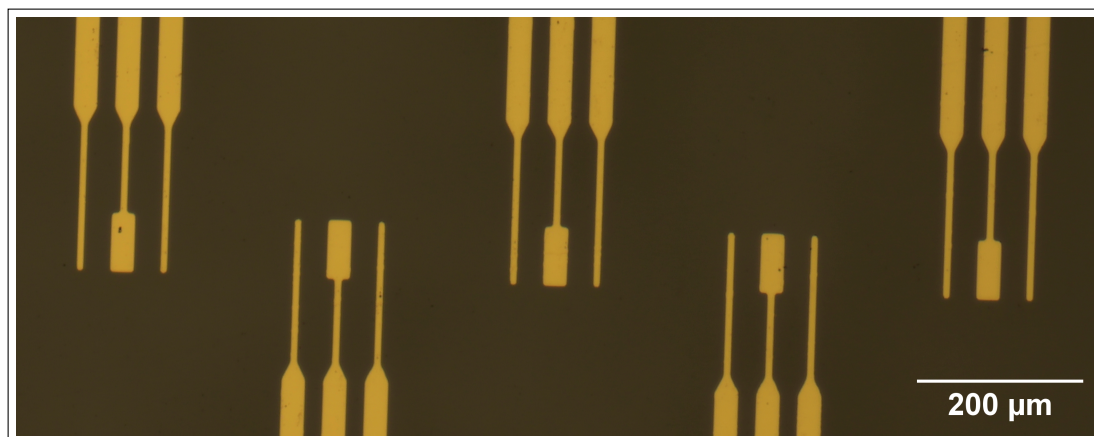


Figure 7.4: Image of the gold electrodes. The image shows five sets of electrodes, each consisting of three individual electrodes. In each set, the gap between electrodes is $30\ \mu\text{m}$.

Zurich Instruments, see next paragraph for details), which is used to drive and record the electrical signals. To achieve this, a circuit board mounted inside a 3D-printed plastic frame was designed using KiCad and Autodesk Inventor, respectively (Fig. 7.5). The glass slide with the gold electrodes is positioned within this frame, aligning the electrode contact pads with the spring contacts soldered on the circuit board (blue rectangle). The frame holds them securely in place. The assembled setup is then connected to the HF2LI device through the coaxial connectors (pink rectangles).

Once all components are connected, the system is placed in a fluorescence microscope (Thunder Imaging System, Leica) equipped with a digital camera for image acquisition. The PDMS stamp is positioned in the central section of the 3D-printed frame (yellow rectangle), and the ten traps within the stamp are aligned with their corresponding electrode pairs under microscopic observation.

To help in this alignment step, a drop of ethanol is added to the glass slide to facilitate the sliding of the PDMS stamp, allowing it to be carefully aligned with the electrodes. Once done, the PDMS stamp is sealed by generating a vacuum in the aspiration channel using an empty syringe. The vacuum causes the ceiling of the channel to collapse, pressing the PDMS against the glass slide and forming a reversible seal. As this seal is reversible, the electrodes can be reused. The vacuum is maintained by keeping the plunger of the syringe extended with the help of a 3D-printed holder (Fig. 7.2 5b).

The buffer solution ($0.22\ \mu\text{m}$ filtered 1% (v/v) Tween 20 in $1\times\text{PBS}$) is then drawn through the channel using another empty syringe to create suction. The flow is maintained by keeping the syringe plunger extended with the aid of a second 3D-printed holder. This is done to ensure the removal of any air bubbles.

After confirming that no bubbles remain, $10\ \mu\text{L}$ of the sample is introduced into the inlet reservoir. By selecting a specific red switch, the corresponding set of electrodes can be activated to send and receive the signal. Only pairs of electrodes associated with a trap containing a single particle in the trapping site are selected.

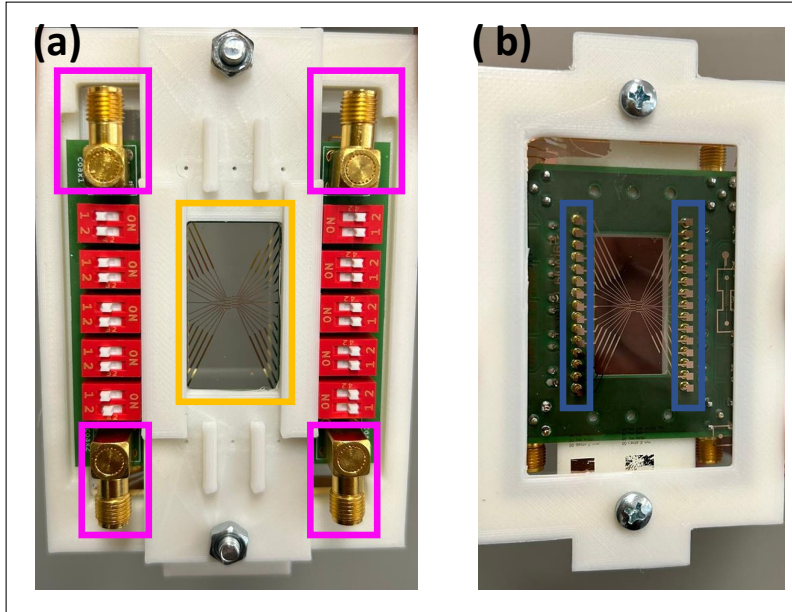


Figure 7.5: Circuit board mounted inside a 3D-printed plastic frame.(a) Top view: Red switches allow selecting the desired gold electrodes, measuring one set of electrodes at a time. Pink rectangles indicate the connectors to the Lock-in Amplifier. (b) Bottom view: Blue rectangles show the spring contacts on the circuit board, which are in contact with the contact pads of the gold electrodes and kept in place by the 3D-printed plastic frame.

7.4 EIS measurement and data analysis

The full EIS process is carried out using a lock-in amplifier (HF2LI, Zurich Instruments). The HF2LI generates the sinusoidal excitation voltage applied to the electrodes and sweeps its frequency within a user-defined range (from almost 0 Hz up to 50 MHz). The resulting electrical current is amplified and converted to a recordable voltage using a current-to-voltage converter (HF2TA, Zurich Instruments), more conventionally named trans-impedance amplifier (TIA). The HF2LI also records this signal and computes both its amplitude and phase relative to the excitation signal.

Using these data, the complex impedance spectrum is computed and displayed via a home made Python script running on the computer also controlling the HF2LI. Thanks to the double-trap geometry (one trap occupied by a cell and the other filled only with the cell medium) and the two independent inputs of the HF2LI, simultaneous double impedance measurements can be performed. This allows direct comparison of the impedance with and without a trapped cell. Such differential measurements are particularly useful to compensate for wiring artifacts and to estimate the contribution of the surrounding medium to the measured signal. The schematic of the connections used for the measurements and data analysis is illustrated in Fig. 7.6.

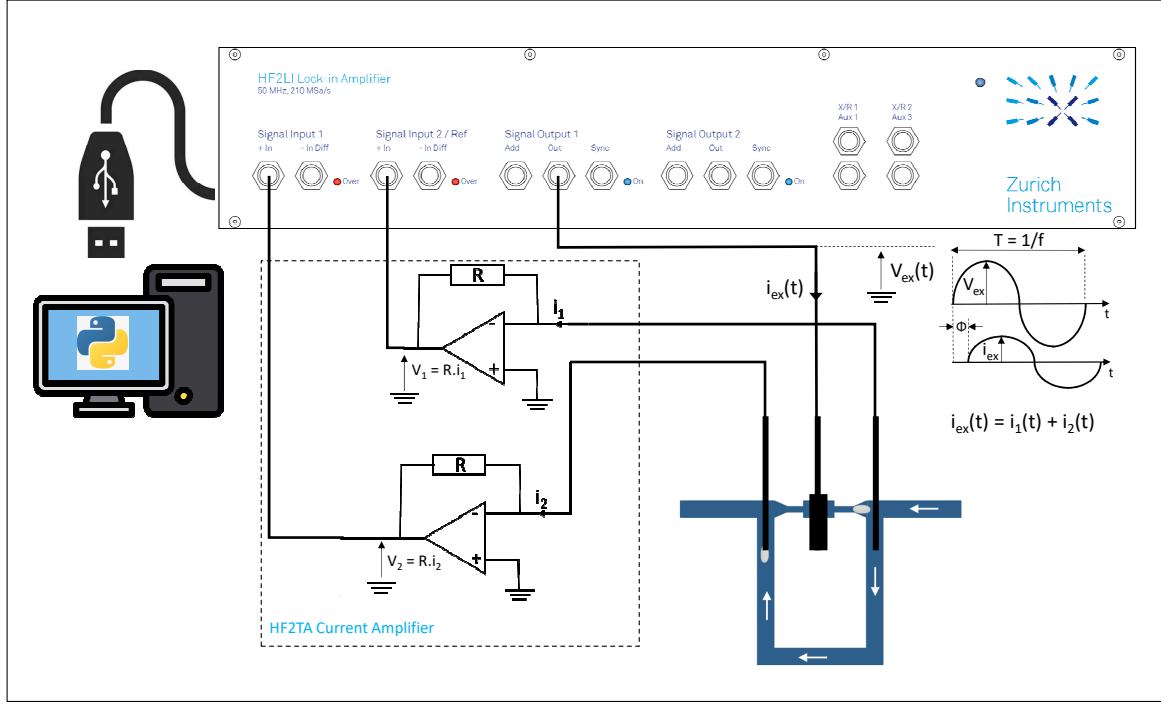


Figure 7.6: Schematic of the circuit connection for EIS measurement and data analysis

Fig. 7.7 gives an electrical equivalent circuit of the area probed by the electrodes. Z_{cell} and Z_{med} are respectively the impedance measured in the trap containing a cell and the impedance of the trap remaining free of a cell, thus only filled with the medium. One can notice that both of these impedances are connected to exactly the same excitation signal (node 'E'). It is also important to notice that the impedance Z_{bypass} coming from the channel allowing the feeding of the next traps do not contribute to the overall measured impedance, despite being in contact with the electrodes at nodes 'M' and 'C'. This comes from the fact that both of these electrodes are connected to a virtual ground (VGND) maintained by the corresponding TIA. At a virtual ground, the potential is set to 0V and then, no current flows through Z_{bypass} , avoiding any parasitic contribution. For each TIA, the value of the resistor R can be selected to adapt the conversion ratio between the input current and the output voltage (transimpedance gain, which can be set from 100 V/A to 100 MV/A, but at the cost of a reduced bandwidth when increased).

7.5 Design optimization

This section presents the optimization performed in the microchannel system and in the gold electrode system.

7.5.1 Trapping system design optimization

The trapping system did not undergo significant design changes, except for the addition of a pillar-based filter at the inlet, similar to the one used in the DLD microdevices. Without this filter, the system clogs a few seconds after injecting the suspension (Fig. 7.8 (c)). Since the traps are connected in series, clogging at the beginning of the

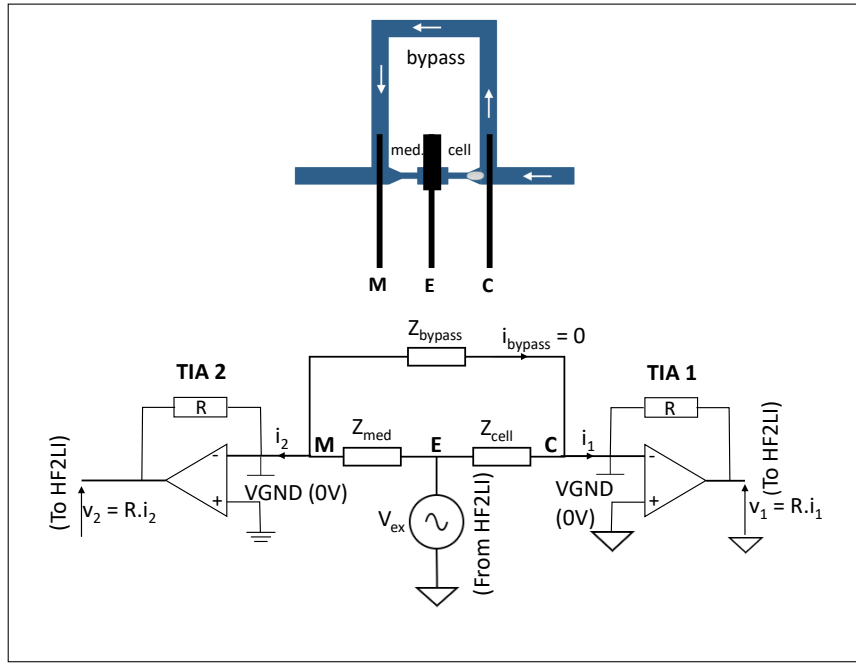


Figure 7.7: Electrical equivalent circuit of the zone sensed by the electrodes

channel can prevent flow to the subsequent traps. With the filter, this issue was considerably reduced; however, clogging still occurred in the first two traps. To address this, highly diluted samples were used, delaying blockage in the initial traps and allowing sufficient time for the remaining eight traps to be successfully filled with a single particle.

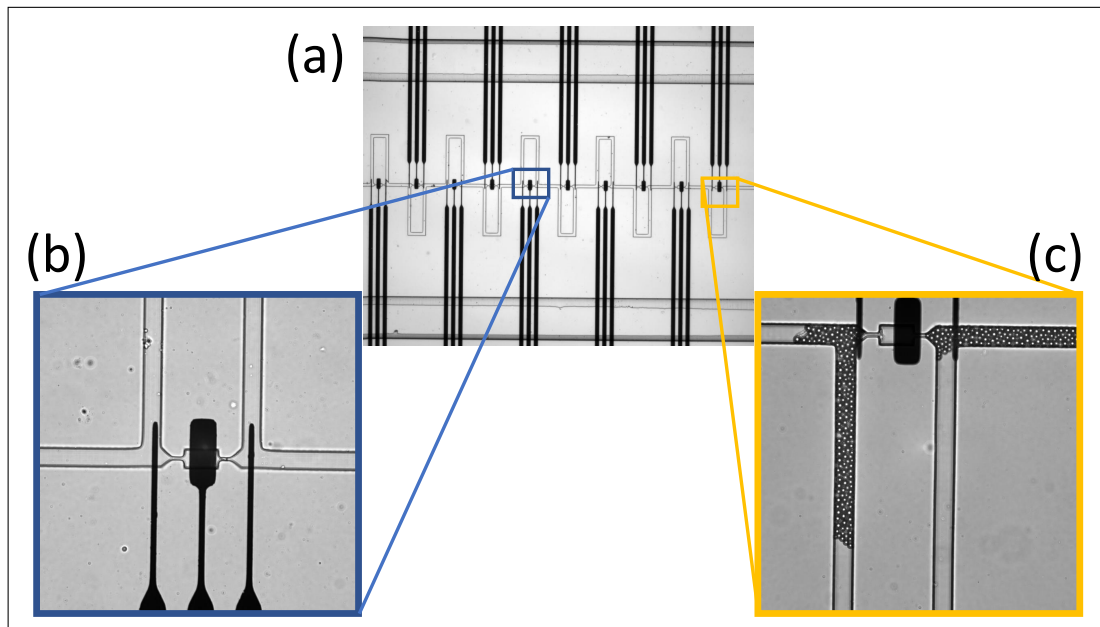


Figure 7.8: Gold electrodes aligned with the traps. (a) Overview of the 10 traps with their corresponding sets of electrodes. (b) Central traps are well aligned with the electrodes. (c) At the edges, a misalignment occurs, and the traps no longer match the gold electrodes.

The second observation is that the pitch between traps was set to 260 μm on the lithog-

raphy mask (and checked by microscopy), but measurements of the resulting PDMS stamp showed that the actual pitch was 258 μm , probably because of mechanical relaxation as it is possible with this soft elastomer. Of course, when attempting alignment (using the center trap as a reference, Fig. 7.8(b)), the traps became completely misaligned along the axial direction (item (c)).

7.5.2 Electrodes design optimization

The first version of the gold electrode design (5 μm width) is shown in Fig. 7.4. As mentioned earlier, the actual pitch of the PDMS traps did not match the expected value, so we decided to correct it through the electrode design. A new mask was drawn to achieve a pitch of 258 μm in the electrodes instead of 260 μm .

Moreover, in this new version, the electrode area was increased and the gap between the electrodes was reduced from 30 μm to 10 μm . The first modification decreases the medium impedance enclosed in the area probed by the electrodes. In both cases, it corresponds to a reduction of parasitic impedance combined with that of interest.

7.6 Conclusion

In this chapter, we presented the development of the microfluidic system for EIS measurements. We described the design and fabrication of the hydrodynamic single-cell trap, demonstrating that a single bead could be successfully captured in 80% of the traps, which represents a good trapping efficiency.

We also presented the fabrication of the first electrode mask. In the new mask design, the electrode pitch was adjusted to match the pitch of the PDMS traps. In addition, a circuit board and 3D-printed frame were fabricated to facilitate the connection to the HF2LI instrument.

At the time of writing this manuscript, no EIS measurement campaign has been conducted due to time constraints. However, we are confident that the current setup is ready to perform the first tests on spores and obtain their electrical signatures.

Résumé du Chapitre 7

Dans ce chapitre, nous présentons le dispositif microfluidique développé pour la caractérisation électrique des spores par spectroscopie d'impédance électrique (EIS, Electrical Impedance Spectroscopy).

La puce développée comporte un substrat en verre au format d'une lame de microscope double et un réseau microfluidique en PDMS produit par soft lithography. Ce réseau permet de piéger de façon hydrodynamique au dizaine de spores de façon individuelle, permettant ainsi la caractérisation d'autant de cellules. Afin de permettre une mesure différentielle, chaque site de piégeage est en fait double : un piège est destiné à capturer une spore et un piège symétrique est destiné à rester uniquement rempli du milieu de suspension.

Le substrat en verre comporte les microélectrodes nécessaires à la mesure d'impédance. Ces électrodes sont réalisées en or structuré par le procédé de lift-off classique. A chaque site de piégeage correspond une paire d'électrodes. L'association du substrat et de l'élément en PDMS est effectué par un système de mise en dépression d'un canal périphérique, ce qui permet un montage réversible. A l'issue d'une expérience il est ainsi possible de récupérer le substrat avec les électrodes en vue de le laver avant réemploi avec un nouveau réseau microfluidique de piégeage.

Afin de rendre ce système facilement intégrable dans le porte lame du microscope utilisé pour l'observation pendant la manipulation, un cadre réalisé par impression 3D a été conçu. Ce système intègre également un circuit imprimé permettant une reprise de contacts aisée entre les pistes présentes sur le substrat en verre et les connecteurs coaxiaux à destination de l'instrument de mesure d'impédance. Ce dernier est un amplificateur de type lock-in (HF2LI, Zurich Instruments) associé à deux amplificateurs transimpédance (HF2TA, Zurich Instruments).

Au cours des premières expériences, nous avons remarqué que le système de piégeage avait tendance à se boucher facilement. Pour remédier à ce problème, un filtre similaire à celui utilisé dans les structures DLD a été ajouté, ce qui a considérablement réduit le colmatage et amélioré les performances de piégeage. Grâce à cette modification, et à une dilution appropriée de l'échantillon, la plupart des pièges ont capturé une seule cellule. Cependant, les deux premiers pièges capturaient encore occasionnellement plus d'une cellule, les rendant inutilisables pour les mesures.

Il est également apparu que l'espacement entre les pièges hydrodynamiques réalisés dans le PDMS n'était pas exactement celui défini par le masque de photolithographie (environ $2\mu\text{m}$ de moins). Nous attribuons cet effet à une probable relaxation

du PDMS après démoulage. Evidemment, cela conduit à l'impossibilité d'aligner l'ensemble des électrodes sur les pièges et l'espacement inter-électrodes a dû être corrigé en conséquence. La nouvelle conception a également permis d'optimiser certains paramètres, tels que l'écartement et la taille des électrodes, en particulier pour réaliser une mesure au plus près des cellules capturées, ce qui limite la contribution du milieu de suspension à la mesure.

Bien que nous n'ayons pas pu effectuer une réelle campagne de mesure en raison de contraintes de temps, l'installation complète est désormais prête à être testée pour mesurer l'impédance électrique des spores.

Conclusion

Conclusion (English)

Throughout this thesis, we have highlighted the issues behind the study of plant pathogens such as *B. cinerea* and its mutant strains.

For this reason, the project was structured around two main objectives. The first objective was to provide an innovative approach to facilitate and accelerate the production of mutant strains. The second objective was to propose new tools, compatible with the workflow of biologists, allowing the collection of novel biophysical signatures of the fungal cell wall.

The first part of this thesis, entitled "**Genetic Purification in Microfluidics**", focuses on the development and implementation of a microfluidic system capable of separating mononuclear spores from multinuclear ones based on their size. Indeed, fungal spores can contain multiple nuclei, which means that a transformant may retain non-mutated copies of its genome in some nuclei. To obtain a homogeneous population, in terms of DNA expression, our strategy was to retrieve spores with only one nucleus, hence ensuring that, when grown in culture media supplemented in antibiotics, only spores containing the modification– including the desired mutation and the resistant gene to the antibiotics– survive.

After demonstrating the correlation between the size of the spores and their number of nuclei, we proposed to isolate the particles below the critical diameter of 6 μm to enrich the sample suspension in mononucleated spores. Due to its high resolution, its ability to be integrated, and its ease of implementation, we have chosen the Deterministic Lateral Displacement approach.

We have developed and tested three different optimized microfluidic devices, each of them associated with a different theoretical critical diameter. After evaluating their performances on model objects (beads), we selected the device referred to as DLD 5 and tested it with spores, following a specific protocol developed to minimize clog-

ging events.

Our quantitative analyses indicate good enrichment: the spores collected at the small outlet—the outlet containing the selected spores—mostly contained one or two nuclei, with very few containing three. We showed that a single run of 15 minutes in our DLD 5, was enough to be able to collect and isolate monocucleated spores for subsequent DNA characterization using PCR. This was consistently observed across the different replicates performed and represents a key contribution of our work.

Not only have we found that our device significantly accelerates the purification process and facilitates the genetic characterization of mutant strains, but it also gives access to mononucleated spores, a particular subpopulation about which very little is known. Indeed, this second point is particularly relevant, from a fundamental point of view, for genomic studies aiming at elucidating the role of nuclear content in fungal development.

Moreover, the setup of the DLD device is not particularly complex, requiring only a microscope with a camera, a system to implement steady injection of the fluids and, of course, the microdevice itself. Therefore, it could be integrated and automated to fit easily with the usual workflow of biologists.

In our continuous effort to integrate as many steps as possible into microsystems, we also proposed the development of the U-shaped trap system, a chip aiming to isolate single spores, promote their growth, and enable the transfer of individual spores to Petri dishes containing the appropriate nutrients, allowing DNA analysis *via* PCR to be performed on each individual colony.

So far, we have tested the first version of the design and demonstrated the feasibility of isolating and culturing single fungal spores in a microfluidic system. However, further improvements are required to (i) increase the trapping efficiency, (ii) reduce clogging, and (iii) optimize the spacing between the traps. These aspects are particularly important because the spores will later need to be transferred to Petri dishes, where adequate spacing is required to facilitate the manual handling of individual spores. Once all these improvements are completed, the next step will be to connect this function to the DLD device, allowing sorting and single-cell isolation to be performed in a single step, creating a continuous workflow.

In the market, the C1 platform from Fluidigm Corp. enables the isolation and subsequent genetic analysis of up to 96 individual cells in parallel, demonstrating the potential of microfluidic technologies for fully integrated biological processing. In the long term, achieving a comparable level of integration in our approach would be highly valuable. Notably, on-chip PCR has already been widely demonstrated, as reported in the review by Ahrberg et al. [235]. Hence, a complete on-chip purification and analysis system could be envisioned as a promising future development.

The second part of this thesis, entitled "**Biophysical characterization**", focuses on the design of microfluidic tools to assess the mechanical and electrical properties of the fungal cell wall of the *B. cinerea* spores.

For the mechanical characterization, the objective was to obtain, for the first time, the

mechanical properties of wild-type *B. cinerea* spores, as well as those of mutant strains. To achieve this, we tried to implement and adjust a microfluidic approach from the literature in parallel with AFM measurements.

The experiments showed that PDMS was not an optimal material for our application, as its rigidity is similar to or even lower than that of the spores, making it unsuitable for deforming objects with a stiffness in the hundreds of MPa range. This highlights the need to explore alternative materials such as PMMA or COC, which have higher Young's moduli. Nonetheless, the PDMS-based device could still be employed in our laboratory for studying cells with lower stiffness values.

Once fabricated with a stiffer material, the next stage will involve performing the first deformation experiments using the microfluidic device. These tests will validate the device's performance and allow qualitative comparison with AFM results, highlighting trends in spore mechanics and providing a reliable, high-throughput method for mechanical characterization.

In parallel, AFM measurements on wild-type and Δ AGS strains provided the first mechanical characterization of the fungal cell wall. While further experiments are needed to strengthen these results, clear differences between the wild-type and mutant were observed. Δ AGS strain shows a decrease in rigidity compared to the wild type, which is consistent with confocal imaging performed at the MAP laboratory, showing that α -glucan is located on the surface of the FCW in the wild type.

These findings are particularly relevant because the fungal cell wall acts as the interface between the fungus and its environment and is recognized by plant receptors. Moreover, cell wall mechanics can influence fungal adhesion to plant surfaces, thereby affecting the efficiency of host colonization. Understanding how specific genes affect cell wall mechanics can provide crucial insights for developing targeted fungicides, helping to prevent *B. cinerea* from attacking crops.

Although AFM provides high-precision reference measurements, the ultimate goal is to implement a high-throughput, user-friendly microfluidic platform capable of efficiently characterizing a wide range of mutant strains.

Finally, for the electrical characterization, although we have not yet performed the first measurements, the setup is fully prepared for testing. This is particularly important because there is currently no electrical information available for *B. cinerea* spores or their mutants. Once validated, the device has the potential to provide insights into spore structures, for example, the fungal cell wall, which could help differentiate mutations. Additionally, it could provide information regarding the number of nuclei, hence offering a complementary way to further enrich spore suspensions in monocultured spores after the DLD device, using electrical approaches.

Electrical impedance measurements could also have broader applications, such as the rapid and precise classification of different spore types. For example, it could be applied to detect airborne fungal spores or to analyze other types of spores, providing a versatile platform for both research and industrial applications [236].

This work has established the first steps toward a comprehensive understanding of *B.*

cinerea spores, and allow opening new perspectives by demonstrating what microfluidics can contribute to the science of fungi.

Conclusion (French)

Tout au long de cette thèse, nous avons mis en évidence les enjeux liés à l'étude des pathogènes végétaux tels que *B. cinerea* et de ses souches mutantes.

Pour cette raison, le projet a été structuré autour de deux objectifs principaux. Le premier objectif était de proposer une approche innovante afin de faciliter et d'accélérer la production de souches mutantes. Le second objectif visait à développer de nouveaux outils, compatibles avec le flux de travail des biologistes, permettant de recueillir de nouvelles signatures biophysiques de la paroi cellulaire fongique.

La première partie de cette thèse, intitulée "**Purification Génétique en Microfluidique**", se concentre sur le développement et la mise en œuvre d'un système microfluidique capable de séparer les spores mononucléaires des spores multinucléaires en fonction de leur taille. En effet, les spores fongiques peuvent contenir plusieurs noyaux, ce qui signifie qu'un transformant peut conserver des copies non mutées de son génome dans certains noyaux. Pour obtenir une population homogène en termes d'expression de l'ADN, notre stratégie consistait à récupérer les spores ne possédant qu'un seul noyau, garantissant ainsi que, lorsqu'elles sont cultivées dans des milieux enrichis en antibiotiques, seules les spores contenant la modification — incluant la mutation désirée et le gène de résistance aux antibiotiques — survivent.

Après avoir démontré la corrélation entre la taille des spores et leur nombre de noyaux, nous avons proposé d'isoler les particules en dessous du diamètre critique de 6 μm afin d'enrichir la suspension d'échantillons en spores mononucléaires. En raison de sa haute résolution, de sa capacité d'intégration et de sa facilité de mise en œuvre, nous avons choisi l'approche de Déplacement Latéral Déterministe (DLD).

Nous avons développé et testé trois dispositifs microfluidiques optimisés différents, chacun associé à un diamètre critique théorique distinct. Après avoir évalué leurs performances sur des objets modèles (perles), nous avons sélectionné le dispositif appelé DLD 5 et l'avons testé avec des spores, suivant un protocole spécifique développé pour minimiser les risques de colmatage.

Nos analyses quantitatives indiquent un bon enrichissement : les spores collectées à la petite sortie — la sortie contenant les spores sélectionnées — contenaient majoritairement un ou deux noyaux, très peu en contenaient trois. Nous avons montré qu'une seule opération de 15 minutes dans notre DLD 5 suffisait pour collecter et isoler des spores mononucléaires en vue de leur caractérisation génétique ultérieure par PCR. Ce résultat a été observé de manière cohérente à travers les différentes répétitions effectuées et constitue une contribution clé de notre travail.

Non seulement nous avons constaté que notre dispositif accélère de manière significative le processus de purification et facilite la caractérisation génétique des souches mutantes, mais il permet également l'accès aux spores mononucléaires, une sous-population particulière sur laquelle on dispose de très peu de connaissances. En ef-

fet, ce second point est particulièrement pertinent, d'un point de vue fondamental, pour les études génomiques visant à élucider le rôle du contenu nucléaire dans le développement fongique.

De plus, la mise en place du dispositif DLD n'est pas particulièrement complexe, nécessitant seulement un microscope avec caméra, un système pour assurer l'injection stable des fluides et, bien sûr, le microdispositif lui-même. Par conséquent, il pourrait être intégré et automatisé pour s'adapter facilement au flux de travail habituel des biologistes.

Dans notre effort continu d'intégration du plus grand nombre possible d'étapes dans les microsystèmes, nous avons également proposé le développement du système de piège en forme de U, une puce visant à isoler les spores individuelles, à favoriser leur croissance et à permettre le transfert des spores individuelles vers des boîtes de Petri contenant les nutriments appropriés, permettant ainsi l'analyse de l'ADN *via* PCR pour chaque colonie individuelle.

Jusqu'à présent, nous avons testé la première version du dispositif et démontré la faisabilité d'isoler et de cultiver des spores fongiques individuelles dans un système microfluidique. Cependant, des améliorations supplémentaires sont nécessaires pour (i) augmenter l'efficacité de piégeage, (ii) réduire le colmatage et (iii) optimiser l'espacement entre les pièges. Ces aspects sont particulièrement importants car les spores devront ensuite être transférées vers des boîtes de Petri, où un espacement adéquat est nécessaire pour faciliter la manipulation manuelle des spores individuelles. Une fois toutes ces améliorations réalisées, l'étape suivante sera de connecter cette fonction au dispositif DLD, permettant de réaliser le tri et l'isolement des cellules uniques en une seule étape, créant ainsi un flux de travail continu.

Sur le marché, la plateforme C1 de Fluidigm Corp. permet l'isolement et l'analyse génétique ultérieure de jusqu'à 96 cellules individuelles en parallèle, démontrant le potentiel des technologies microfluidiques pour un traitement biologique entièrement intégré. À long terme, atteindre un niveau d'intégration comparable dans notre approche serait extrêmement précieux. Notamment, la PCR sur puce a déjà été largement démontrée, comme le rapporte la revue de Ahrberg et al. [235]. Ainsi, un système complet de purification et d'analyse sur puce pourrait être envisagé comme un développement prometteur pour l'avenir.

La deuxième partie de cette thèse, intitulée "**Caractérisation Biophysique**", se concentre sur la conception d'outils microfluidiques pour évaluer les propriétés mécaniques et électriques de la paroi cellulaire fongique des spores de *B. cinerea*.

Pour la caractérisation mécanique, l'objectif était d'obtenir, pour la première fois, les propriétés mécaniques des spores de *B. cinerea* de type sauvage, ainsi que celles des souches mutantes. Pour ce faire, nous avons essayé de mettre en œuvre et d'ajuster une approche microfluidique issue de la littérature, en parallèle avec des mesures par AFM.

Les expériences ont montré que le PDMS n'était pas un matériau optimal pour notre application, car sa rigidité est similaire, voire inférieure, à celle des spores, le rendant inadéquat pour déformer des objets dont la rigidité se situe dans la gamme de centaines

de MPa. Cela souligne la nécessité d'explorer des matériaux alternatifs tels que le PMMA ou le COC, qui possèdent des modules de Young plus élevés. Néanmoins, le dispositif à base de PDMS pourrait encore être utilisé dans notre laboratoire pour l'étude de cellules présentant des valeurs de rigidité plus faibles.

Une fois fabriqué avec un matériau plus rigide, l'étape suivante consistera à réaliser les premiers essais de déformation à l'aide du dispositif microfluidique. Ces tests permettront de valider les performances du dispositif et de comparer qualitativement les résultats avec ceux de l'AFM, mettant en évidence les tendances mécaniques des spores et fournissant une méthode fiable et à haut débit pour la caractérisation mécanique.

Parallèlement, les mesures AFM sur les souches de type sauvage et Δ AGS ont fourni la première caractérisation mécanique de la paroi cellulaire fongique. Bien que des expériences supplémentaires soient nécessaires pour consolider ces résultats, des différences nettes entre le type sauvage et le mutant ont été observées. La souche Δ AGS montre une diminution de rigidité par rapport au type sauvage, ce qui est cohérent avec l'imagerie confocale réalisée au laboratoire MAP, montrant que l' α -glucane est localisé à la surface de la paroi cellulaire fongique chez le type sauvage.

Ces résultats sont particulièrement pertinents car la paroi cellulaire fongique constitue l'interface entre le champignon et son environnement et est reconnue par les récepteurs des plantes. De plus, la mécanique de la paroi cellulaire peut influencer l'adhésion du champignon aux surfaces végétales, affectant ainsi l'efficacité de la colonisation de l'hôte. Comprendre comment des gènes spécifiques influencent la mécanique de la paroi cellulaire peut fournir des informations cruciales pour le développement de fongicides ciblés, contribuant à empêcher *B. cinerea* d'attaquer les cultures.

Bien que l'AFM fournisse des mesures de référence de haute précision, l'objectif ultime est de mettre en place une plateforme microfluidique à haut débit et conviviale, capable de caractériser efficacement une large gamme de souches mutantes.

Enfin, pour la caractérisation électrique, bien que nous n'ayons pas encore réalisé les premières mesures, l'installation est entièrement prête pour les tests. Cela est particulièrement important car il n'existe actuellement aucune information électrique disponible pour les spores de *B. cinerea* ni pour leurs mutants. Une fois validé, le dispositif pourrait fournir des informations sur la structure des spores, par exemple la paroi cellulaire fongique, ce qui pourrait aider à différencier les mutations. De plus, il pourrait fournir des informations sur le nombre de noyaux, offrant ainsi un moyen complémentaire pour enrichir davantage les suspensions de spores mononucléaires après le dispositif DLD, en utilisant des approches électriques.

Les mesures d'impédance électrique pourraient également avoir des applications plus larges, telles que la classification rapide et précise de différents types de spores. Par exemple, elles pourraient être appliquées à la détection de spores fongiques aériennes ou à l'analyse d'autres types de spores, fournissant ainsi une plateforme polyvalente pour la recherche et les applications industrielles [236].

Ces travaux ont permis de poser les premiers jalons d'une compréhension globale des spores de *B. cinerea* et ouvrent de nouvelles perspectives en démontrant ce que la

microfluidique peut apporter à la science des champignons.

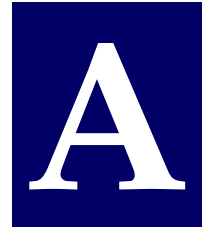
List of communications

- **Poster presentation:** Fabiola Elena Chacon Chale, Magalie Faivre, Jean-François Chateaux, Christophe Bruel. Microfluidic sorting system for Botrytis cinerea spores genetic purification. GDR MicroNanoFluidics, Nov 2024, Nice, France.

URL: <https://hal.science/hal-05233450v1>

- **Oral presentation at an international conference:** Fabiola Elena Chacon Chale, Magalie Faivre, Jean-François Chateaux, Christophe Bruel. Speeding up Genetic Purification in the Production of Botrytis Cinerea Mutant Spores Using Deterministic Lateral Displacement. Lab-on-a-chip and Microfluidics Europe 2025, SelectBio, Jun 2025, Montpellier, France

URL: <https://hal.science/hal-05233412v1>



DLD experiments

Comparison of the sample stream width at the small outlet of the three DLD devices

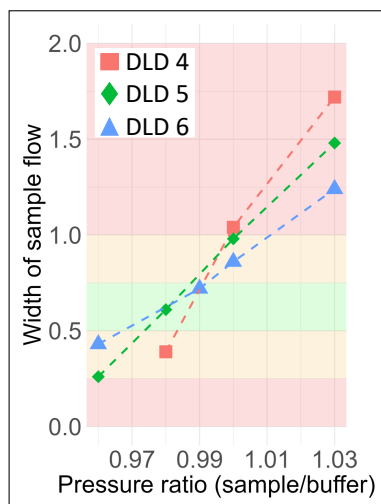


Figure A.1: Sample stream width at the small outlet of the three DLD devices: DLD 4 (red squares), DLD 5 (green diamonds), and DLD 6 (blue triangles) as a function of the pressure ratio (sample to buffer) for a fixed buffer inlet pressure of 900 mbar, which was selected because it allows for a higher throughput (among the three pressures tested). The colored bands represent different regions: the green band indicates the optimal parameters; the lower and upper orange bands correspond to the safe margin regions; and the upper red region leads to contamination.

Axis ratio analysis of spores collected from the inlet suspension

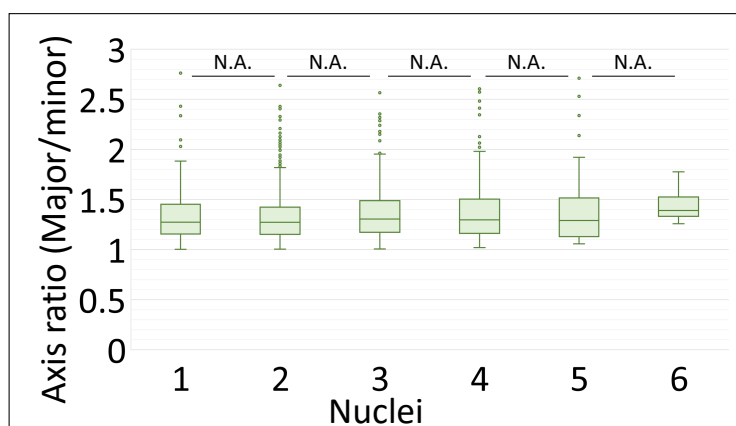


Figure A.2: Bright-field images of spores were fitted to ellipses, from which the major and minor axes were obtained. These values were then used to calculate the axis ratio (major to minor) for each nuclei number group. Values close to 1 indicate nearly spherical shapes, whereas higher values correspond to elongated objects. All nuclei groups exhibited axis ratios below 2, indicating moderate elongation and shapes that remain relatively close to spherical.

AFM experiments

Determination of the inverse optical lever sensitivity (InvOLS)

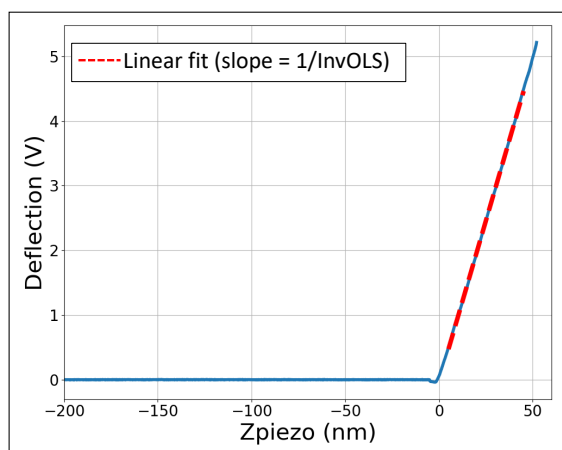


Figure B.1: Before starting the AFM measurements, the actual spring constant (k_c) of the cantilever must be calibrated. To do so, the inverse optical lever sensitivity (InvOLS) is determined by fitting the slope of the linear region of a force curve obtained on a rigid substrate (a glass slide in this case). The InvOLS value ($1/\text{slope}$) is then used to convert the photodiode voltage signal into the actual cantilever deflection in nanometers.

Force–separation curve acquired on a glass slide

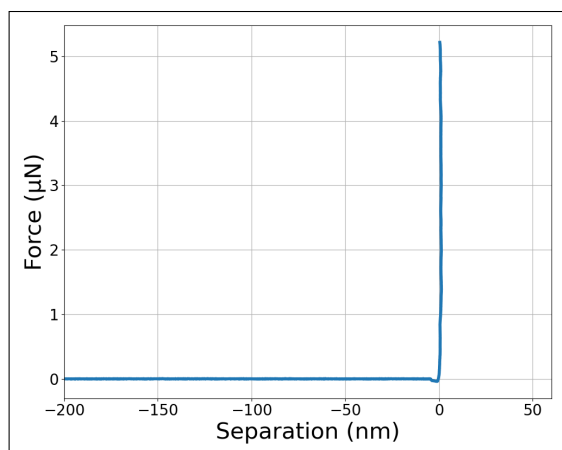


Figure B.2: The calibration was performed on a glass substrate to obtain the actual cantilever spring constant. This figure shows the force–separation curve recorded on a hard surface, such as the glass slide, where no indentation is observed and the contact region appears vertical. In contrast, for soft samples, a mechanical model—such as the Hertz–Sneddon model—is required to fit the indentation region and calculate the Young's modulus.

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Résumé

B. cinerea est un phytopathogène nécrotrophe qui peut infecter plus de 200 espèces de plantes dans le monde entier et qui est considéré comme le deuxième champignon pathogène le plus important. Les biologistes produisent souvent des souches génétiquement modifiées afin d'étudier l'influence de gènes spécifiques sur la pathogenèse fongique. Cependant, comme les spores de *B. cinerea* sont généralement multi-nucléées, l'obtention d'une population purifiée dans laquelle tous les noyaux contiennent la mutation souhaitée est un processus long et fastidieux qui peut prendre jusqu'à plusieurs mois. Une stratégie pour accélérer le processus de purification consiste à isoler des spores mono-nucléées, ce qui garantit un ADN homogène.

Pour ce faire, nous examinons d'abord la possible corrélation entre la taille des spores et le nombre de noyaux qu'elles contiennent, démontrant ainsi qu'une spore contenant un noyau unique a tendance à être plus petite qu'une spore contenant de multiples noyaux. Nous proposons donc un outil microfluidique mettant en œuvre une séparation basée sur la taille, exploitant le déplacement latéral déterministe (DLD), dans le but d'obtenir un enrichissement d'une suspension de spores en spores mono-nucléées. Nous avons d'abord caractérisé la puce DLD grâce à une suspension de billes commerciales, afin d'optimiser le design du dispositif, les paramètres d'écoulement et la composition du milieu suspendant, avant de quantifier son efficacité de tri. Finalement, nous avons testé le dispositif avec des spores de *B. cinerea* et étudié sa capacité à enrichir une suspension contenant initialement un mélange de spores mono-nucléées et multi-nucléées. Après traitement, l'échantillon est enrichi en spores mono-nucléées, ce qui augmente la probabilité d'obtenir un ADN homogène et démontre le potentiel de la puce à accélérer la purification génétique des champignons.

En outre, les signatures biophysiques de la paroi cellulaire fongique (FCW) ont été étudiées pour donner un aperçu de ses propriétés mécaniques et électriques, en utilisant la microscopie à force atomique (AFM) et la spectroscopie d'impédance électrique (EIS), respectivement. Pour les expériences AFM, nous avons utilisé un cantilever sans pointe pour comparer la réponse à la déformation de la souche sauvage et de la souche mutante Δ AGS, cette dernière ayant une mutation impactant le FCW. Cela nous a permis d'évaluer les différences de rigidité mécanique entre les deux souches. Pour les mesures d'impédance, nous avons fabriqué un dispositif microfluidique pour le piégeage de spores uniques sur des paires intégrées d'électrodes en or. Des tests préliminaires ont été effectués avec des billes de polystyrène pour valider le dispositif. À notre connaissance, cette question n'a jamais été abordée auparavant sur cette souche de spores fongiques et notre objectif est de savoir comment la paroi

fongique peut être affectée par des mutations perturbant les gènes impliqués dans sa synthèse ainsi que de trouver une signature corrélée au nombre de noyaux.

Abstract

B. cinerea is a necrotrophic plant pathogen that can infect over 200 plant species worldwide and is considered the second most significant fungal pathogen. Genetically modified strains are often produced by biologists to study the influence of specific genes on fungal pathogenesis. However, because *B. cinerea*'s spores are multinucleated, obtaining a purified population in which all nuclei contain the desired mutation is a long and tedious process that can take up to several months. One strategy to speed up the purification process is to isolate mononucleated spores, hence ensuring homogeneous DNA.

To do so, we first examined a possible correlation between spore size and nuclei number, thus demonstrating that spores with a single nucleus tend to be smaller than those with multiple nuclei. Hence, we propose a microfluidic tool implementing a size-based sorting approach to enrich a spore suspension with mononucleated spores, using deterministic lateral displacement (DLD). We first characterize the DLD chip using a commercial bead suspension to optimize the device design, flow parameters, and suspending media composition. We then quantify its sorting efficiency with the same bead suspension. Finally, we test the chip with *B. cinerea* spores and observe its ability to enrich a suspension initially containing a mix of mono- and multinucleated spores. After processing, the sample is enriched with mononucleated spores, increasing the likelihood of obtaining homogeneous DNA and demonstrating the chip's potential to accelerate fungal genetic purification.

In addition, biophysical signatures of the fungal cell wall (FCW) were investigated to provide insights into its mechanical and electrical properties, using Atomic Force Microscopy (AFM) and Electrical Impedance Spectroscopy (EIS), respectively. For the AFM experiments, we employed a tipless cantilever to compare the indentation response of the wild-type strain and the mutant strain Δ AGS; the latter impacting the FCW. This allowed us to assess differences in mechanical stiffness between the two strains. For the impedance measurements, we fabricated a microfluidic device for single spore trapping over integrated pairs of gold electrodes. Preliminary tests were performed using polystyrene beads to validate the setup. To our knowledge, this has not been previously addressed on this strain of fungal spores and our goal is to get knowledge of how the fungal wall can be affected by mutations disrupting genes involved in its synthesis as well as finding a signature correlated to the number of nuclei.